Thin gel limit

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Abstract

Key words

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1 Introduction

1.1 Notation and definitions

$$E_\varepsilon[u,v] := \int_0^1 \int_0^1 \frac{\gamma}{2}\left(u_x^2 + \varepsilon^{-2}u_y^2 + \varepsilon^2 v_x^2 + v_y^2 - 2\right) + H(J)\,dx\,dy, \tag{1.1}$$

where $\gamma := \frac{GV_m}{K_B T}$, with G the shear modulus, V_m the molecular volume of the solvent to which the gel film is exposed, K_B the Boltzmann's constant and T the temperature;

$$u = u(x,y),\ v = v(x,y),\ J = u_x v_y - u_y v_x; \quad u, v \in H^1(Q),\ Q := (0,1)^2; \tag{1.2}$$

H is the Flory-Huggins mixing energy density suitably normalized and written in Lagrangian coordinates:

$$H(J) = Jf(\phi_0/J) + \phi_0(1-\chi), \quad f(\phi) := (1-\phi)\ln(1-\phi) + \chi\phi(1-\phi), \tag{1.3}$$

that is,

$$H(J) := (J-\phi_0)\ln\left(1 - \frac{\phi_0}{J}\right) + \phi_0\chi\left(1 - \frac{\phi_0}{J}\right) + \phi_0(1-\chi), \quad J > \phi_0, \tag{1.4}$$

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where ϕ_0 is the initial polymer volume fraction, and χ is the Flory chemical interaction parameter between the polymer and the solvent. The constant $\phi_0(1-\chi)$ is added here so that $H(J) \geq 0$ (See Lemma 2), which helps slightly in the analysis. We may assume that

$$\gamma = 0.13/136.6 \approx 10^{-4}, \quad \chi = 0.348, \quad \phi_0 = 0.2, \tag{1.5}$$

as in the measurements in . We may start with the simpler problem in which the following Dirichlet conditions are imposed at the gel/substrate interface:

$$u(x,0) = x \quad \text{and} \quad v(x,0) = 0 \quad \text{for all} \quad x \in (0,1). \tag{1.6}$$

With an abuse of notation, let us denote by $((u^\varepsilon, v^\varepsilon))_\varepsilon$ a sequence of maps in $\mathcal{A}$, each of them a minimizer of E_ε, being ε itself an element of a sequence of epsilons that tends to zero.

Comparing against the optimal homogeneous vertical extension (u_0, v_0):

$$u_0(x,y) = x, \quad v_0(x,y) = \lambda_{\text{uni}} y, \quad 0 < x, y < 1, \tag{1.7}$$

where λ_{uni} is the unique minimizer (see Lemma 3) of

$$\lambda \in [\phi_0, \infty) \mapsto W \begin{pmatrix} 1 & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad W(\mathbf{F}) := \frac{\gamma}{2}\left(|\mathbf{F}|^2 - 3\right) + H(\det \mathbf{F}), \tag{1.8}$$

we obtain the energy bound

$$E_\varepsilon[u^\varepsilon, v^\varepsilon] \leq E_\varepsilon[u^0, v^0] = E_0, \quad \text{with} \quad E_0 := \frac{\gamma}{2}(\lambda_{\text{uni}}^2 - 1) + H(\lambda_{\text{uni}}). \tag{1.9}$$

(For the values of γ, ϕ_0 and χ in (1.5), the value of λ_{uni} is approx. 1.99) Defining

$$\mathbf{F}_\varepsilon := \begin{pmatrix} u_x^\varepsilon & \varepsilon^{-1} u_y^\varepsilon \\ \varepsilon v_x^\varepsilon & v_y^\varepsilon \end{pmatrix} \tag{1.10}$$

we have that $\mathbf{F}_\varepsilon^T \mathbf{F}_\varepsilon$ si symmetric and positive definite, hence $E_\varepsilon[u^\varepsilon, v^\varepsilon]$ can be written as

$$E_\varepsilon[u^\varepsilon, v^\varepsilon] := \int_0^1 \int_0^1 \gamma \left(\frac{\lambda_1^2 + \lambda_2^2}{2} - 1 \right) + H(J_\varepsilon)\, dx\, dy,$$

where λ_1 and λ_2 are the principal strectches (the square roots of the eigenvalues of $\mathbf{F}_\varepsilon^T \mathbf{F}_\varepsilon$ which are positive). Since we are studying the swelling of the gel, let us impose the pointwise constraint $J_\varepsilon \geq 1$. By Cauchy's inequality,

$$\frac{\lambda_1^2 + \lambda_2^2}{2} \geq \lambda_1 \lambda_2 = \sqrt{\lambda_1^2 \lambda_2^2} = \sqrt{\det \mathbf{F}_\varepsilon^T \mathbf{F}_\varepsilon} = \det \mathbf{F}_\varepsilon = J_\varepsilon \geq 1,$$

so that $E_\varepsilon[u^\varepsilon, v^\varepsilon] \geq 0$.

All in all, we may work in the space

$$\begin{aligned}\mathcal{A} := \big\{(u,v) \in H^1(Q,\mathbb{R}^2) : J(x,y) \geq 1 \text{ for a.e.} \,(x,y) \in Q;\\ u(x,0) = x, \quad v(x,0) = 0 \quad \text{for a.e.}\, x \in (0,1);\\ \text{and} \quad 0 \leq E_\varepsilon[u,v] \leq E_0\big\}\,. \end{aligned} \tag{1.11}$$

Since $H(J) \geq 0$, one consequence of the energy bound is that for all ε

$$||\nabla u^\varepsilon||^2_{L^2(Q)} \leq \int_Q |u^\varepsilon_x|^2 + \varepsilon^{-2}|u^\varepsilon_y|^2 \leq \frac{2}{\gamma} E_\varepsilon[u^\varepsilon, v^\varepsilon] + 2 \leq C. \tag{1.12}$$

The maps v^ε_y are also bounded in $L^2(Q)$. Hence, by the Dirichlet boundary condition and Poincaré's inequality, we may assume, without loss of generality that

$$u^\varepsilon \overset{H^1}{\rightharpoonup} u \quad \text{and} \quad v^\varepsilon_y \overset{L^2}{\rightharpoonup} \beta \tag{1.13}$$

for some maps $u \in H^1(Q)$ and $\beta \in L^2(Q)$. Since the trace operator on ∂Q is continuous with respect the weak convergence in H^1, the limit map u is the identity on the gel/substrate interface.

$$u(x,0) = x \quad \text{for a.e. } x \in (0,1). \tag{1.14}$$

using (1.12) and the lowersemicontinuity of $\int u_y^2$, it can be seen that

$$\int_Q u_y^2 \leq \liminf_{\varepsilon\to 0} \int_Q |u^2_y|^2 \leq \liminf_{\varepsilon\to 0} C\varepsilon^2 = 0. \tag{1.15}$$

Therefore, $u_y = 0$ a.e. and u is independent of y. Combining this with the boundary condition, we find that

$$u(x,y) = x \quad \text{for a.e. } (x,y) \in Q. \tag{1.16}$$

Consequently, $u_x(x,y) = 1$ for a.e. $(x,y) \in Q$. Using the lowersemicontinuity of $\int u_x^2$ we find that

$$\int_Q \frac{\gamma}{2}\left(1 + \beta(x,y)^2\right) = \int_Q \frac{\gamma}{2}\left(u_x^2 + \beta^2\right) \leq \liminf_{\varepsilon\to 0} \int_Q \frac{\gamma}{2}\left(|u^\varepsilon_x|^2 + |v^\varepsilon_y|^2\right). \tag{1.17}$$

Hence,

$$\begin{aligned}&\int_Q \frac{\gamma}{2}\left(\beta(x,y)^2 - 1\right) + \liminf_{\varepsilon\to 0}\int_Q H(J_\varepsilon)\\ &\leq \liminf_{\varepsilon\to 0} \int_Q \frac{\gamma}{2}\left(|u^\varepsilon_x|^2 + |v^\varepsilon_y|^2 - 2\right) + \liminf_{\varepsilon\to 0} H(J_\varepsilon)\\ &\leq \liminf_{\varepsilon\to 0} \int_Q \frac{\gamma}{2}\left(|u^\varepsilon_x|^2 + |v^\varepsilon_y|^2 - 2\right) + H(J_\varepsilon) \leq \liminf_{\varepsilon\to 0} E[u^\varepsilon, v^\varepsilon] \leq E_0. \end{aligned} \tag{1.18}$$

If life were good and v^ε converged strongly in H^1 to some v in H^1, the Jacobians J_ε would converge to

$$\begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ v_x & v_y \end{vmatrix} = v_y = \beta$$

weakly in L^1. This, and perhaps other hypotheses on the limit Jacobian not leaving the domain of definition of H, would yield, using the convexity of H,

$$\int_Q H(\beta(x,y)) \le \liminf_{\varepsilon\to 0} \int_Q H(J_\varepsilon). \tag{1.19}$$

Since λ_{uni} is the unique minimizer for the auxiliary function w defined in Lemma 3, we would finally see that

$$\begin{aligned} E_0 = \int_Q \frac{\gamma}{2}(\lambda_{\text{uni}} - 1) + H(\lambda_{\text{uni}}) &\le \int_Q \frac{\gamma}{2}(\beta(x,y)^2 - 1) + H(\beta(x,y)) \\ &\le \liminf_{\varepsilon\to 0} E_\varepsilon[u^\varepsilon, v^\varepsilon] \le \limsup_{\varepsilon\to 0} E_\varepsilon[u^\varepsilon, v^\varepsilon] \le E_0. \end{aligned} \tag{1.20}$$

This would imply that

$$\beta(x,y) = \lambda_{\text{uni}} \quad \text{for a.e.} \quad (x,y) \in Q \tag{1.21}$$

and that

$$E_\varepsilon[u^\varepsilon, v^\varepsilon] \xrightarrow{\varepsilon\to 0} E_0. \tag{1.22}$$

Clearly we do not have strong H^1 convergence for v^ε. The question is whether, in spite of that, we can prove that $J_\varepsilon \rightharpoonup \beta$ in $L^1(Q)$.

The Jacobian can be written as a divergence and integrated, since we are in $2D$ and (u,v) is in H^1. Let φ be in $C^1_c(Q)$,

$$\int_Q J_\varepsilon(x,y)\varphi(x,y) = \int u^\varepsilon_x v^\varepsilon_y \varphi - \int u^\varepsilon_y v^\varepsilon_x \varphi = \int v^\varepsilon u^\varepsilon_y \varphi_x - \int v^\varepsilon u^\varepsilon_x \varphi_y. \tag{1.23}$$

Since $u^\varepsilon_y \rightharpoonup 0$ and u^ε_x converges weakly to the constant function 1, having a strong L^2 convergence of v^ε to the map

$$v(x,y) := \int_0^y \beta(x,t)\,dt \tag{1.24}$$

would give:

$$\begin{aligned} \int_Q J_\varepsilon(x,y)\varphi(x,y) &= -\int_Q v(x,y)\varphi_y(x,y) \\ &= \int_Q v_y(x,y)\varphi(x,y) = \int_Q \beta(x,y)\varphi(x,y). \end{aligned} \tag{1.25}$$

If (J_ε) happened to be equiintegrable (if for example, the argument of Müller or of Coifman-Lions-Meyers-Semmes worked in spite of the dependence of ε), we could pass from that distributional convergence to the weak convergence in L^1 (that is, the convergence of v^ε would not need to be in H^1). For the time being, let us suppose that (J_ε) is equiintegrable. (If worse comes to worst, we can even impose an a priori upper bound for J in the definition of the space $\mathcal{A}$, just as we arbitrarily impose that $J \ge 1$. After all, the swelling factor is expected to be pointwise less than 3.29, the value observed in the free-swelling experiment.)

One question is whether we could use the arguments in any proof Rellich-Kondrachov in order to obtain strong convergence using, on the one hand, less ingredients (controlling only v_y^ε and v_x^ε), but , on the other hand, the boundary contidition $v \equiv 0$ on the bottom and that the boundary is flat. That is, whether we can find some argument involving only vertical slices and the vertical partial derivative, which somohow yields a control that is uniform in x. If one of the proof of Rellich-Kondrachov were by contradiction, perhaps the boundary condition $v = 0$ and the simple geometry we are considering might be compatible with such an argument.

Something else to consider is that the map

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto \frac{\gamma}{2}(a^2+b^2+c^2+d^2-2) + H\left(\begin{vmatrix} a & b \\ c & d \end{vmatrix}\right) \tag{1.26}$$

is rotation-invariant. Minimizers, hence, can be studied asumming $b = c = 0$. Partial derivatives are

$$\gamma a + dH'(ad), \quad \gamma d + aH'(ad).$$

Regarding the map as as function of (a, d), its only critical point, then, is given by the equations

$$adH'(ad) = -\gamma ad, \quad \begin{vmatrix} a & b \\ -H'(ad) & \gamma \end{vmatrix} = 0.$$

Let us denote by λ^* the unique solution of

$$H'(J^*) = -\gamma, \quad J^* = (\lambda^*)^2, \tag{1.27}$$

(which exists, and is indeed unique, by Lemmas 1 and 2). Let us add the hypothesis that

$$\text{The parameters}\, \gamma, \phi_0, \text{and}\, \chi \,\text{are such that}\, \lambda^* > 1. \tag{1.28}$$

(For the values in (1.5), the value of λ^* is approx. 1.65.) Since $ad = J \geq 1$, we can divide the first of the equations above by add and find that $H'(ad) = -\gamma$ and $ad = J^*$. Substituting this into the second equation we find that the vector the only critical points is the equibiaxial strain $a = d = \lambda^*$.

By Cauchy's inequality,

$$\frac{\gamma}{2}(a^2+d^2-2) + H(ad) \geq \gamma(J-1) + H(J) \quad \text{with} \quad J = ad.$$

Regarding the expression at the right-hand side as a function of J we see, from Lemmas 1-2, that it tends to infinity as $J \to \infty$, that it is convex, that it is decreasing for $J < J^*$ and increasing for $J > J^*$. In particular, it has a unique minimizer at $J = J^*$.

All in all, the only critical point of $(a, d) \mapsto \frac{\gamma}{2}(a^2+d^2-2) + H(ad)$ is $(a, d) = (\lambda^*, \lambda^*)$ and the set of minimizers of (1.26) consists of all the rotations of $(\lambda^*)\mathbf{I}$, where $\mathbf{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. Two such rotations are

$$\begin{pmatrix} 1 & \pm\sqrt{J^*-1} \\ \mp\sqrt{J^*-1} & 1 \end{pmatrix}$$

. In particular, they are better energetically than the vertical extension $a = 1$, $b = c = 0$, $d = \lambda_{\text{uni}}$. One of the things that need to be justified is why fine mixtures with gradients of the form

$$u_x^\varepsilon = 1, \quad u_y^\varepsilon = \pm\varepsilon\sqrt{J^* - 1}, \quad v_x^\varepsilon = \mp\varepsilon^{-1}\sqrt{J^* - 1}, \quad v_y^\varepsilon = 1$$

do not show up in this particular thin gel limit. Necessarily the boundary condition $v \sim 0$, or the injectivity of maps (u, v), or their continuity in the x direction, or the expansive nature of the Jacobians, or something else, will need to be invoked.

The following can be said about the first term in (1.23):

$$\left| \int_Q v^\varepsilon u_y^\varepsilon \varphi_x \right| \le ||\varphi_x||_{L^\infty} ||v^\varepsilon||_{L^2(Q)} \underbrace{||u_y^\varepsilon||_{L^2(Q)}}_{\le \sqrt{\frac{2}{\gamma}E_0 + 2\varepsilon}} \tag{1.29}$$

and

$$\begin{aligned} \int_Q v^\varepsilon(x,y)^2 d(x,y) &= \int_Q \left| \int_0^y v_y^\varepsilon(x,t)\,dt \right|^2 d(x,y) \\ &\le \int_Q \left(\int_0^y v_y^\varepsilon(x,t)\,dt \right)^2 d(x,y) \\ &\le \int_{x=0}^1 \int_{y=0}^1 \left(\int_0^y |v_y^\varepsilon(x,t)|\,dt \right)^2 dy\,dx \\ &\le \int_{x=0}^1 \left(\int_0^1 |v_y^\varepsilon(x,t)|\,dt \right)^2 dx \\ &\le \int_{x=0}^1 \int_{t=0}^1 |v_y^\varepsilon(x,t)|^2\,dt\,dx \le \frac{2}{\gamma}E_0 + 2. \end{aligned} \tag{1.30}$$

Therefore, what we would like to prove is that

$$\int_Q v^\varepsilon u_x^\varepsilon \varphi_y \xrightarrow{\varepsilon \to 0} \int_Q v(x,y)\varphi_y(x,y),$$

with v defined in (1.24). A first estimate is that:

$$\begin{aligned} &\left| \int_Q v^\varepsilon u^\varepsilon \varphi_y - \int_Q v(x,y)\varphi_y(x,y) \right| \\ &\le \left| \int_Q u_x^\varepsilon v \varphi_y - \int_Q v(x,y)\varphi_y(x,y) \right| + \left| \int_Q (v^\varepsilon - v) u_x^\varepsilon \varphi_y \right|, \end{aligned} \tag{1.31}$$

but this is just to recall that what we need to prove is that

$$\int_Q (v^\varepsilon - v) u_x^\varepsilon \varphi_y \xrightarrow{\varepsilon \to 0} 0. \tag{1.32}$$

Equation (1.30) shows that, up to a subsequence, v^ε converges weakly in L^2. A first challenge that could try and overcome is to show that the weak limit is the map v defined in (1.24). The problem with an argument like

$$\begin{aligned}\int_Q v^\varepsilon \varphi &= \int_{x=0}^1 \int_{y=0}^1 \left(\int_{t=0}^y v_y^\varepsilon(x,t)\,dt \right) \varphi(x,y)\,dy\,dx \\ &= \int_{x=0}^1 \int_{y=0}^1 \left(\int_{t=0}^1 v_y^\varepsilon(x,t)\chi_{t\le y}\,dt \right) \varphi(x,y)\,dy\,dx \\ &= \int_{x=0}^1 \int_{t=0}^1 v_y^\varepsilon(x,t) \left(\int_{t=0}^1 \chi_{t\le y}\,\varphi(x,y)\,dy \right) dt\,dx = \int_Q v_y^\varepsilon \Phi, \end{aligned} \tag{1.33}$$

with

$$\Phi(x,t) := \int_{y=t}^1 \varphi(x,y)\,dy \tag{1.34}$$

is that the test function $\tilde{\varphi}$ is not of compact support (it does not on the top boundary, before $t=1$).

Now, since in the end we want to prove that β is constant, perhaps we could work form the begining with the vertical averages $\int_0^1 v_y^\varepsilon$ rather than with the maps v_y^ε themselves. Because of the boundary condition, these vertical averages are the upper traces $v^\varepsilon(x,1)$.

Another piece of information that we have, althought perhaps does not help, is that

$$\int_0^1 v^\varepsilon(x,1)^2\,dx = \int_0^1 \left(\int_0^1 |v^\varepsilon(x,y)|\,dy \right)^2 dx \le \int_0^1 \int_0^1 (v_y^\varepsilon)^2\,dy\,dx \le C, \tag{1.35}$$

so the upper traces have a weak limit in L^2 (up to a subsequence). The same can be said for the averages $x \mapsto \int_0^1 v^\varepsilon(x,y)dy$:

$$\begin{aligned}\int_0^1 |\tilde{v}^\varepsilon(x)|^2 dx &\le \int_0^1 \left(\int_0^1 |v^\varepsilon(x,y)|\,dy \right)^2 dx \\ &\le \int_0^1 \left(\int_0^1 \left| \int_0^1 v_y^\varepsilon(x,t)\,dt \right| dy \right)^2 dx \\ &\le \int_{x=0}^1 \int_{t=0}^1 \left| v_y^\varepsilon(x,t) \right|^2 dt\,dx \le \frac{2}{\gamma} E_0. \end{aligned} \tag{1.36}$$

2 Properties of $H(J)$

LEMMA 2.1. *$H(J)$ is convex in the whole range $J > \phi_0$ if and only if $\chi < \frac{1}{2}$.*

Proof. Writting $\phi = \phi_0/J$, we have that

$$\begin{aligned} H'(J) &= f(\phi) + Jf'(\phi)\cdot(-\phi_0 J^{-2}), \\ &= f(\phi) - \phi f'(\phi) = \phi + \ln(1-\phi) + \chi\phi^2. \end{aligned} \tag{2.1}$$

Observe that

$$H''(J) = f'(\phi) - f'(\phi) - \phi f''(\phi)\cdot(-\phi_0 J^{-2}) = \frac{\phi^2}{J} f''(\phi). \tag{2.2}$$

Hence, it suffices to show that $f''(\phi) > 0$ for all $0 < \phi < 1$ if and only if $\chi < \frac{1}{2}$.

Now,

$$f'(\phi) = -(1 + \ln(1-\phi)) + \chi(1 - 2\phi), \tag{2.3}$$

$$f''(\phi) = \frac{1}{1-\phi} - 2\chi = \frac{1 - 2\chi + 2\chi\phi}{1-\phi}. \tag{2.4}$$

The denominator is positive when $0 < \phi < 1$. The numerator is affine in ϕ, with extreme values $1 - 2\chi$ and 1. The values in the interval $\phi \in (0, 1)$ will all be positive if and only if $1 - 2\chi > 0$, that is, if and only if $\chi < \frac{1}{2}$. □

LEMMA 2.2. *$H(J)$ is decreasing in (ϕ_0, ∞),*

$$H(\phi_0^-) = \phi_0(1-\chi), \quad H(\infty) = 0, \quad H'(\phi_0^-) = -\infty, \quad H'(\infty) = 0.$$

Proof. From (2.1) it can be seen that

$$\lim_{J\to\infty} H'(J) = \lim_{\phi\to 0^+} \phi + \ln(1-\phi) + \chi\phi^2 = 0. \tag{2.5}$$

Since H is strictly convex, H is necessarily strictly decreasing in all its domain. In addition,

$$\lim_{J\to\phi_0^+} H'(J) = \lim_{\phi\to 1^-} \phi + \ln(1-\phi) + \chi\phi^2 = -\infty. \tag{2.6}$$

As for the extrem values of H, it suffices to observe that

$$\lim_{J\to\phi_0^+} H'(J) = \phi_0(1-\chi) + \phi_0 \cdot \lim_{\phi\to 1^-} f(\phi) = \phi_0(1-\chi) + 0 = 0, \tag{2.7}$$

$$\lim_{J\to\infty} H(J) = \phi_0(1-\chi) + \lim_{J\to\infty} Jf(\phi_0/J) = \phi_0(1-\chi) + \lim_{\phi\to 0^-} \frac{\phi_0}{\phi} f(\phi), \tag{2.8}$$

$$= \phi_0(1-\chi) + \phi_0 \cdot \lim_{\phi\to 0^-} f'(\phi) = \phi_0(1-\chi) - \phi_0(1-\chi) = 0. \tag{2.9}$$

□

LEMMA 2.3. *The expression in (1.8), which defines the optimal vertical extension, has a unique minimizer.*

Proof. Set

$$w(\lambda) := W\begin{pmatrix}1 & 0 & 0\\ 0 & \lambda & 0\\ 0 & 0 & 1\end{pmatrix} = \frac{\gamma}{2}(\lambda^2 - 1) + H(\lambda).$$

By Lemmas 1-2, its derivative

$$w'(\lambda) = \gamma\lambda + H'(\lambda)$$

is an increasing function of λ, with values that range from $-\infty$ to ∞. The optimal extension λ_{uni} is, then, the unique solution to

$$\gamma\lambda + H'(\lambda) = 0.$$

□

3 Nonlinear analysis for perturbed surface

$$\begin{aligned} u(x,y) &= \frac{y_1}{L} = \frac{Lx + u_1(Lx, \lambda dy)}{L} = x + L^{-1}u_1(Lx, \lambda dy),\\ v(x,y) &= \frac{y_1}{d} = \frac{\lambda dy + u_2(Lx, \lambda dy)}{d} = \lambda y + d^{-1}u_2(Lx, \lambda dy). \end{aligned}$$

then

$$\begin{aligned} u_\varepsilon(x,y) &= x + L_\varepsilon^{-1}u_1(L_\varepsilon x, \lambda d_\varepsilon y),\\ v_\varepsilon(x,y) &= \lambda y + d_\varepsilon^{-1}u_2(L_\varepsilon x, \lambda d_\varepsilon y), \end{aligned}$$

where

$$\frac{d_\varepsilon}{L_\varepsilon} = \varepsilon.$$

Questions:

- Is $E_\varepsilon(u,v) < E_0$?.
- Does $\limsup_{\varepsilon\to 0} E_0 - E_\varepsilon(u,v) > 0$?.

3.1 Second Variation Functional computing

Since

$$E_\varepsilon[u,v] := \int_0^1\int_0^1 \frac{\gamma}{2}\left(u_x^2+\varepsilon^{-2}u_y^2+\varepsilon^2 v_x^2+v_y^2-2\right)+H(J)\,dx\,dy, \tag{3.1}$$

where

$$u=u(x,y),\ v=v(x,y),\ J=u_xv_y-u_yv_x;\quad u,v\in H^1(Q),\ Q:=(0,1)^2,$$

and $H(\cdot)$ function is defined as in (1.3). Then computing the first variation, we obtain:

$$\begin{aligned}
\delta E_\varepsilon[(u,v),(\dot u,\dot v)] &= \left\langle E'_\varepsilon[u,v]|_{(u,v)=(u_0,v_0)},(\dot u,\dot v)\right\rangle\\
&= \int_0^1\int_0^1 \frac{\gamma}{2}\left(2(u_0)_x\dot u_x+2\varepsilon^{-2}(u_0)_y\dot u_y+2\varepsilon^2(v_0)_x\dot v_x+2(v_0)_y\dot v_y\right)\\
&\quad + H'(J)\begin{pmatrix}(v_0)_y & -(v_0)_x\\ -(u_0)_y & (u_0)_x\end{pmatrix}\cdot\begin{pmatrix}\dot u_x & \dot u_y\\ \dot v_x & \dot v_y\end{pmatrix}dx\,dy,\\
&= \int_0^1\int_0^1 \gamma\left((u_0)_x\dot u_x+\varepsilon^{-2}(u_0)_y\dot u_y+\varepsilon^2(v_0)_x\dot v_x+(v_0)_y\dot v_y\right)\\
&\quad + H'(J)\begin{pmatrix}(v_0)_y & -(v_0)_x\\ -(u_0)_y & (u_0)_x\end{pmatrix}\cdot\begin{pmatrix}\dot u_x & \dot u_y\\ \dot v_x & \dot v_y\end{pmatrix}dx\,dy,
\end{aligned}$$

$$\begin{aligned}
\langle E'_\varepsilon,(\dot u,\dot v)\rangle &= \int_0^1\int_0^1 \gamma\begin{pmatrix}u_x & \varepsilon^{-2}u_y\\ \varepsilon^2 v_x & v_y\end{pmatrix}\cdot\begin{pmatrix}\dot u_x & \dot u_y\\ \dot v_x & \dot v_y\end{pmatrix}\\
&\quad + H'(J)\begin{pmatrix}v_y & -v_x\\ -u_y & u_x\end{pmatrix}\cdot\begin{pmatrix}\dot u_x & \dot u_y\\ \dot v_x & \dot v_y\end{pmatrix}dx\,dy,\\
&= \int_0^1\int_0^1\left[\gamma\begin{pmatrix}u_x & \varepsilon^{-2}u_y\\ \varepsilon^2 v_x & v_y\end{pmatrix}+H'(J)\begin{pmatrix}v_y & -v_x\\ -u_y & u_x\end{pmatrix}\right]\cdot\begin{pmatrix}\dot u_x & \dot u_y\\ \dot v_x & \dot v_y\end{pmatrix}dx\,dy,
\end{aligned}$$

Therefore,

$$\begin{aligned}
\langle\langle E''_\varepsilon,(\dot u,\dot v)\rangle,(\dot u,\dot v)\rangle &= \int_0^1\int_0^1\left[\gamma\begin{pmatrix}\dot u_x & \varepsilon^{-2}\dot u_y\\ \varepsilon^2\dot v_x & \dot v_y\end{pmatrix}+H''(J)\dot J\begin{pmatrix}v_y & -v_x\\ -u_y & u_x\end{pmatrix}\right.\\
&\quad\left.+H'(J)\begin{pmatrix}\dot v_y & -\dot v_x\\ -\dot u_y & \dot u_x\end{pmatrix}\right]\cdot\begin{pmatrix}\dot u_x & \dot u_y\\ \dot v_x & \dot v_y\end{pmatrix}dx\,dy.
\end{aligned}$$

By virtue of the previous computation, we can calculate the second variation as follows

$$\begin{aligned}
\langle\langle E''_\varepsilon,(\dot u,\dot v)\rangle,(\dot u,\dot v)\rangle &= \int_0^1\int_0^1\gamma\begin{pmatrix}\dot u_x^2 & \varepsilon^{-2}\dot u_y^2\\ \varepsilon^2\dot v_x^2 & \dot v_y^2\end{pmatrix}\\
&\quad +H''(J)\left[\begin{pmatrix}v_y & -v_x\\ -u_y & u_x\end{pmatrix}\cdot\begin{pmatrix}\dot u_x & \dot u_y\\ \dot v_x & \dot v_y\end{pmatrix}\right]\begin{pmatrix}v_y & -v_x\\ -u_y & u_x\end{pmatrix}\cdot\begin{pmatrix}\dot u_x & \dot u_y\\ \dot v_x & \dot v_y\end{pmatrix}\\
&\quad + H'(J)(\dot v_y\dot u_x-\dot v_x\dot u_y-\dot u_y\dot v_x+\dot u_x\dot v_y)\,dx\,dx.
\end{aligned}$$

Evaluating at $(u_0, v_0) = (x, \lambda y)$ yields

$$\left\langle \left\langle E''_\varepsilon|_{(u,v)=(u_0,v_0)}, (\dot{u}, \dot{v}) \right\rangle, (\dot{u}, \dot{v}) \right\rangle = \int_0^1 \int_0^1 \gamma \begin{pmatrix} \dot{u}_x^2 & \varepsilon^{-2}\dot{u}_y^2 \\ \varepsilon^2 \dot{v}_x^2 & \dot{v}_y^2 \end{pmatrix}$$
$$+ H''(\lambda) \cdot (\lambda \dot{u}_x + \dot{v}_y)^2 + H'(\lambda)(2\dot{v}_y \dot{u}_x - 2\dot{v}_x \dot{u}_y)\, dx\, dy.$$

In addition, using the notation that you sugested: u instead of that here we call $\dot{u}$ and v instead of that we call $\dot{v}$, the second variation E''_ε, evaluated at (u_0, v_0), where $u_0(x, y) = x$ and $v_0(x, y) = \lambda y$ is the cuadratic form

$$(u, v) \in (H^1(\Omega))^2 \mapsto \int_0^1 \int_0^1 \gamma(u_x^2 + \varepsilon^{-2}u_y^2 + \varepsilon^2 v_x^2 + v_y^2)$$
$$+ H''(\lambda) \cdot (\lambda u_x + v_y)^2 + 2H'(\lambda)(v_y u_x - v_x u_y)\, dx\, dy.$$

By setting,

$$a = H''(\lambda), \tag{3.2}$$
$$b = 2H'(\lambda), \tag{3.3}$$

we have

$$(u, v) \in (H^1(\Omega))^2 \mapsto E''_\varepsilon[u, v] := \int_0^1 \int_{-1}^1 \gamma(u_x^2 + \varepsilon^{-2}u_y^2 + \varepsilon^2 v_x^2 + v_y^2)$$
$$+ a(\lambda u_x + v_y)^2 + b(v_y u_x - v_x u_y)\, dx\, dy, \tag{3.4}$$

Or well, it can be see (integration by parts) as the operator

$$\begin{pmatrix} u \\ v \end{pmatrix} \in (H^1(\Omega))^2 \mapsto$$
$$\begin{pmatrix} -\gamma u_{xx} - \gamma\varepsilon^{-2}u_{yy} - \lambda^2 H''(\lambda)u_{xx} - 2\lambda H''(\lambda)v_{yx} - 2H'(\lambda)v_{yx} + 2H'(\lambda)v_{xy} \\ -\gamma\varepsilon^2 v_{xx} - \gamma v_{yy} - H''(\lambda)v_{yy} - 2\lambda H''(\lambda)u_{xy} - 2H'(\lambda)u_{xy} + 2H'(\lambda)u_{yx} \end{pmatrix} \in (H^1(\Omega))^2$$

3.2 Fourier transform analysis for second variation of E_ε using sine and cosine type functions

In the sequel, we denote E''_ε the second variation of our energy E_ε in (3.1). Based on the insight provided in the linear stability analysis by Huang and Kang [7], we study the functional in the restricted class of vector-valued functions $(u(x, y), v(x, y))$ having the following form

$$u(x, y) = \varepsilon p(y) \cos(k\pi x),$$
$$v(x, y) = s(y) \sin(k\pi x),$$

then energy should be:

$$E''_\varepsilon = \frac{\gamma}{2}\int_0^1 \left[(s')^2 + (p')^2 + \varepsilon^2 k^2\pi^2(p^2+s^2)\right]\,dy$$
$$+ a\int_0^1 (s' - \lambda\varepsilon k\pi p)^2\,dy + b\varepsilon k\pi\int_0^1 (-sp)'\,dy.$$

By setting

$$\varepsilon k\pi = 1, \tag{3.5}$$

then our functions are

$$u(x,y) = \varepsilon p(y)\cos(x/\varepsilon),$$
$$v(x,y) = s(y)\sin(x/\varepsilon),$$

and the energy should be:

$$E''_\varepsilon = \frac{\gamma}{2}\int_0^1 \left[(s')^2 + (p')^2 + (p^2+s^2)\right]\,dy$$
$$+ a\int_0^1 (s' - \lambda p)^2\,dy + b\int_0^1 (-sp)'\,dy. \tag{3.6}$$

3.2.1 Computing the Euler-Lagrange equations for (3.6)

From the first variation of (3.6) with respect to p and s respectively we obtain

$$\begin{cases} -\gamma p'' + (\gamma + 2a\lambda^2)p - 2a\lambda s' = 0, \\ -(\gamma+2a)s'' + \gamma s + 2a\lambda p' = 0. \end{cases} \tag{3.7}$$

Terms on the boundary in the integration by parts are given by

$$\begin{cases} p = s = 0, \quad \text{on} \quad y = 0, \\ \gamma p' - bs = 0, \quad \text{on} \quad y = 1, \\ (\gamma+2a)s' - (2a\lambda + b)p = 0, \quad \text{on} \quad y = 1. \end{cases} \tag{3.8}$$

3.3 Eigenvalues problem

We are seek for Υ such that the system

$$\begin{cases} -\gamma p'' + (\gamma + 2a\lambda_{\text{uni}}^2 - \Upsilon)p - 2a\lambda_{\text{uni}} s' = 0, \\ -(\gamma+2a)s'' + (\gamma - \Upsilon)s + 2a\lambda_{\text{uni}} p' = 0. \end{cases} \tag{3.9}$$

with the boundary condition

$$\begin{cases} p = s = 0, \quad \text{on} \quad y = 0, \\ \gamma p' - bs = 0, \quad \text{on} \quad y = 1, \\ (\gamma+2a)s' - (2a\lambda_{\text{uni}} + b)p = 0, \quad \text{on} \quad y = 1. \end{cases} \tag{3.10}$$

have solution.

3.3.1 Computing the solution of system ODE (3.9)

For system (3.9), by setting $c_1 = \gamma + 2a\lambda_{\text{uni}}^2 - \Upsilon$, $c_3 = -2a\lambda_{\text{uni}}$, $c_5 = -(\gamma + 2a)$, system can be rewritten as follows:

$$\begin{cases} c_1 p - \gamma p'' + c_3 s' = 0, \\ (\gamma - \Upsilon)s + c_5 s'' - c_3 p' = 0. \end{cases}$$

Let us seek solutions of the form

$$\begin{aligned} p(y) &= p_0 \cosh(\mu y), \\ s(y) &= s_0 \sinh(\mu y). \end{aligned}$$

Then, the equations become

$$\begin{cases} c_1 p_0 - \gamma\mu^2 p_0 + c_3 \mu s_0 = 0, \\ (\gamma - \Upsilon)s_0 + c_5\mu^2 s_0 - c_3\mu p_0 = 0. \end{cases}$$

The determinant of this 2×2 system on p_0, s_0 is

$$\begin{vmatrix} c_1 - \gamma\mu^2 & c_3\mu \\ -c_3\mu & \gamma - \Upsilon + c_5\mu^2 \end{vmatrix}.$$

For a nontrivial solution to exist, it is necessary that

$$-\Upsilon c_1 + \Upsilon\gamma\mu^2 + c_1\gamma + c_1 c_5 \mu^2 - \gamma^2\mu^2 - \gamma c_5 \mu^4 + c_3^2\mu^2 = 0,$$

i.e.

$$\mu^2 = \frac{-(\Upsilon\gamma + c_1 c_5 - \gamma^2 + c_3^2) \pm \sqrt{(\Upsilon\gamma + c_1 c_5 - \gamma^2 + c_3^2)^2 - 4(-\gamma c_5)(c_1\gamma - c_1\Upsilon)}}{-2\gamma c_5}.$$

$$\begin{aligned} -(\Upsilon\gamma + c_1 c_5 - \gamma^2 + c_3^2) &= -2[a(\Upsilon - \gamma(\lambda_{\text{uni}}^2 + 1)) + \gamma(\Upsilon - \gamma)] \\ &= 2\gamma(\gamma + a\lambda_{\text{uni}}^2 + a) - 2(a + \gamma)\Upsilon \end{aligned}$$

By simplicity $\lambda_{\text{uni}} := \lambda$, then

$$\begin{aligned} (\Upsilon\gamma + c_1 c_5 - \gamma^2 + c_3^2)^2 &= [2\gamma(\gamma + a\lambda^2 + a) - 2(a + \gamma)\Upsilon]^2 \\ &= 4\gamma^4 + 8a\gamma^3\lambda^2 + 4a^2\gamma^2\lambda^4 + 8a\gamma^3 + 8a^2\lambda^2\gamma^2 + 4a^2\gamma^2 - 16a\gamma^2\Upsilon - 8a^2\gamma\lambda^2\Upsilon \\ &\quad - 8a^2\gamma\Upsilon - 8\gamma^3\Upsilon - 8a\lambda^2\gamma^2\Upsilon + 4a^2\Upsilon^2 + 8a\gamma\Upsilon^2 + 4\gamma^2\Upsilon^2, \end{aligned}$$

$$\begin{aligned} -4(-\gamma c_5)(c_1\gamma - c_1\Upsilon) &= -4\gamma^4 - 8a\gamma^3\lambda^2 + 4\gamma^3\Upsilon - 8a\gamma^3 - 16a^2\gamma^2\lambda^2 + 8a\gamma^2\Upsilon + 4\gamma^3\Upsilon \\ &\quad + 8a\gamma^2\lambda^2\Upsilon - 4\gamma^2\Upsilon^2 + 8a\gamma^2\Upsilon + 16a^2\gamma\lambda^2\Upsilon - 8a\gamma\Upsilon^2, \end{aligned}$$

$$(\Upsilon\gamma + c_1c_5 - \gamma^2 + c_3^2)^2 - 4(-\gamma c_5)(c_1\gamma - c_1\Upsilon) = a^2[(4\Upsilon^2 + \Upsilon\gamma(8\lambda_{\text{uni}}^2 - 8) + \gamma^2(4\lambda_{\text{uni}}^4 - 8\lambda_{\text{uni}}^2 + 4)]$$
$$= 4a^2(\Upsilon + \gamma(\lambda_{\text{uni}}^2 - 1))^2$$

$$\begin{aligned}\mu^2 &= \frac{2\gamma(\gamma + a\lambda_{\text{uni}}^2 + a) - 2(a+\gamma)\Upsilon \pm 2a(\Upsilon + \gamma(\lambda_{\text{uni}}^2 - 1))}{2\gamma(\gamma+2a)}\\ &= \frac{(\gamma + a\lambda_{\text{uni}}^2 + a) \pm a(\lambda_{\text{uni}}^2 - 1)}{(\gamma+2a)} + \frac{-2a\Upsilon - 2\gamma\Upsilon \pm 2a\Upsilon}{2\gamma(\gamma+2a)},\end{aligned}$$

now,

$$\begin{cases}\mu_{3,4}^2 = \dfrac{\gamma + 2a\lambda_{\text{uni}}^2 - \Upsilon}{(\gamma+2a)},\\ \mu_{1,2}^2 = \dfrac{\gamma - \Upsilon}{\gamma}.\end{cases}$$

We now have cases depending of Υ:

(a) $\Upsilon < \gamma(1-\lambda^2)$.

(b) $\Upsilon = \gamma(1-\lambda^2)$.

(c) $\gamma(1-\lambda^2) < \Upsilon < 0$.

(d) $\Upsilon = 0$.

(e) $0 < \Upsilon < \gamma$.

(f) $\Upsilon = \gamma$.

(g) $\gamma < \Upsilon < \gamma + 2a\lambda^2$.

(h) $\Upsilon = \gamma + 2a\lambda^2$.

(i) $\Upsilon > \gamma + 2a\lambda^2$.

3.4 Case $\Upsilon < \gamma$.

$$\begin{cases}\mu_{3,4} = \pm\sqrt{\dfrac{\gamma + 2a\lambda_{\text{uni}}^2 - \Upsilon}{(\gamma+2a)}},\\ \mu_{1,2} = \pm\sqrt{\dfrac{\gamma - \Upsilon}{\gamma}}.\end{cases} \tag{3.11}$$

When $\mu = \mu_1$, the system

$$\begin{pmatrix} c_1 - \gamma\mu^2 & c_3\mu \\ -c_3\mu & \gamma - \Upsilon + c_5\mu^2 \end{pmatrix}\begin{pmatrix} p_0 \\ s_0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

for p_0, s_0 becomes:

$$\begin{pmatrix} 2a\lambda_{\text{uni}}^2 & -2a\lambda_{\text{uni}}\mu_1 \\ 2a\lambda_{\text{uni}}\mu_1 & -2a\mu_1^2 \end{pmatrix} \begin{pmatrix} p_0 \\ s_0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

whose solutions are the multiples of

$$\begin{pmatrix} p_0 \\ s_0 \end{pmatrix} = \begin{pmatrix} \mu_1 \\ \lambda_{\text{uni}} \end{pmatrix}.$$

When $\mu = \mu_3$, the system

$$\begin{pmatrix} c_1 - \gamma\mu^2 & c_3\mu \\ -c_3\mu & \gamma - \Upsilon + c_5\mu^2 \end{pmatrix} \begin{pmatrix} p_0 \\ s_0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

for p_0, s_0 becomes:

$$\begin{pmatrix} 2a\mu_3^2 & -2a\lambda_{\text{uni}}\mu_3 \\ 2a\lambda_{\text{uni}}\mu_3 & -2a\lambda_{\text{uni}}^2 \end{pmatrix} \begin{pmatrix} p_0 \\ s_0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

whose solutions are the multiples of

$$\begin{pmatrix} p_0 \\ s_0 \end{pmatrix} = \begin{pmatrix} \lambda_{\text{uni}} \\ \mu_3 \end{pmatrix}.$$

All in all,

$$\begin{cases} p = \mu_1 \cosh(\mu_1 y), \\ s = \lambda_{\text{uni}} \sinh(\mu_1 y), \end{cases}$$

and

$$\begin{cases} p = \lambda_{\text{uni}} \cosh(\mu_3 y), \\ s = \mu_3 \sinh(\mu_3 y), \end{cases}$$

are two l.i. solutions.

Now seek solutions of the form

$$\begin{aligned} p(y) &= p_0 \sinh(\mu y), \\ s(y) &= s_0 \cosh(\mu y). \end{aligned}$$

This leads to exactly the same algebraic system in p_0, s_0. Thusm the following other two l.i. solutions appear:

$$\begin{cases} p = \mu_1 \sinh(\mu_1 y), \\ s = \lambda_{\text{uni}} \cosh(\mu_1 y), \end{cases}$$

and

$$\begin{cases} p = \lambda_{\text{uni}} \sinh(\mu_3 y), \\ s = \mu_3 \cosh(\mu_3 y). \end{cases}$$

Since the function space of solutions of the homogeneous system has dimension four, the any solution is of the form

$$p(y) = A_1\mu_1 \cosh(\mu_1 y) + A_2\mu_1 \sinh(\mu_1 y) + A_3\lambda_{\text{uni}} \cosh(\mu_3 y) + A_4\lambda_{\text{uni}} \sinh(\mu_3 y), \tag{3.12}$$

$$s(y) = A_1\lambda_{\text{uni}} \sinh(\mu_1 y) + A_2\lambda_{\text{uni}} \cosh(\mu_1 y) + A_3\mu_3 \sinh(\mu_3 y) + A_4\mu_3 \cosh(\mu_3 y), \tag{3.13}$$

where A_i are constants for $i \in \{1, 2, 3, 4\}$.

3.4.1 Computing the equations for amplitude A_n using the boundary conditions

In the other hand, by substituting explicit expressions for p and s given in (3.12)-(3.13) in the boundary conditions (3.10) we have the following equations:

$$A_1\mu_1 + \lambda_{\text{uni}}A_3 = 0, \tag{3.14}$$

$$\lambda_{\text{uni}}A_2 + \mu_3 A_4 = 0, \tag{3.15}$$

$$\begin{aligned}(\gamma - \Upsilon - b\lambda_{\text{uni}})\sinh(\mu_1)A_1 + (\gamma - \Upsilon - b\lambda_{\text{uni}})\cosh(\mu_1)A_2 + (\gamma\lambda_{\text{uni}} - b)\mu_3\sinh(\mu_3)A_3 \\ + (\gamma\lambda_{\text{uni}} - b)\mu_3\cosh(\mu_3)A_4 = 0,\end{aligned} \tag{3.16}$$

$$\begin{aligned}(\gamma\lambda_{\text{uni}} - b)\mu_1\cosh(\mu_1)A_1 + (\gamma\lambda_{\text{uni}} - b)\mu_1\sinh(\mu_1)A_2 + (\gamma - \Upsilon - b\lambda_{\text{uni}})\cosh(\mu_3)A_3 \\ + (\gamma - \Upsilon - b\lambda_{\text{uni}})\sinh(\mu_3)A_4 = 0.\end{aligned} \tag{3.17}$$

In order to compute more precisely the unstable condition , we can substituting (3.14) and (3.15) into (3.16) and we obtain

$$\begin{aligned}&[-\frac{\lambda_{\text{uni}}}{\mu_1}(\gamma - \Upsilon - b\lambda_{\text{uni}})\sinh(\mu_1) + (\gamma\lambda_{\text{uni}} - b)\mu_3\sinh(\mu_3)]A_3 \\ &+ [-\mu_3\lambda_{\text{uni}}^{-1}(\gamma - \Upsilon - b\lambda_{\text{uni}})\cosh(\mu_1) + (\gamma\lambda_{\text{uni}} - b)\mu_3\cosh(\mu_3)]A_4 = 0.\end{aligned} \tag{3.18}$$

Similarly, substituting (3.14) and (3.15) into (3.17) and therefore

$$\begin{aligned}&[-\lambda_{\text{uni}}(\gamma\lambda_{\text{uni}} - b)\cosh(\mu_1) + (\gamma - \Upsilon - b\lambda_{\text{uni}})\cosh(\mu_3)]A_3 \\ &+ [-\mu_1\mu_3\lambda_{\text{uni}}^{-1}(\gamma\lambda_{\text{uni}} - b)\sinh(\mu_1) + (\gamma - \Upsilon - b\lambda_{\text{uni}})\sinh(\mu_3)]A_4 = 0.\end{aligned} \tag{3.19}$$

Finally, from (3.18) and (3.19) we dedude that the eigenvalue condition implies

$$\begin{aligned}&[-\frac{\lambda_{\text{uni}}}{\mu_1}(\gamma - \Upsilon - b\lambda_{\text{uni}})\sinh(\mu_1) + (\gamma\lambda_{\text{uni}} - b)\mu_3\sinh(\mu_3)] \cdot [-\mu_1\mu_3\lambda_{\text{uni}}^{-1}(\gamma\lambda_{\text{uni}} - b)\sinh(\mu_1) \\ &+(\gamma - \Upsilon - b\lambda_{\text{uni}})\sinh(\mu_3)] \\ &- [-\mu_3\lambda_{\text{uni}}^{-1}(\gamma - \Upsilon - b\lambda_{\text{uni}})\cosh(\mu_1) + (\gamma\lambda_{\text{uni}} - b)\mu_3\cosh(\mu_3)] \cdot [-\lambda_{\text{uni}}(\gamma\lambda_{\text{uni}} - b)\cosh(\mu_1) \\ &+ (\gamma - \Upsilon - b\lambda_{\text{uni}})\cosh(\mu_3)] = 0,\end{aligned}$$

therefore

$$\begin{aligned}&-2\mu_3(\gamma - \Upsilon - b\lambda_{\text{uni}})(\gamma\lambda_{\text{uni}} - b) + \left[-\frac{\lambda_{\text{uni}}}{\mu_1}(\gamma - \Upsilon - b\lambda)^2 - \frac{\mu_1\mu_3^2}{\lambda_{\text{uni}}}(\lambda_{\text{uni}}\gamma - b)^2\right]\sinh(\mu_1)\sinh(\mu_3) \\ &+ \left[\mu_3\lambda_{\text{uni}}(\gamma\lambda_{\text{uni}} - b)^2 + \frac{\mu_3}{\lambda_{\text{uni}}}(\gamma - \Upsilon - b\lambda_{\text{uni}})^2\right]\cosh(\mu_1)\cosh(\mu_3) = 0.\end{aligned} \tag{3.20}$$

Let be Υ, the solution of (3.20).Then, by setting $\gamma = 0.001$, $\chi = 0.348$, $\phi_0 = 0.2$ with $\lambda_{\text{uni}} = 1.96$ and using the fact that

$$b = 2H'(\lambda_{\text{uni}}) = 2\left(\frac{\phi_0}{\lambda_{\text{uni}}} + \ln\left(1 - \frac{\phi_0}{\lambda_{\text{uni}}}\right) + \chi\frac{\phi_0^2}{\lambda_{\text{uni}}^2}\right) = -0.004515805685$$

and

$$a = H''(\lambda_{\text{uni}}) = \frac{\phi_0^2}{\lambda_{\text{uni}}^3} \frac{\left(1 - 2\chi + 2\chi \dfrac{\phi_0}{\lambda_{\text{uni}}}\right)}{\left(1 - \dfrac{\phi_0}{\lambda_{\text{uni}}}\right)} = 0.00221866,$$

With help of Python we can conclude that the only solution for (3.20) is $\Upsilon = -0.002842 = \gamma(1 - \lambda^2)$ (see Figure 1 below).

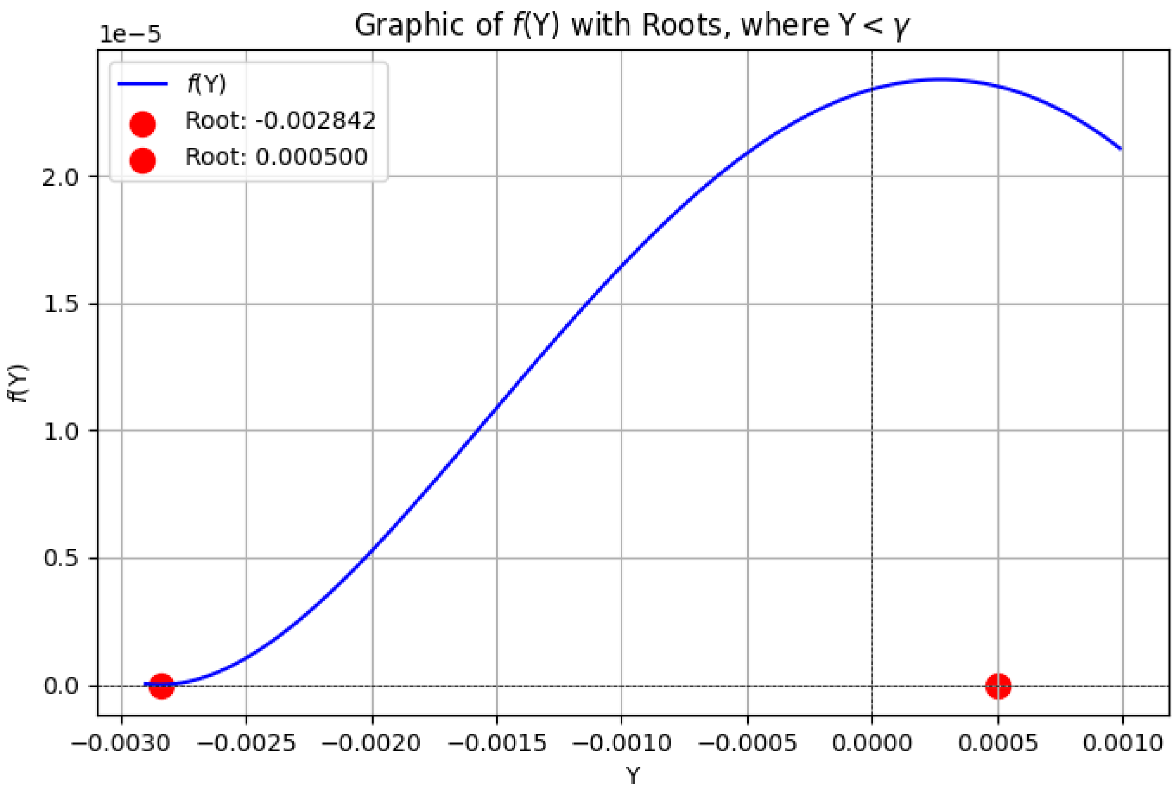


Figure 1: Case $\Upsilon > \gamma$

When we apply the boundary condition we find that $A_i = 0$.

3.5 Case $\Upsilon = \gamma(1 - \lambda^2)$.

If we have $\Upsilon = \gamma(1 - \lambda^2)$ then system (3.9) becomes into the following

$$\begin{cases} (\gamma + 2a)\lambda^2 p - \gamma p'' - 2a\lambda s' = 0, \\ \gamma\lambda^2 s - (\gamma + 2a)s'' + 2a\lambda p' = 0. \end{cases} \tag{3.21}$$

By change of variable

$$\begin{cases} x_1 = p, \\ x_2 = p', \\ x_3 = s, \\ x_4 = s', \end{cases}$$

which implies that

$$\begin{cases} x_1' = x_2, \\ x_3' = x_4. \end{cases} \tag{3.22}$$

then by virtue of (3.22) in (3.21) we obtain

$$\begin{cases} x_2' = \dfrac{(\gamma+2a)\lambda^2}{\gamma}x_1 - \dfrac{2a\lambda}{\gamma}x_4, \\ x_4' = \dfrac{\gamma\lambda^2}{(\gamma+2a)}x_3 + \dfrac{2a\lambda}{(\gamma+2a)}x_2. \end{cases} \tag{3.23}$$

Finally, (3.21) can be rewritten as the linear ODE (3.22)-(3.23), i.e.

$$\mathbf{X}' = \mathbb{M}\mathbf{X},$$

where

$$\mathbf{X} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix},$$

and

$$\mathbb{M} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ \frac{(\gamma+2a)\lambda^2}{\gamma} & 0 & 0 & -\frac{2a\lambda}{\gamma} \\ 0 & 0 & 0 & 1 \\ 0 & \frac{2a\lambda}{(\gamma+2a)} & \frac{\gamma\lambda^2}{(\gamma+2a)} & 0 \end{pmatrix}.$$

by the elemental operations by rows $\frac{\gamma\lambda^2}{(\gamma+2x)}F_3, -\frac{\gamma}{(\gamma+2a)}F_2, -\lambda^2 F_1$ we obtain

$$\mathbb{M} = \begin{pmatrix} 0 & -\lambda^2 & 0 & 0 \\ -\lambda^2 & 0 & 0 & \frac{2a\lambda}{(\gamma+2a)} \\ 0 & 0 & 0 & \frac{\gamma\lambda^2}{(\gamma+2a)} \\ 0 & \frac{2a\lambda}{(\gamma+2a)} & \frac{\gamma\lambda^2}{(\gamma+2a)} & 0 \end{pmatrix},$$

then $\mathbb{M}$ si symmetric therefore we have four eigenvalues linearly independent. In the other hand,

$$\det(\mathbb{M} - \mu\mathbb{I}) = \begin{vmatrix} -\mu & 1 & 0 & 0 \\ \frac{(\gamma+2a)\lambda^2}{\gamma} & -\mu & 0 & -\frac{2a\lambda}{\gamma} \\ 0 & 0 & -\mu & 1 \\ 0 & \frac{2a\lambda}{(\gamma+2a)} & \frac{\gamma\lambda^2}{(\gamma+2a)} & -\mu \end{vmatrix} = (\mu+\lambda)^2(\mu-\lambda)^2.$$

We have repeated eigenvalues $\mu = \pm\lambda$. Now, we can to proceed to find the eigenvectors.

If $\mu = \lambda$ we have

$$\left(\begin{array}{cccc|c} -\lambda & 1 & 0 & 0 & 0 \\ \frac{(\gamma+2a)\lambda^2}{\gamma} & -\lambda & 0 & -\frac{2a\lambda}{\gamma} & 0 \\ 0 & 0 & -\lambda & 1 & 0 \\ 0 & \frac{2a\lambda}{(\gamma+2a)} & \frac{\gamma\lambda^2}{(\gamma+2a)} & -\lambda & 0 \end{array}\right)$$

by RREF we have

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & -\frac{1}{\lambda} & 0 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & -\frac{1}{\lambda} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

then the eigenvector have the form

$$\mathbf{K}_1 = \begin{pmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{pmatrix} = \begin{pmatrix} \frac{1}{\lambda} \\ 1 \\ \frac{1}{\lambda} \\ 1 \end{pmatrix} \alpha,$$

with $\alpha \in \mathbb{R}$.

Similarly, If $\mu = -\lambda$ we have

$$\left(\begin{array}{cccc|c} \lambda & 1 & 0 & 0 & 0 \\ \frac{(\gamma+2a)\lambda^2}{\gamma} & \lambda & 0 & -\frac{2a\lambda}{\gamma} & 0 \\ 0 & 0 & \lambda & 1 & 0 \\ 0 & \frac{2a\lambda}{(\gamma+2a)} & \frac{\gamma\lambda^2}{(\gamma+2a)} & \lambda & 0 \end{array}\right)$$

by RREF we have

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & -\frac{1}{\lambda} & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & \frac{1}{\lambda} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

then the eigenvector have the form

$$\mathbf{K}_3 = \begin{pmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{pmatrix} = \begin{pmatrix} \frac{1}{\lambda} \\ -1 \\ -\frac{1}{\lambda} \\ 1 \end{pmatrix} \beta,$$

with $\beta \in \mathbb{R}$. Now, we have the first two solutions:

$$\mathbf{X}_1 = \begin{pmatrix} \frac{1}{\lambda} \\ 1 \\ \frac{1}{\lambda} \\ 1 \end{pmatrix} e^{\lambda y}, \quad \mathbf{X}_2 = \begin{pmatrix} \frac{1}{\lambda} \\ -1 \\ -\frac{1}{\lambda} \\ 1 \end{pmatrix} e^{-\lambda y}. \tag{3.24}$$

Computation of vector $\mathbf{P}_1$ for $\mu_1 = \lambda$:

$$\left(\begin{array}{cccc|c} -\lambda & 1 & 0 & 0 & \frac{1}{\lambda} \\ \frac{(\gamma+2a)\lambda^2}{\gamma} & -\lambda & 0 & -\frac{2a\lambda}{\gamma} & 1 \\ 0 & 0 & -\lambda & 1 & \frac{1}{\lambda} \\ 0 & \frac{2a\lambda}{(\gamma+2a)} & \frac{\gamma\lambda^2}{(\gamma+2a)} & -\lambda & 1 \end{array}\right)$$

by RREF we have

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & -\frac{1}{\lambda} & \frac{\gamma}{a\lambda^2} \\ 0 & 1 & 0 & -1 & \frac{a+\gamma}{a\lambda} \\ 0 & 0 & 1 & -\frac{1}{\lambda} & -\frac{1}{\lambda^2} \\ 0 & 0 & 0 & 0 & 0 \end{array}\right),$$

then

$$\mathbf{P}_1 = \begin{pmatrix} p_1 \\ p_2 \\ p_3 \\ p_4 \end{pmatrix} = \begin{pmatrix} \frac{\gamma}{a\lambda^2} \\ \frac{a+\gamma}{a\lambda} \\ -\frac{1}{\lambda^2} \\ 0 \end{pmatrix} + \begin{pmatrix} \frac{1}{\lambda} \\ 1 \\ \frac{1}{\lambda} \\ 1 \end{pmatrix} \alpha,$$

with $\alpha \in \mathbb{R}$. If $\alpha = 0$ we obtain

$$\mathbf{P}_1 = \begin{pmatrix} \frac{\gamma}{a\lambda^2} \\ \frac{a+\gamma}{a\lambda} \\ -\frac{1}{\lambda^2} \\ 0 \end{pmatrix}.$$

For the computation of vector $\mathbf{P}_2$ for $\mu_1 = -\lambda$:

$$\left(\begin{array}{cccc|c} \lambda & 1 & 0 & 0 & \frac{1}{\lambda} \\ \frac{(\gamma+2a)\lambda^2}{\gamma} & \lambda & 0 & -\frac{2a\lambda}{\gamma} & 1 \\ 0 & 0 & \lambda & 1 & \frac{1}{\lambda} \\ 0 & \frac{2a\lambda}{(\gamma+2a)} & \frac{\gamma\lambda^2}{(\gamma+2a)} & \lambda & 1 \end{array}\right)$$

by RREF we have

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & -\frac{1}{\lambda} & -\frac{\gamma}{a\lambda^2} \\ 0 & 1 & 0 & 1 & \frac{a+\gamma}{a\lambda} \\ 0 & 0 & 1 & \frac{1}{\lambda} & -\frac{1}{\lambda^2} \\ 0 & 0 & 0 & 0 & 0 \end{array}\right),$$

then

$$\mathbf{P}_2 = \begin{pmatrix} -\frac{\gamma}{a\lambda^2} \\ \frac{a+\gamma}{a\lambda} \\ -\frac{1}{\lambda^2} \\ 0 \end{pmatrix} + \begin{pmatrix} \frac{1}{\lambda} \\ -1 \\ -\frac{1}{\lambda} \\ 1 \end{pmatrix} \beta,$$

with $\beta \in \mathbb{R}$. If $\beta = 0$ we obtain

$$\mathbf{P}_2 = \begin{pmatrix} -\frac{\gamma}{a\lambda^2} \\ \frac{a+\gamma}{a\lambda} \\ -\frac{1}{\lambda^2} \\ 0 \end{pmatrix}.$$

Now, we obtain the last two solutions:

$$\mathbf{X}_2 = \begin{pmatrix} \frac{1}{\lambda} \\ 1 \\ \frac{1}{\lambda} \\ 1 \end{pmatrix} ye^{\lambda y} + \begin{pmatrix} \frac{\gamma}{a\lambda^2} \\ \frac{a+\gamma}{a\lambda} \\ -\frac{1}{\lambda^2} \\ 0 \end{pmatrix} e^{\lambda y}, \quad \mathbf{X}_4 = \begin{pmatrix} \frac{1}{\lambda} \\ -1 \\ -\frac{1}{\lambda} \\ 1 \end{pmatrix} ye^{-\lambda y} + \begin{pmatrix} -\frac{\gamma}{a\lambda^2} \\ \frac{a+\gamma}{a\lambda} \\ -\frac{1}{\lambda^2} \\ 0 \end{pmatrix} e^{-\lambda y}. \tag{3.25}$$

From the previous analysis, we have

$$\mathbf{X} = A_1\mathbf{X}_1 + A_2\mathbf{X}_2 + A_3\mathbf{X}_3 + A_4\mathbf{X}_4, \tag{3.26}$$

where $\{\mathbf{X}_i\}_{i=1}^4$ are defined in (3.24)-(3.25) and $\{A_i\}_{i=1}^4 \in \mathbb{R}$. Finally, recovering the solutions we have that

$$\begin{pmatrix} p \\ s \end{pmatrix} = A_1\frac{1}{\lambda}\begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{\lambda y} + A_2\left[\begin{pmatrix} \frac{1}{\lambda} \\ \frac{1}{\lambda} \end{pmatrix} y + \begin{pmatrix} \frac{\gamma}{a\lambda^2} \\ -\frac{1}{\lambda^2} \end{pmatrix}\right] e^{\lambda y} + A_3\frac{1}{\lambda}\begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-\lambda y} + A_4\left[\begin{pmatrix} \frac{1}{\lambda} \\ -\frac{1}{\lambda} \end{pmatrix} y + \begin{pmatrix} -\frac{\gamma}{a\lambda^2} \\ -\frac{1}{\lambda^2} \end{pmatrix}\right] e^{-\lambda y}. \tag{3.27}$$

When we apply the boundary condition we find that $A_i = 0$.

3.6 Case $\Upsilon = 0$.

For system (3.9), by setting $c_1 = \gamma + 2a\lambda^2$, $c_3 = -2a\lambda$, $c_5 = -(\gamma + 2a)$, system can be rewritten as follows:

$$\begin{cases} c_1 p - \gamma p'' + c_3 s' = 0, \\ \gamma s + c_5 s'' - c_3 p' = 0. \end{cases}$$

Let us seek solutions fo the form

$$\begin{aligned} p(y) &= p_0 \cosh(\mu y), \\ s(y) &= s_0 \sinh(\mu y). \end{aligned}$$

Then, the equations become

$$\begin{cases} c_1 p_0 - \gamma\mu^2 p_0 + c_3\mu s_0 = 0, \\ \gamma s_0 + c_5\mu^2 s_0 - c_3\mu p_0 = 0. \end{cases}$$

The determinant of this 2×2 system on p_0, s_0 is

$$\begin{vmatrix} c_1 - \gamma\mu^2 & c_3\mu \\ -c_3\mu & \gamma + c_5\mu^2 \end{vmatrix}.$$

For nontrivial solution to exist it is necessary that

$$c_1\gamma + c_1c_5\mu^2 - \gamma^2\mu^2 - \gamma c_5\mu^4 + c_3^2\mu^2 = 0,$$

i.e.

$$\mu^2 = \frac{-(c_1c_5 - \gamma^2 + c_3^2) \pm \sqrt{(c_1c_5 - \gamma^2 + c_3^2)^2 - 4(-\gamma c_5)(c_1\gamma)}}{-2\gamma c_5}.$$

Since,

$$-(c_1c_5 - \gamma^2 + c_3^2) = 2\gamma(\gamma + a\lambda^2 + a),$$

$$(c_1c_5 - \gamma^2 + c_3^2)^2 - 4(-\gamma c_5)(c_1\gamma) = [2\gamma(\gamma + a\lambda^2 + a)]^2 - 4[\gamma(\gamma + 2a)][\gamma(\gamma + 2a\lambda^2)] = 2^2\gamma^2a^2(\lambda^2 - 1)^2,$$

then

$$\mu^2 = \frac{2\gamma(\gamma + a\lambda^2 + a) \pm 2\gamma a(\lambda^2 - 1)}{2\gamma(\gamma + 2a)} = \frac{(\gamma + a\lambda^2 + a) \pm a(\lambda^2 - 1)}{(\gamma + 2a)},$$

now,

$$\begin{cases} \mu_{3,4}^2 = \dfrac{\gamma + 2a\lambda^2}{(\gamma + 2a)}, \\ \mu_{1,2}^2 = 1, \end{cases}$$

Let be

$$\eta^2 = \frac{\gamma + 2a\lambda^2}{(\gamma + 2a)}, \tag{3.28}$$

then

$$\begin{cases} \mu_{3,4} = \pm\sqrt{\dfrac{\gamma + 2a\lambda^2}{(\gamma + 2a)}} = \pm\eta, \\ \mu_{1,2} = \pm 1. \end{cases} \tag{3.29}$$

When $\mu = 1$, the system

$$\begin{pmatrix} c_1 - \gamma\mu^2 & c_3\mu \\ -c_3\mu & \gamma + c_5\mu^2 \end{pmatrix} \begin{pmatrix} p_0 \\ s_0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

for p_0, s_0 becomes:

$$\begin{pmatrix} 2a\lambda^2 & -2a\lambda \\ 2a\lambda & -2a \end{pmatrix} \begin{pmatrix} p_0 \\ s_0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

whose solutions are the multiples of

$$\begin{pmatrix} p_0 \\ s_0 \end{pmatrix} = \begin{pmatrix} 1 \\ \lambda \end{pmatrix}.$$

When $\mu = \eta$ by using definition of η in (3.28) we compute that

$$\begin{aligned} c_1 - \gamma\mu^2 &= \gamma + 2a\lambda^2 - \gamma\left(\frac{\gamma + 2a\lambda^2}{\gamma + 2a}\right) \\ &= \frac{(\gamma + 2a\lambda^2)\cdot 2a}{\gamma + 2a} = 2a\eta^2, \\ -c_3\mu &= -2a\lambda\eta, \\ \gamma + c_5\mu^2 &= \gamma - (\gamma + 2a)\cdot\frac{\gamma + 2a\lambda^2}{\gamma + 2a} = -2a\lambda^2, \end{aligned}$$

so the system becomes

$$\begin{pmatrix} 2a\eta^2 & -2a\lambda\eta \\ 2a\lambda\eta & -2a\lambda^2 \end{pmatrix} \begin{pmatrix} p_0 \\ s_0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

whose solutions are the multiples of

$$\begin{pmatrix} p_0 \\ s_0 \end{pmatrix} = \begin{pmatrix} \lambda \\ \eta \end{pmatrix}.$$

All in all,

$$\begin{cases} p = \cosh(y), \\ s = \lambda \sinh(y), \end{cases}$$

and

$$\begin{cases} p = \lambda \cosh(\eta y), \\ s = \eta \sinh(\eta y), \end{cases}$$

are two l.i. solutions.

Now seek solutions fo the form

$$p(y) = p_0 \sinh(\mu y),$$
$$s(y) = s_0 \cosh(\mu y).$$

This leads to exctly the same algebraic system in p_0, s_0. Thusm the following other two l.i. solutions appear:

$$\begin{cases} p = \sinh(y), \\ s = \lambda \cosh(y), \end{cases}$$

and

$$\begin{cases} p = \lambda \sinh(\eta y), \\ s = \eta \cosh(\eta y). \end{cases}$$

Since the function space of solutions of the homogeneous system has dimension four, the any solution is of the form

$$p(y) = A_1 \cosh(y) + A_2 \sinh(y) + A_3 \lambda \cosh(\eta y) + A_4 \lambda \sinh(\eta y), \tag{3.30}$$
$$s(y) = A_1 \lambda \sinh(y) + A_2 \lambda \cosh(y) + A_3 \eta \sinh(\eta y) + A_4 \eta \cosh(\eta y), \tag{3.31}$$

where A_i are constants for $i \in \{1, 2, 3, 4\}$.

3.6.1 Computing the equations for amplitude A_n using the boundary conditions

In the other hand, by substituting explicit expressions for p and s given in (3.30)-(3.31) in the boundary conditions (3.8) we have the following equations:

$$A_1 + \lambda A_3 = 0, \tag{3.32}$$
$$\lambda A_2 + \eta A_4 = 0, \tag{3.33}$$

$$(\gamma-b\lambda)\sinh(1)A_1+(\gamma-b\lambda)\cosh(1)A_2+(\gamma\lambda-b)\eta\sinh(\eta)A_3+(\gamma\lambda-b)\eta\cosh(\eta)A_4=0, \quad (3.34)$$
$$(\gamma\lambda-b)\cosh(1)A_1+(\gamma\lambda-b)\sinh(1)A_2+(\gamma-b\lambda)\cosh(\eta)A_3+(\gamma-b\lambda)\sinh(\eta)A_4=0. \quad (3.35)$$

The equations (3.32)-(3.35) can be rewritten as the following 4×4 system

$$\sum_{n=1}^{4} D^{p,s}_{mn}A_n=0, \tag{3.36}$$

where

$$[D^{p,s}_{mn}]=\begin{bmatrix} 1 & 0 & \lambda & 0 \\ 0 & \lambda & 0 & \eta \\ (\gamma-b\lambda)\sinh(1) & (\gamma-b\lambda)\cosh(1) & (\gamma\lambda-b)\eta\sinh(\eta) & (\gamma\lambda-b)\eta\cosh(\eta) \\ (\gamma\lambda-b)\cosh(1) & (\gamma\lambda-b)\sinh(1) & (\gamma-b\lambda)\cosh(\eta) & (\gamma-b\lambda)\sinh(\eta) \end{bmatrix}. \tag{3.37}$$

Then, the unstable condition

$$\det[D^{p,s}_{mn}(k)]=0, \tag{3.38}$$

implies that

$$\begin{aligned}
&b^2\eta\lambda^2\sinh(1)^2-\eta\gamma^2\lambda^2\cosh^2(1)-b^2\eta\lambda^2\cosh(1)^2+\eta\gamma^2\lambda^2\sinh(1)^2-b^2\eta^2\sinh(1)\sinh(\eta)\\
&-b^2\lambda^4\sinh(1)\sinh(\eta)-\gamma^2\lambda^2\sinh(1)\sinh(\eta)-b^2\eta\lambda^2\cosh(\eta)^2\\
&-\eta\gamma^2\lambda^2\cosh(\eta)^2+b^2\eta\lambda^2\sinh(\eta)^2+\eta\gamma^2\lambda^2\sinh(\eta)^2+\cosh(1)\eta\gamma^2\cosh(\eta)-\eta^2\gamma^2\lambda^2\sinh(1)\sinh(\eta)\\
&+b\cosh^2(1)\eta\gamma\lambda^3+b\eta\gamma\lambda\cosh(\eta)^2-b\eta\gamma\lambda^3\sinh^2(1)-b\eta\gamma\lambda\sinh(\eta)^2\\
&+2b\gamma\lambda^3\sinh(1)\sinh(\eta)+2b^2\cosh(1)\eta\lambda^2\cosh(\eta)+b\eta\gamma\lambda^3\cosh(\eta)^2+\cosh(1)\eta\gamma^2\lambda^4\cosh(\eta)\\
&-b\eta\gamma\lambda^3\sinh(\eta)^2+b\cosh^2(1)\eta\gamma\lambda-b\eta\gamma\lambda\sinh^2(1)-2b\cosh(1)\eta\gamma\lambda\cosh(\eta)-2b\cosh(1)\eta\gamma\lambda^3\cosh(\eta)\\
&+2b\eta^2\gamma\lambda\sinh(1)\sinh(\eta)=0.
\end{aligned}$$

In order to compute more precisely the unstable condition (3.38), we can substituting (3.32) and (3.33) into (3.34) and we obtain

$$[-\lambda(\gamma-b\lambda)\sinh(1)+(\gamma\lambda-b)\eta\sinh(\eta)]A_3+[-\eta\lambda^{-1}(\gamma-b\lambda)\cosh(1)+(\gamma\lambda-b)\eta\cosh(\eta)]A_4=0. \tag{3.39}$$

Similarly, substituting (3.32) and (3.33) into (3.35) and therefore

$$[-\lambda(\gamma\lambda-b)\cosh(1)+(\gamma-b\lambda)\cosh(\eta)]A_3+[-\eta\lambda^{-1}(\gamma\lambda-b)\sinh(1)+(\gamma-b\lambda)\sinh(\eta)]A_4=0. \tag{3.40}$$

Finally, from (3.39) and (3.40) we dedude that the condition (3.38) implies

$$\begin{aligned}
&[-\lambda(\gamma-b\lambda)\sinh(1)+(\gamma\lambda-b)\eta\sinh(\eta)]\cdot[-\eta\lambda^{-1}(\gamma\lambda-b)\sinh(1)+(\gamma-b\lambda)\sinh(\eta)]\\
&-[-\eta\lambda^{-1}(\gamma-b\lambda)\cosh(1)+(\gamma\lambda-b)\eta\cosh(\eta)]\cdot[-\lambda(\gamma\lambda-b)\cosh(1)+(\gamma-b\lambda)\cosh(\eta)]=0,
\end{aligned}$$

therefore

$$\begin{aligned}
&-2\eta(\gamma-b\lambda)(\gamma\lambda-b)+\left[-\lambda(\gamma-b\lambda)^2-\frac{\eta^2}{\lambda}(\lambda\gamma-b)^2\right]\sinh(1)\sinh(\eta)\\
&+\left[\eta\lambda(\gamma\lambda-b)^2+\frac{\eta}{\lambda}(\gamma-b\lambda)^2\right]\cosh(1)\cosh(\eta)=0.
\end{aligned} \tag{3.41}$$

But the right hand-side of (3.41) is not zero, therefore $A_i = 0$ for all $i \in \{1, 2, 3, 4\}$.

3.7 Exploring case $\Upsilon = \gamma$:

Then we have,

$$c_1 = 2a\lambda^2 \qquad c_3 = -2a\lambda \qquad c_5 = -(\gamma + 2a)$$

therefore,

$$\begin{cases} 2a\lambda p'' - \gamma p'' - 2a\lambda s' = 0 \\ -(\gamma + 2a)s'' + 2a\lambda p' = 0 \end{cases}$$

Let

$$x_1 = p, \quad x_2 = p', \quad x_3 = s, \quad x_4 = s'$$

$$\Rightarrow \quad \begin{cases} x_1' = x_2 \\ x_3' = x_4 \end{cases} \qquad (1)$$

Equation (2):

$$\begin{cases} 2a\lambda^2 x_1 - \gamma x_2' - 2a\lambda x_4 = 0 \\ -(\gamma + 2a)x_4' + 2a\lambda x_2 = 0 \end{cases}$$

Which gives:

$$\begin{cases} x_2' = \frac{2a\lambda^2}{\gamma} x_1 - \frac{2a\lambda}{\gamma} x_4 \\ x_4' = \frac{2a\lambda}{\gamma + 2a} x_2 \end{cases} \qquad (2)$$

Define:

$$M = \begin{pmatrix} 0 & 1 & 0 & 0 \\ \frac{2a\lambda^2}{\gamma} & 0 & 0 & -\frac{2a\lambda}{\gamma} \\ 0 & 0 & 0 & 1 \\ 0 & \frac{2a\lambda}{\gamma+2a} & 0 & 0 \end{pmatrix}, \quad X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}$$

Then the ODE is now:

$$X' = MX.$$

Eigenvalues:

$$\mu_{1,2} = 0, \qquad \mu_{3,4} = \pm\sqrt{\frac{2a}{(\gamma + 2a)}}\lambda$$

If $\mu_1 = 0$, let $\gamma = c$, $\lambda = d$,

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ \frac{2ad^2}{c} & 0 & 0 & -\frac{2ad}{c} \\ 0 & 0 & 0 & 1 \\ 0 & \frac{2ad}{c+2a} & 0 & 0 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Then,

$$\begin{pmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \alpha \\ 0 \end{pmatrix}, \quad \alpha \in \mathbb{R}$$

Computation of P for $\mu_1 = 0$

$$\begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ \frac{2ad^2}{c} & 0 & 0 & -\frac{2ad}{c} & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & \frac{2ad}{c+2a} & 0 & 0 & 0 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & 0 & 0 & 0 & 1/d \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

Then,

$$\begin{pmatrix} p_1 \\ p_2 \\ p_3 \\ p_4 \end{pmatrix} = \begin{pmatrix} 1/\lambda \\ 0 \\ \beta \\ 1 \end{pmatrix}, \quad \beta \in \mathbb{R}.$$

If $\beta = 0$:

$$X_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} ye^{0y} + \begin{pmatrix} 1/\lambda \\ 0 \\ 0 \\ 1 \end{pmatrix} e^{0y},$$

$$X_1 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} e^{0y},$$

Then,

$$X = c_1 \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} + c_2 \left[\begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} y + \begin{pmatrix} 1/\lambda \\ 0 \\ 0 \\ 1 \end{pmatrix} \right] + c_3 X_3 + c_4 X_4$$

Now:

$$\begin{pmatrix} -\mu & 1 & 0 & 0 & 0 \\ \frac{2a\lambda^2}{\gamma} & -\mu & 0 & -\frac{2a\lambda}{\gamma} & 0 \\ 0 & 0 & -\mu & 1 & 0 \\ 0 & \frac{2a\lambda}{\gamma+2a} & 0 & -\mu & 0 \end{pmatrix} \xrightarrow{\text{RREF}} (I|0)$$

If $\mu = +\sqrt{\frac{2a}{\gamma+2a}}\lambda = \theta\lambda$, let $\theta = t$, $r = c$, $\lambda = d$,

$$\begin{pmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix},$$

$$X_3 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} e^{\theta\lambda y} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

If $\mu = -\sqrt{\frac{2a}{\gamma+2a}}\lambda = -\theta\lambda \Rightarrow$ By RREF obtain $(I|0)$.

$$\Rightarrow \begin{pmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad X_4 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} e^{-\theta\lambda y} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

Finally,

$$X = c_1 \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} + c_2 \left[\begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} y + \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right].$$

Thus,

$$\begin{cases} p = \frac{c_2}{\lambda}, \\ s = c_1 + yc_2. \end{cases}$$

Verifying the solution:

$$2a\lambda^2 \frac{c_2}{\lambda} - 0 - 2a\lambda c_2 = 2a\lambda c_2 - 2a\lambda c_2 = 0$$
$$-(\gamma + 2a)c_2 + 2a\lambda \cdot 0 = 0$$

At $y = 0$, $p = s = 0$ and the other boundary conditions we have:

$$\begin{cases} c_2/\lambda = 0 \\ c_1 = 0 \\ -b(c_1 + c_2) = 0 \\ (\gamma + 2a)c_2 - (2a\lambda + b)\frac{c_2}{\lambda} = 0 \end{cases}$$

$$\begin{pmatrix} 0 & 1/\lambda & 0 & 0 \\ 1 & 0 & 0 & 0 \\ -b & -b & 0 & 0 \\ 0 & \left[\gamma + 2a - 2a - \frac{b}{\lambda}\right] & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Hence,

$$c_1 = c_2 = 0.$$

3.8 Case: $\Upsilon = \gamma + 2a\lambda^2$

$$\mu_{3,4} = 0,$$
$$\mu_{1,2} = \sqrt{1 - \frac{(\gamma + 2a\lambda^2)}{\gamma}} = \pm\sqrt{\frac{2a}{\gamma}} i\lambda.$$

When $\mu = \sqrt{\frac{2a}{\gamma}} i\lambda$ we have

$$c_1 = \gamma + 2a\lambda^2 - (\gamma + 2a\lambda^2) = 0$$
$$c_3 = -2a\lambda \quad ; \quad c_5 = -(\gamma + 2a)$$

$$\begin{cases} -\gamma p'' - 2a\lambda s' = 0 \\ -2a\lambda^2 s - (\gamma + 2a)s'' + 2a\lambda p' = 0 \end{cases}$$

Let $x_1 = p, \quad x_2 = p', \quad x_3 = s, \quad x_4 = s'$

$$\Rightarrow \begin{cases} x_1' = x_2 \\ x_3' = x_4 \end{cases}$$

$$\begin{cases} -\gamma x_2' - 2a\lambda x_4 = 0 \\ -2a\lambda^2 x_3 - (\gamma + 2a)x_4' + 2a\lambda x_2 = 0 \end{cases}$$

$$x_2' = -\frac{2a\lambda}{\gamma} x_4 \quad ; \quad x_4' = \frac{2a\lambda}{\gamma + 2a} x_2 - \frac{2a\lambda^2}{\gamma + 2a} x_3$$

$$M = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2a\lambda}{\gamma} \\ 0 & 0 & 0 & 1 \\ 0 & \frac{2a\lambda}{\gamma+2a} & -\frac{2a\lambda^2}{\gamma+2a} & 0 \end{pmatrix}$$

$$X' = MX \quad ; \quad X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}$$

$$\mu_{1,2} = \pm \sqrt{\frac{2a}{\gamma}} i\lambda, \quad \mu_{3,4} = 0$$

If $\mu_1 = \sqrt{\frac{2a}{\gamma}} i\lambda$, then $\mu_1 = \theta i\lambda$

$$\begin{pmatrix} -\mu_1 & 1 & 0 & 0 \\ 0 & -\mu_1 & 0 & -\frac{2a\lambda}{\gamma} \\ 0 & 0 & -\mu_1 & 1 \\ 0 & \frac{2a\lambda}{\gamma+2a} & -\frac{2a\lambda^2}{\gamma+2a} & -\mu_1 \end{pmatrix}$$

Let $\gamma = c, \quad \lambda = d, \quad t = \theta$

$$\begin{pmatrix} -tid & 1 & 0 & 0 \\ 0 & -tid & 0 & -\frac{2ad}{c} \\ 0 & 0 & -tid & 1 \\ 0 & \frac{2ad}{c+2a} & -\frac{2ad^2}{c+2a} & -tid \end{pmatrix} \xrightarrow{\text{RRef}} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Computation of P for $\mu_3 = 0$,

$$\begin{pmatrix} 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & -\frac{2a\lambda}{\gamma} & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & \frac{2a\lambda}{\gamma+2a} & -\frac{2a\lambda^2}{\gamma+2a} & 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & -\frac{2ad}{c} & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & \frac{2ad}{c+2a} & -\frac{2ad^2}{c+2a} & 0 & 0 \end{pmatrix} \xrightarrow{\text{RRef}} \begin{pmatrix} 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1/d \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \Rightarrow \begin{pmatrix} p_1 \\ p_2 \\ p_3 \\ p_4 \end{pmatrix} = \begin{pmatrix} \beta \\ 1 \\ 1/d \\ 0 \end{pmatrix}$$

where $\beta \in \mathbb{R}$.

If $\beta = 0$, we have

$$X_4 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} ye^{0y} + \begin{pmatrix} 0 \\ 1/\lambda \\ 0 \\ 0 \end{pmatrix} e^{0y},$$

$$X_3 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} e^{0y}.$$

$$\begin{pmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow X_1 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} e^{\sqrt{\frac{2a}{\gamma}}i\lambda} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

If $\mu_2 = -\sqrt{\frac{2a}{\gamma}}i\lambda$, then $\mu_2 = -\theta i\lambda$

$$\begin{pmatrix} tid & 1 & 0 & 0 \\ 0 & tid & 0 & -\frac{2ad}{c} \\ 0 & 0 & tid & 1 \\ 0 & \frac{2ad}{c+2a} & -\frac{2ad^2}{c+2a} & tid \end{pmatrix} \xrightarrow{\text{RRef}} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Then

$$\begin{pmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow X_2 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} e^{-\sqrt{\frac{2a}{\gamma}}i\lambda} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

If $\mu_3 = 0$, then

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2a\lambda}{\gamma} \\ 0 & 0 & 0 & 1 \\ 0 & \frac{2a\lambda}{Y+2a} & -\frac{2a\lambda^2}{\gamma+2a} & 0 \end{pmatrix} \sim \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2cd}{c} \\ 0 & 0 & 0 & 1 \\ 0 & \frac{2ad}{c+2a} & -\frac{2ad^2}{c+2a} & 0 \end{pmatrix}$$

$$\xrightarrow{\text{RRef}} \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad ; \quad \begin{pmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{pmatrix} = \begin{pmatrix} \alpha \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad \alpha \in \mathbb{R},$$

$$X = c_3 \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + c_4 \left[\begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} y + \begin{pmatrix} 0 \\ 1/\lambda \\ 0 \\ 0 \end{pmatrix} \right] + c_1 \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} = 0$$

$$\begin{cases} p = c_1 + c_2 y \\ s = c_2/\lambda \end{cases}$$

are the solution of

$$\begin{cases} -\gamma p'' - 2a\lambda s' = 0 \\ -2a\lambda^2 s - (\gamma + 2a)s'' + 2a\lambda p' = 0 \end{cases}$$

Verifying,

$$-\gamma(0) - 2a\lambda(0) = 0$$

$$-2a\lambda^2(c_2/\lambda) - (\gamma + 2a)(0) + 2a\lambda c_2 = 0 \quad \Rightarrow \quad -2a\lambda c_2 + 2a\lambda c_2 = 0.$$

The general solution is

$$\begin{cases} p(y) = A_1(\frac{1}{\gamma\theta_1})\cos(\theta_1 y) - A_2(\frac{1}{\gamma\theta_1})\sin(\theta_1 y) + A_3 + A_4 y, \\ s(y) = A_1(\frac{1}{2a\gamma})\sin(\theta_1 y) + A_2(\frac{1}{2a\gamma})\cos(\theta_1 y) + \frac{A_4}{\lambda}. \end{cases}$$

3.9 Case $\gamma < \Upsilon < \gamma + 2a\lambda^2$.

We have,

$$\begin{cases} \mu_{3,4} = \pm\sqrt{\dfrac{\gamma + 2a\lambda_{\text{uni}}^2 - \Upsilon}{(\gamma + 2a)}} = \pm\mu_3, \\ \mu_{1,2} = \pm\sqrt{\dfrac{-\gamma + \Upsilon}{\gamma}}i = \pm\theta_1 i, \end{cases} \tag{3.42}$$

and an analogous equation to (3.20) is obtained as follows

$$
-2\mu_3(\gamma-\Upsilon-b\lambda_{\text{uni}})(\gamma\lambda_{\text{uni}}-b)+\left[-\frac{\lambda_{\text{uni}}}{\theta_1 i}(\gamma-\Upsilon-b\lambda)^2-\frac{\theta_1 i\mu_3^2}{\lambda_{\text{uni}}}(\lambda_{\text{uni}}\gamma-b)^2\right]i\sin(\theta_1)\sinh(\mu_3)
$$
$$
+\left[\mu_3\lambda_{\text{uni}}(\gamma\lambda_{\text{uni}}-b)^2+\frac{\mu_3}{\lambda_{\text{uni}}}(\gamma-\Upsilon-b\lambda_{\text{uni}})^2\right]\cos(\theta_1)\cosh(\mu_3)=0.
$$

i.e.,

$$
-2\mu_3(\gamma-\Upsilon-b\lambda_{\text{uni}})(\gamma\lambda_{\text{uni}}-b)+\left[-\frac{\lambda_{\text{uni}}}{\theta_1}(\gamma-\Upsilon-b\lambda)^2+\frac{\theta_1\mu_3^2}{\lambda_{\text{uni}}}(\lambda_{\text{uni}}\gamma-b)^2\right]\sin(\theta_1)\sinh(\mu_3)
$$
$$
+\left[\mu_3\lambda_{\text{uni}}(\gamma\lambda_{\text{uni}}-b)^2+\frac{\mu_3}{\lambda_{\text{uni}}}(\gamma-\Upsilon-b\lambda_{\text{uni}})^2\right]\cos(\theta_1)\cosh(\mu_3)=0. \tag{3.43}
$$

With help of Python we can conclude that the only two solutions for (3.43) are $\Upsilon = 0.002399$ and $\Upsilon = 0.017410$ (see Figure 2 below).

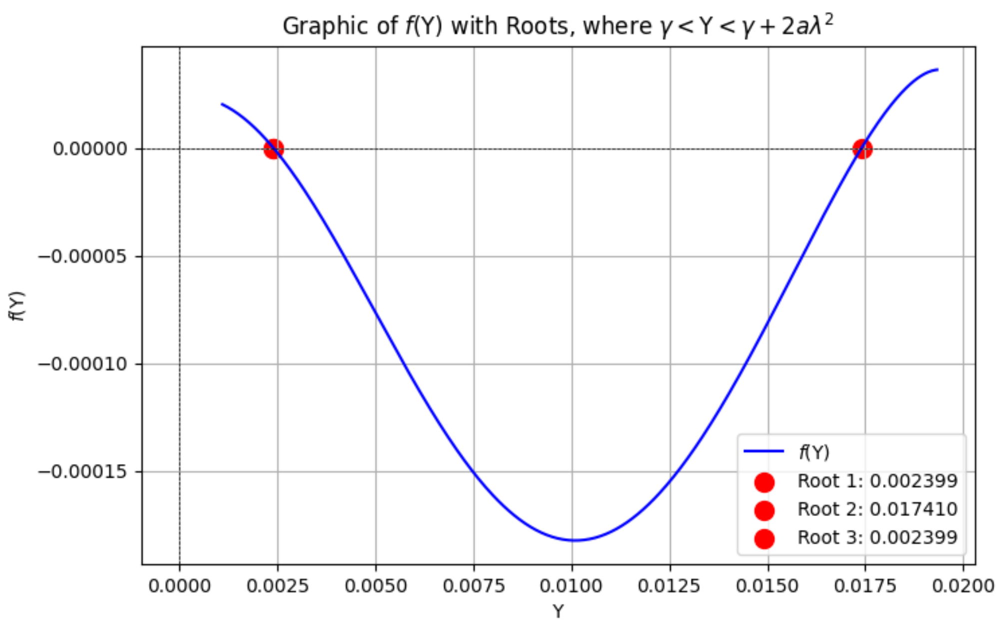


Figure 2: Case $\gamma < \Upsilon < \gamma + 2a\lambda^2$

We need check which Υ are such that the constants A_i of the solutions such that when apply the boundary conditions we obtain $A_i \neq 0$.

Then the system of ODE can be rewritten as follows:

$$
\left.\begin{array}{l} x_1 = p \\ x_2 = p' \end{array}\right\} \qquad \left.\begin{array}{l} x_1' = x_2 \\ x_3' = x_4 \end{array}\right\} \quad (1)
$$

$$
\left.\begin{array}{l} x_3 = s \\ x_4 = s' \end{array}\right\}
$$

$$c_1 = \gamma + 2a\lambda^2 - \Upsilon \qquad c_3 = -2a\lambda \qquad c_5 = -(\gamma + 2a)$$

$$\begin{aligned} c_1 x_1 - \gamma x_2' + c_3 x_4 = 0 \\ (\gamma - \Upsilon) x_3 + c_5 x_4' - c_3 x_2 = 0 \end{aligned}$$

$$\Rightarrow \left.\begin{aligned} x_2' &= \frac{c_1}{\gamma} x_1 + \frac{c_3}{\gamma} x_4 \\ x_4' &= -\frac{(\gamma - \Upsilon)}{c_5} x_3 + \frac{c_3}{c_5} x_2 \end{aligned}\right\} \quad (2)$$

$$\Rightarrow M = \begin{pmatrix} 0 & 1 & 0 & 0 \\ c_1/\gamma & 0 & 0 & c_3/\gamma \\ 0 & 0 & 0 & 1 \\ 0 & c_3/c_5 & (\gamma - \Upsilon)/c_5 & 0 \end{pmatrix}, \quad X' = MX, \quad X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}. \tag{3.44}$$

$$\det(M - \mu I) = \begin{vmatrix} -\mu & 1 & 0 & 0 \\ c_1/\gamma & -\mu & 0 & c_3/\gamma \\ 0 & 0 & -\mu & 1 \\ 0 & c_3/c_5 & (\gamma - \Upsilon)/c_5 & -\mu \end{vmatrix} = \left(\mu^2 - \frac{\gamma + 2a\lambda^2 - \Upsilon}{\gamma + 2a}\right)\left(\mu^2 - \frac{\gamma - \Upsilon}{\gamma}\right).$$

Let be $\theta_1 = \sqrt{\frac{-\gamma+\Upsilon}{\gamma}}, \mu_3 = \sqrt{\frac{\gamma+2a\lambda^2-\Upsilon}{\gamma+2a}}, \gamma = 0.001, \lambda = 1.96, a = 0.00251615165$ and $b = -0.00451580568$. The eigenvalues to analyze in this regime are $\Upsilon_0 = 0.0024$ and $\Upsilon_1 = 0.017410$.

(i) For $\Upsilon_0 = 0.0024$: We obtain

$\theta_1 = \frac{\sqrt{35}}{5} = 1.183215956$

$\mu_3 = 1.724145637$

and we have in (3.43) that

$$\begin{aligned} -2.318639974 \times 10^{-8} = -2\mu_3(\gamma - \Upsilon - b\lambda)(\gamma\lambda - b) + \Big[& -\frac{\lambda}{\theta_1}(\gamma - \Upsilon - b\lambda)^2 \\ & + \frac{\theta_1 \mu_3^2}{\lambda}(2\gamma - b)^2 \Big] \sin(\theta_1)\sinh(\mu_3) \\ & + \left[\mu_3 \lambda(\gamma\lambda - b)^2 + \frac{\mu_3}{\lambda}(\gamma - \Upsilon - b\lambda)^2\right] \cos(\theta_1)\cosh(\mu_3) \end{aligned}$$

Consider the $\mu-$eigenvalues matrix M_μ given by

$$M_\mu = \begin{pmatrix} -\mu & 1 & 0 & 0 \\ c_1/\gamma & -\mu & 0 & c_3/\gamma \\ 0 & 0 & -\mu & 1 \\ 0 & c_3/c_5 & (\gamma - \Upsilon)/c_5 & -\mu \end{pmatrix}. \tag{3.45}$$

Now we analyze the ODE for $\Upsilon_0 = 0.0024$

(a) If $\mu = \theta_1 i$ in (3.45), we have

$$\begin{pmatrix} -1.183215956i & 1 & 0 & 0 \\ 17.93209697 & -1.183215956i & 0 & -9.863314526 \\ 0 & 0 & -1.183215956i & 1 \\ 0 & 1.635082652 & 0.232083819 & -1.183215956i \end{pmatrix}$$

$$\xrightarrow{RRef} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

$$\Rightarrow X_1 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} e^{\theta_1 i} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

(b) $\mu = -\theta_1 i$ in (3.45), we have

Similarly with $\mu = -\theta_1 i \Rightarrow M_\mu \xrightarrow{RRef} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

$$\Rightarrow \begin{pmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow X_2 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} e^{-\theta_1 i} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix},$$

(c) $\mu = 1.724145637 = \mu_3$ in (3.45), we have

$$\begin{pmatrix} -1.724145636 & 1 & 0 & 0 \\ 17.93209697 & -1.724145636 & 0 & -1.863314526 \\ 0 & 0 & -1.724145636 & 1 \\ 0 & 1.635082652 & 0.232083819 & -1.724145636 \end{pmatrix}$$

$$\xrightarrow{RRef} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow X_3 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} e^{\mu_3} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

(d) Similarly we obtain identity matrix with $\mu = -\mu_3$, then again

$$\Rightarrow X_4 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} e^{-\mu_3} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

Finally, $X = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$ Therefore, $\begin{pmatrix} p \\ s \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ are the solutions.

(ii) For $\Upsilon_1 = 0.017410$:

We have $\theta_1 = 13.15674731 \quad \mu_3 = 0.6959143101$

and we have in (3.43) that

$$\begin{aligned} 0.001422653425 = & -2\mu_3(\gamma - \Upsilon - b\lambda)(\gamma\lambda - b) + \Big[-\frac{\lambda}{\theta_1}(\gamma - \Upsilon - b\lambda)^2 \\ & + \frac{\theta_1 \mu_3^2}{\lambda}(2\gamma - b)^2 \Big] \sin(\theta_1)\sinh(\mu_3) \\ & + \Big[\mu_3\lambda(\gamma\lambda - b)^2 + \frac{\mu_3}{\lambda}(\gamma - \Upsilon - b\lambda)^2 \Big] \cos(\theta_1)\cosh(\mu_3). \end{aligned}$$

(a) If $\mu = \theta_1 i$ in (3.45), we have

$$\begin{pmatrix} -13.15674731i & 1 & 0 & 0 \\ 2.922076492 & -13.15674731i & 0 & -9.863314526 \\ 0 & 0 & -13.15674731i & 1 \\ 0 & 1.635082652 & 2.726353918 & -13.15674731i \end{pmatrix}$$

$$\xrightarrow{RRef} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow X_1 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} e^{\theta_1 i} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

(b) Similarly if $\mu = -\theta_1 i$, we obtain $X_2 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} e^{-\theta_1 i} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$.

(c) If $\mu = \mu_3 = 0.6959143101$ we obtain $X_3 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} e^{\mu_3} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$.

(d) If $\mu = -\mu_3$ we obtain $X_4 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} e^{-\mu_3} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$.

Finally, $X = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$. Therefore $\begin{pmatrix} p \\ s \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$.

In general matrix of the form

$$M^* = \begin{pmatrix} a & 1 & 0 & 0 \\ b & a & 0 & d \\ 0 & 0 & a & 1 \\ 0 & e & f & a \end{pmatrix}$$

where $a, b, d, e, f \neq 0$, when we transform the matrix to the RREF obtain identity $Id_{4\times 4}$, which implies that eigenvectors are zero and consequently $p = s = 0$.

In general, the solutions are of the form:

$$p(y) = A_1\theta_1 \cos(\theta_1 y) - A_2\theta_1 \sin(\theta_1 y) + A_3\lambda_{\text{uni}} \cosh(\mu_3 y) + A_4\lambda_{\text{uni}} \sinh(\mu_3 y), \tag{3.46}$$

$$s(y) = A_1\lambda_{\text{uni}} \sin(\theta_1 y) + A_2\lambda_{\text{uni}} \cos(\theta_1 y) + A_3\mu_3 \sinh(\mu_3 y) + A_4\mu_3 \cosh(\mu_3 y), \tag{3.47}$$

where A_i are constants for $i \in \{1, 2, 3, 4\}$, and when apply the boundary conditions, we obtain that $A_i = 0$, therefore $p(y) = s(y) = 0$.

Important Question: Which Υ are such that constants A_1, A_2, A_3, A_3 that appear in the solutions are not zero once apply the boundary conditions?

Boundary Conditions for the transformed ODE system (3.44)

For the ODE system (3.44) the original boundary conditions

$$\begin{cases} p = s = 0 \quad \text{on}\, y = 0, \\ \gamma p' - bs = 0 \quad \text{on}\, y = 1, \\ (\gamma + 2a)s' - (2a\lambda + b)p = 0 \quad \text{on}\, y = 1, \end{cases}$$

become into

1. $x_1(0) = x_3(0) = 0$,
2. $\gamma x_2(1) - b x_3(1) = 0$,
3. $(\gamma + 2a)x_4(1) - (2a\lambda + b)x_1(1) = 0$.

3.10 Case $\Upsilon > \gamma + 2a\lambda^2$.

We have,

$$\begin{cases} \mu_{3,4} = \pm\sqrt{\dfrac{-\gamma - 2a\lambda_{\text{uni}}^2 + \Upsilon}{(\gamma + 2a)}}i = \pm\theta_3 i, \\ \mu_{1,2} = \pm\sqrt{\dfrac{-\gamma + \Upsilon}{\gamma}}i = \pm\theta_1 i, \end{cases} \tag{3.48}$$

and an analogous equation to (3.20) is obtained as follows

$$\begin{aligned} &-2\theta_3 i(\gamma - \Upsilon - b\lambda_{\text{uni}})(\gamma\lambda_{\text{uni}} - b) + i\left[-\frac{\lambda_{\text{uni}}}{\theta_1}(\gamma - \Upsilon - b\lambda)^2 - \frac{\theta_1\theta_3^2}{\lambda_{\text{uni}}}(\lambda_{\text{uni}}\gamma - b)^2\right]\sin(\theta_1)\sin(\theta_3) \\ &+ \theta_3 i\left[\lambda_{\text{uni}}(\gamma\lambda_{\text{uni}} - b)^2 + \frac{1}{\lambda_{\text{uni}}}(\gamma - \Upsilon - b\lambda_{\text{uni}})^2\right]\cos(\theta_1)\cos(\theta_3) = 0. \end{aligned}$$

i.e.,

$$\begin{aligned} &-2\theta_3(\gamma - \Upsilon - b\lambda_{\text{uni}})(\gamma\lambda_{\text{uni}} - b) + \left[-\frac{\lambda_{\text{uni}}}{\theta_1}(\gamma - \Upsilon - b\lambda)^2 - \frac{\theta_1\theta_3^2}{\lambda_{\text{uni}}}(\lambda_{\text{uni}}\gamma - b)^2\right]\sin(\theta_1)\sin(\theta_3) \\ &+ \theta_3\left[\lambda_{\text{uni}}(\gamma\lambda_{\text{uni}} - b)^2 + \frac{1}{\lambda_{\text{uni}}}(\gamma - \Upsilon - b\lambda_{\text{uni}})^2\right]\cos(\theta_1)\cos(\theta_3) = 0. \end{aligned} \tag{3.49}$$

With help of Python we can conclude that there exists many infinity solutions for (3.49) which are $\Upsilon = \gamma + 2a\lambda^2 = 0.0020332$ which was already checked.

Then the eigenvalues Υ to check are $\Upsilon = 0.045388, 0.123082, 0.147973, 0.208719, 0.294559, \cdots$ (see Figure 3 below).

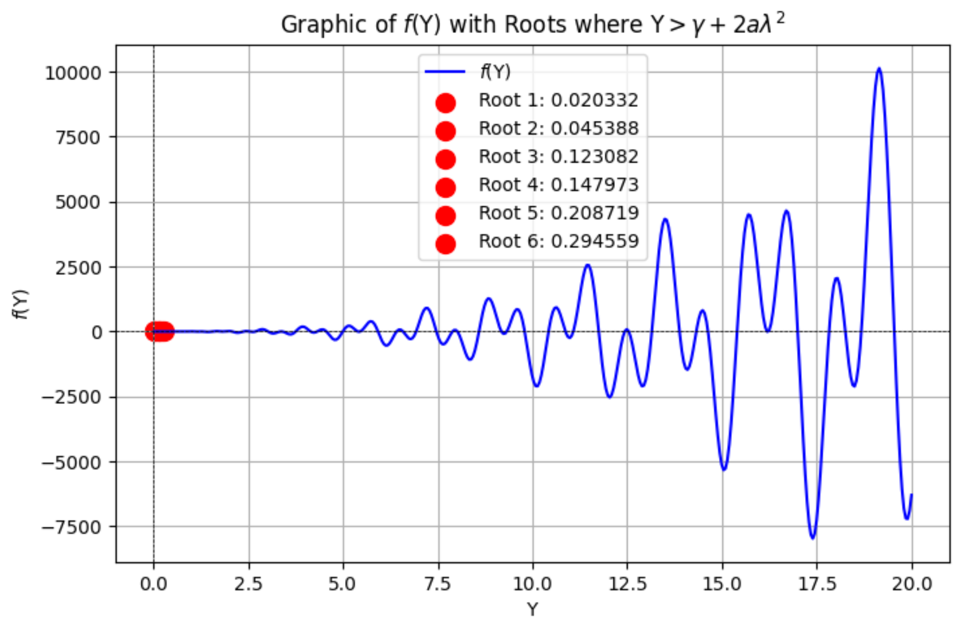


Figure 3: Case $\Upsilon > \gamma + 2a\lambda^2$

We need check which Υ are such that $A_i \neq 0$ with the boundary conditions. The solutions are of the form:

$$p(y) = A_1\theta_1 \cos(\theta_1 y) - A_2\theta_1 \sin(\theta_1 y) + A_3\lambda_{\text{uni}} \cos(\theta_3 y) + A_4\lambda_{\text{uni}} \sin(\theta_3 y), \tag{3.50}$$

$$s(y) = A_1\lambda_{\text{uni}} \sin(\theta_1 y) + A_2\lambda_{\text{uni}} \cos(\theta_1 y) - A_3\theta_3 \sin(\theta_3 y) + A_4\theta_3 \cos(\theta_3 y), \tag{3.51}$$

where A_i are constants for $i \in \{1, 2, 3, 4\}$.

4 New computations for $k \neq \dfrac{1}{\pi\varepsilon}$.

Given the functional,

$$Q(u, v) = \int \gamma(u_x^2 + \varepsilon^{-2}u_y^2 + \varepsilon^2 v_x^2 + v_y^2) + H''(\lambda)(\lambda u_x + v_y)^2 + 2H'(\lambda)(v_y u_x - v_x u_y).$$

Set $a_{11} = 4\pi^2(\gamma + \lambda^2 H''(\lambda))$ $\quad a_{22} = \gamma + H''(\lambda)$ $\quad a_{12} = -\pi(\lambda H''(\lambda) + H'(\lambda))$
$b_{11} = \varepsilon^{-2}\gamma$ $\quad b_{22} = 4\pi^2\gamma\varepsilon^2$ $\quad b_{12} = -\pi H'(\lambda)$

Write $u(x, y) = \displaystyle\sum_{k\in\mathbb{Z}} u_k(y)e^{2i\pi kx}$, $\quad v(x, y) = \displaystyle\sum_{k\in\mathbb{Z}} v_k(y)e^{2i\pi kx}$
where $u_{-k} = \overline{u_k}$ and $v_{-k} = \overline{v_k}$. Therefore, we have

$$\int u_x^2 = 4\pi^2 \sum_{k\in\mathbb{Z}} k^2 \int_0^1 |u_k|^2\,dy, \qquad \int v_x^2 = 4\pi^2 \sum_{k\in\mathbb{Z}} k^2 \int_0^1 |v_k|^2\,dy,$$
$$\int u_y^2 = \sum_{k\in\mathbb{Z}} \int_0^1 |u_k'|^2\,dy, \qquad \int v_y^2\,dy = \sum_{k\in\mathbb{Z}} \int_0^1 |v_k'|^2\,dy,$$
$$\int u_x v_y = 2i\pi \sum_{k\in\mathbb{Z}} k \int_0^1 u_k\overline{v_k'}, \qquad \int v_x u_y = 2i\pi \sum_{k\in\mathbb{Z}} k \int_0^1 v_k\overline{u_k'}.$$

Now, Q can be rewritten as follows

$$\begin{aligned} Q(u,v) &= \int (\gamma+\lambda^2 H''(\lambda))u_x^2 + \gamma\varepsilon^{-2}u_y^2 + \gamma\varepsilon^2 v_x^2 + (\gamma+H''(\lambda))v_y^2 \\ &\quad + 2(\lambda H''(\lambda)+H'(\lambda))u_xv_y - 2H'(\lambda)u_yv_x \\ &= \sum_{k\in\mathbb{Z}} \Bigg(k^2 a_{11}\int |u_k|^2\,dy + b_{11}\int |u_k'|^2\,dy + k^2 b_{22}\int |v_k|^2 + a_{22}\int |v_k'|^2 \\ &\quad -4ika_{12}\int u_k\overline{v_k'} + 4ikb_{12}\int v_k\overline{u_k'} \Bigg). \end{aligned}$$

Then,

$$\begin{aligned} Q(u,v) &= \sum_{k>1} k^2\cdot 2a_{11}\int |u_k|^2 + b_{11}\int_0^1 |u_0'|^2\,dy + a_{22}\int |v_0'|^2 \\ &\quad + \sum_{k>1} 2b_{11}\int |u_k'|^2 + 2a_{22}\int |v_k'|^2 + 2k^2 b_{22}\int |v_k|^2 \\ &\quad - \sum_{k>1} 4ika_{12}\left(\int u_k\overline{v_k'} - u_{-k}\overline{v_{-k}'}\right) + 4ikb_{12}\left(\int v_k\overline{u_k'} - v_{-k}\overline{u_{-k}'}\right). \end{aligned}$$

Consider the following auxiliary computation:

$$\begin{aligned} u\overline{v'} - \overline{u}v' &= (\Re u + i\Im u)(\Re v' - i\Im v') - (\Re u - i\Im u)(\Re v' + i\Im v') \\ &= 2i[-(\Re u)(\Im v') + (\Im u)(\Re v')]. \end{aligned} \tag{4.1}$$

Therefore, by virtue of (4.1), we have

$$\begin{aligned} Q(u,v) &= b_{11}\int_0^1 |u_0'|^2 + a_{22}\int_0^1 |v_0'|^2 + 2\sum_{k>1}\int_0^1 a_{11}(\Re ku_k)^2 + a_{11}(\Im ku_k)^2 \\ &\quad + a_{22}(\Im v_k')^2 + a_{22}(\Re v_k')^2 + 4a_{12}(\Re ku_k)(-\Im v_k') + 4a_{12}(\Im ku_k)(\Re v_k') \\ &\quad + 4b_{12}\Re ku_k')(-\Im kv_k) + 4b_{12}(\Im ku_k')(\Re kv_k) \\ &\quad + b_{11}(\Re u_k')^2 + b_{11}(\Im u_k')^2 + b_{22}(\Re kv_k)^2 + b_{22}(\Im kv_k)^2. \end{aligned}$$

Now, given $L>0$, define

$$Q_L(p,s) := \frac{1}{2}\int_{y=0}^{L} a_{11}p^2 + a_{22}(s')^2 + 4a_{12}ps' + b_{11}(p')^2 + b_{22}s^2 + 4b_{12}p's\,dy. \tag{4.2}$$

It can be seen that

$$Q(u,v) = \int_0^1 b_{11}|u_0'|^2 + a_{22}|v_0'|^2 + 4\sum_{k>1} \frac{1}{k} Q_k\left(y \mapsto k\Re u_k\left(\frac{y}{k}\right), y \mapsto -k\Im v_k\left(\frac{y}{k}\right)\right)$$
$$+ \frac{1}{k} Q_k\left(y \mapsto k\Im u_k\left(\frac{y}{k}\right), y \mapsto k\Re v_k\left(\frac{y}{k}\right)\right).$$

Indeed, if p_k and s_k are defined by

$$p_k(\tilde{y}) = k\Re u_k\left(\frac{\tilde{y}}{k}\right), \quad s_k(\tilde{y}) = -k\Im v_k\left(\frac{\tilde{y}}{k}\right),$$
$$\text{for } 0 \le \tilde{y} \le k, \quad \text{where,} \quad \tilde{y} = ky,$$

then, $p_k' = \Re u_k'(y)$ and $s_k' = -\Im v_k'(y)$ and

$$4\sum_{k>1} \frac{1}{k} \int_0^k [a_{11}p_k^2 + a_{22}s_k'^2 + 4a_{12}p_k s_k' + b_{11}p_k'^2 + b_{22}s_k^2 + 4b_{12}p_k' s_k] d\tilde{y}$$
$$= 2\sum_{k>1} \int_0^1 a_{11}(k\Re u_k(y))^2 + a_{22}(\Im v_k'(y))^2 + 4a_{12}(\Re k u_k(y))(-\Im v_k'(y)) + 4b_{12}(\Re k u_k'(y))(-\Im v_k(y))$$
$$+ b_{11}(k\Re u_k'(y))^2 + b_{22}(\Im v_k(y))^2 \, dy.$$

We obtain something similar if $(p_k, s_k) = (k\Im u_k(\frac{\tilde{y}}{k}), k\Re v_k(\frac{\tilde{y}}{k}))$.

Now, in order to obtain the Euler-Lagrange equation for (4.2), we have

$$\langle Q_L'(p,s), (\dot{p}, \dot{s})\rangle = \int_0^L a_{11}p\dot{p} + a_{22}s'\dot{s}' + 2a_{12}\dot{p}s' + 2a_{12}p\dot{s}'$$
$$+ b_{11}p'\dot{p}' + 2b_{22}s\dot{s} + 2b_{12}\dot{p}'s + 2b_{12}p'\dot{s}\, dy.$$

therefore, we have

$$\begin{cases} -b_{11}p'' + 2(a_{12} - b_{12})s' + a_{11}p = 0, \\ -a_{22}s'' + 2(b_{12} - a_{12})p' + b_{22}s = 0. \end{cases}$$

with the boundary conditions

$$\begin{cases} p(0) = s(0) = 0, \\ b_{11}p'(L) + 2b_{12}s(L) = 0, \\ a_{22}s'(L) + 2a_{12}p(L) = 0. \end{cases} \tag{4.3}$$

For the eigenvalues Υ, we have

$$\begin{cases} -b_{11}p'' + 2(a_{12} - b_{12})s' + a_{11}p = \Upsilon p, \\ -a_{22}s'' + 2(b_{12} - a_{12})p' + b_{22}s = \Upsilon s. \end{cases}$$

Let us seek solutions of the form

$$\begin{aligned} p(y) &= p_0 \cosh(\mu y), \\ s(y) &= s_0 \sinh(\mu y). \end{aligned}$$

Then, the equations system become

$$\begin{cases} -b_{11} p_0 \mu^2 + 2(a_{12} - b_{12}) s_0 \mu + (a_{11} - \Upsilon) p_0 = 0, \\ -a_{22} s_0 \mu^2 + 2(b_{12} - a_{12}) p_0 \mu + (b_{22} - \Upsilon) s_0 = 0. \end{cases}$$

The determinant of this 2×2 system on p_0, s_0 is

$$\begin{vmatrix} -b_{11}\mu^2 + a_{11} - \Upsilon & -2(b_{12} - a_{12})\mu \\ 2(b_{12} - a_{12})\mu & -a_{22}\mu^2 + b_{22} - \Upsilon \end{vmatrix}.$$

For nontrivial solution to exist, it is necessary that the previous determinant be zero, i.e.

$$\begin{aligned} & a_{22} b_{11} \mu^4 + (-b_{11} b_{22} + b_{11} \Upsilon - a_{11} a_{22} + a_{22} \Upsilon + 4(b_{12} - a_{12})^2) \mu^2 \\ & \quad + (a_{11} - \Upsilon)(b_{22} - \Upsilon) = 0. \end{aligned} \tag{4.4}$$

Let be

$$\begin{aligned} a &= a_{22} b_{11}, \\ b(\Upsilon) &= (-b_{11} b_{22} + b_{11} \Upsilon - a_{11} a_{22} + a_{22} \Upsilon + 4(b_{12} - a_{12})^2), \\ c(\Upsilon) &= (a_{11} - \Upsilon)(b_{22} - \Upsilon), \end{aligned} \tag{4.5}$$

Then, the discriminant for the previous biquadratic equation (4.4) is

$$\begin{aligned} \Delta(\Upsilon) &= (b(\Upsilon))^2 - 4ac(\Upsilon) \\ &= (-b_{11} b_{22} - a_{11} a_{22} + 4(b_{12} - a_{12})^2 + (b_{11} + a_{22})\Upsilon)^2 - 4a_{22} b_{11} (a_{11} - \Upsilon)(b_{22} - \Upsilon). \end{aligned}$$

Therefore,

$$\begin{aligned} \Delta(\Upsilon) = {} & b_{11}^2 b_{22}^2 + a_{11}^2 a_{22}^2 + 16(b_{12} - a_{12})^4 + (a_{22} + b_{11})^2 \Upsilon^2 + 2a_{11} a_{22} b_{11} b_{22} \\ & - 8a_{11} a_{22} (b_{12} - a_{12})^2 + 8(a_{22} + b_{11})(b_{12} - a_{12})^2 \Upsilon - 2b_{11} b_{22} (a_{22} + b_{11}) \Upsilon \\ & - 8b_{11} b_{22} (b_{12} - a_{12})^2 - 2a_{11} a_{22} (b_{11} + a_{22}) \Upsilon - 4a_{22} b_{11} (a_{11} b_{22} - a_{11} \Upsilon - b_{22} \Upsilon + \Upsilon^2). \end{aligned}$$

We can check that $\Delta(\Upsilon)$ is not perfect square, but we can rewrite this expression as a quadratic polynomial

$$\Delta(\Upsilon) = C_2 \Upsilon^2 + C_1 \Upsilon + C_0 = C_2 (\Upsilon - \Upsilon_0)(\Upsilon - \Upsilon_1), \tag{4.6}$$

where,

$$\begin{aligned} C_2 &= (a_{22} + b_{11})^2 - 4a_{22} b_{11} = (a_{22} - b_{11})^2, \\ C_1 &= 8(a_{22} + b_{11})(b_{12} - a_{12})^2 - 2b_{11} b_{22} (a_{22} + b_{11}) - 2a_{11} a_{22} (b_{11} + a_{22}) - 4a_{22} b_{11} (-a_{11} - b_{22}), \\ C_0 &= b_{11}^2 b_{22}^2 + a_{11}^2 a_{22}^2 + 16(b_{12} - a_{12})^4 - 2a_{11} a_{22} b_{11} b_{22} - 8a_{11} a_{22} (b_{12} - a_{12})^2 - 8b_{11} b_{22} (b_{12} - a_{12})^2, \end{aligned}$$

and

$$\Upsilon_0 = \frac{-C_1 - \sqrt{C_1^2 - 4C_2C_0}}{2C_2} = -0.0012393615593015068, \tag{4.7}$$

$$\Upsilon_1 = \frac{-C_1 + \sqrt{C_1^2 - 4C_2C_0}}{2C_2} = -0.0000052580389911456956. \tag{4.8}$$

From the previous, we conclude that the roots for (4.4) are of the form

$$\begin{cases} \mu_{3,4} = \pm\sqrt{\dfrac{-b(\Upsilon) + \sqrt{\Delta(\Upsilon)}}{2a}}, \\ \mu_{1,2} = \pm\sqrt{\dfrac{-b(\Upsilon) - \sqrt{\Delta(\Upsilon)}}{2a}}. \end{cases} \tag{4.9}$$

Let be Υ_2 the root of $b(\Upsilon)$ defined in (4.5)

$$\Upsilon_2 = \frac{b_{11}b_{22} + a_{11}a_{22} - 4(b_{12} - a_{12})^2}{(b_{11} + a_{22})} = 0.00016281127970815256,$$

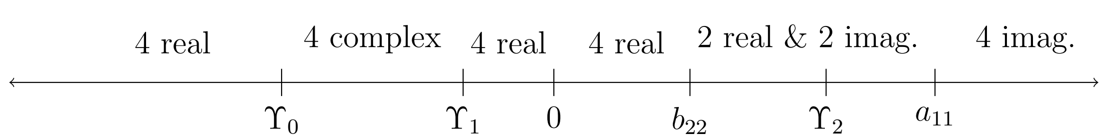
we have that $\Upsilon < \Upsilon_2$ iff $b(\Upsilon) < 0$ and $\Upsilon > \Upsilon_2$ iff $b(\Upsilon) > 0$, where $(b_{11} + a_{22}) \neq 0$. In the other hand since

$$\begin{aligned} C_2 &= 9.506129001535475, \\ C_1 &= 0.011831514459209217, \\ C_0 &= 0.0000000619477486491719 9, \\ a_{11} &= 0.37596051495949284, \\ b_{22} &= 0.000012791007303811812, \end{aligned}$$

now, we can study the nature of the roots for (4.4) as follows:

Nature of the μ_i-roots		
Case	$\Upsilon-$value	Nature of Roots μ_i
1	$(-\infty, \Upsilon_0)$	4 real roots
2	Υ_0	4 real roots (2 pair of repeated roots)
3	(Υ_0, Υ_1)	4 complex roots
4	Υ_1	4 real roots (2 pair of repeated roots)
5	(Υ_1, b_{22})	4 real roots
6	b_{22}	$\mu_{1,2} = 0$ and 2 real roots
7	(b_{22}, a_{11})	2 real roots and 2 pure imaginary roots
8	a_{11}	$\mu_{3,4} = 0$ and 2 pure imaginary roots
9	$(a_{11}, +\infty)$	4 pure imaginary roots

Table 1: Behaviour of the roots μ_i



4.1 Case $\Upsilon_1 < \Upsilon < b_{22}$

The roots are of the form

$$\begin{cases} \mu_{3,4} = \pm\sqrt{\dfrac{-b(\Upsilon) + \sqrt{\Delta(\Upsilon)}}{2a}}, \\ \mu_{1,2} = \pm\sqrt{\dfrac{-b(\Upsilon) - \sqrt{\Delta(\Upsilon)}}{2a}}, \end{cases}$$

When $\mu = \mu_1$, the system

$$\begin{pmatrix} -b_{11}\mu_1^2 + a_{11} - \Upsilon & -2(b_{12} - a_{12})\mu_1 \\ 2(b_{12} - a_{12})\mu_1 & -a_{22}\mu_1^2 + b_{22} - \Upsilon \end{pmatrix} \begin{pmatrix} p_0 \\ s_0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

When the matrix

$$M = \begin{pmatrix} -b_{11}\mu_1^2 + a_{11} - \Upsilon & -2(b_{12} - a_{12})\mu_1 \\ 2(b_{12} - a_{12})\mu_1 & -a_{22}\mu_1^2 + b_{22} - \Upsilon \end{pmatrix}$$

has determinant zero, there exists a non-trivial solution (p_0, s_0) (unique up to scalar multiples) such that

$$M \begin{pmatrix} p_0 \\ s_0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

where

$$\begin{aligned} m_{11} &= -b_{11}\mu_1^2 + a_{11} - \Upsilon, \\ m_{12} &= -2(b_{12} - a_{12})\mu_1, \\ m_{21} &= 2(b_{12} - a_{12})\mu_1, \\ m_{22} &= -a_{22}\mu_1^2 + b_{22} - \Upsilon. \end{aligned}$$

Since $\det M = 0$, the rows (or columns) are linearly dependent. A convenient choice for a vector in the nullspace is

$$\boxed{(p_0, s_0) = \big(m_{12},\ -m_{11}\big)}.$$

Substituting the expressions we have

$$({}^{\mu_1}p_0, {}^{\mu_1}s_0) = \Big(-2(b_{12} - a_{12})\mu_1,\ b_{11}\mu_1^2 - a_{11} + \Upsilon \Big)$$

then

$$\boxed{({}^{\mu_1}p_0, {}^{\mu_1}s_0) = \left(\mu_1,\ \frac{b_{11}\mu_1^2 - a_{11} + \Upsilon}{-2(b_{12} - a_{12})}\right)}. \tag{4.10}$$

This is a non-zero solution; any non-zero scalar multiple of it is also a valid eigenvector.

Similarly, when $\mu = \mu_3$, then

$$({}^{\mu_3}p_0, {}^{\mu_3}s_0) = \Big(-2(b_{12} - a_{12})\mu_3,\ b_{11}\mu_3^2 - a_{11} + \Upsilon \Big)$$

Since $\Upsilon \neq b_{11}\mu_3^2 - a_{11}$ we have

$$\boxed{({}^{\mu_3}p_0, {}^{\mu_3}s_0) = \left(\frac{-2(b_{12}-a_{12})}{b_{11}\mu_3^2 - a_{11} + \Upsilon}\mu_3^2,\ \mu_3\right) = (p_0^*\mu_3^2, \mu_3)}. \tag{4.11}$$

Since the function space of solutions of the homogeneous system has dimension four, the any solution is of the form

$$p(y) = A_1\mu_1\cosh(\mu_1 y) + A_2\mu_1\sinh(\mu_1 y) + A_3({}^{\mu_3}p_0)\cosh(\mu_3 y) + A_4({}^{\mu_3}p_0)\sinh(\mu_3 y), \tag{4.12}$$
$$s(y) = A_1({}^{\mu_1}s_0)\sinh(\mu_1 y) + A_2({}^{\mu_1}s_0)\cosh(\mu_1 y) + A_3\mu_3\sinh(\mu_3 y) + A_4\mu_3\cosh(\mu_3 y), \tag{4.13}$$

where A_i are constants for $i \in \{1, 2, 3, 4\}$.

4.1.1 Computing the equations for amplitude A_n using the boundary conditions

In the other hand, by substituting explicit expressions for p and s given in (4.12)-(4.13) in the boundary conditions (4.3) we have the following equations:

$$A_1\mu_1 + ({}^{\mu_3}p_0)A_3 = 0, \tag{4.14}$$

$$({}^{\mu_1}s_0)A_2 + \mu_3 A_4 = 0, \tag{4.15}$$

$$\begin{aligned}&(b_{11}\mu_1^2 + 2b_{12}({}^{\mu_1}s_0))\sinh(\mu_1 L)A_1 + (b_{11}\mu_1^2 + 2b_{12}({}^{\mu_1}s_0))\cosh(\mu_1 L)A_2 \\ &\quad + (b_{11}({}^{\mu_3}p_0) + 2b_{12})\mu_3\sinh(\mu_3 L)A_3 + (b_{11}({}^{\mu_3}p_0) + 2b_{12})\mu_3\cosh(\mu_3 L)A_4 = 0,\end{aligned} \tag{4.16}$$

$$\begin{aligned}&(a_{22}({}^{\mu_1}s_0) + 2a_{12})\mu_1\cosh(\mu_1 L)A_1 + (a_{22}({}^{\mu_1}s_0) + 2a_{12})\mu_1\sinh(\mu_1 L)A_2 \\ &\quad + (a_{22}\mu_3^2 + 2a_{12}({}^{\mu_3}p_0))\cosh(\mu_3 L)A_3 + (a_{22}\mu_3^2 + 2a_{12}({}^{\mu_3}p_0))\sinh(\mu_3 L)A_4 = 0.\end{aligned} \tag{4.17}$$

In order to reduce the number of equations we have now

$$\begin{aligned}&\left[-\frac{({}^{\mu_3}p_0)}{\mu_1}(b_{11}\mu_1^2 + 2b_{12}({}^{\mu_1}s_0))\sinh(\mu_1 L) + (b_{11}({}^{\mu_3}p_0) + 2b_{12})\mu_3\sinh(\mu_3 L)\right]A_3 \\ &\quad + \left[-\frac{\mu_3}{({}^{\mu_1}s_0)}(b_{11}\mu_1^2 + 2b_{12}({}^{\mu_1}s_0))\cosh(\mu_1 L) + (b_{11}({}^{\mu_3}p_0) + 2b_{12})\mu_3\cosh(\mu_3 L)\right]A_4 = 0.\end{aligned}$$

and

$$\begin{aligned}&\left[-\frac{({}^{\mu_3}p_0)}{\mu_1}(a_{22}({}^{\mu_1}s_0) + 2a_{12})\mu_1\cosh(\mu_1 L) + (a_{22}\mu_3^2 + 2a_{12}({}^{\mu_3}p_0))\cosh(\mu_3 L)\right]A_3 \\ &\quad + \left[-\frac{\mu_3}{({}^{\mu_1}s_0)}(a_{22}({}^{\mu_1}s_0) + 2a_{12})\mu_1\sinh(\mu_1 L) + (a_{22}\mu_3^2 + 2a_{12}({}^{\mu_3}p_0))\sinh(\mu_3 L)\right]A_4 = 0.\end{aligned}$$

Therefore,

$$\left[-\frac{({}^{\mu_3}p_0)}{\mu_1}(b_{11}\mu_1^2 + 2b_{12}({}^{\mu_1}s_0))\sinh(\mu_1 L) + (b_{11}({}^{\mu_3}p_0) + 2b_{12})\mu_3\sinh(\mu_3 L)\right] \times$$
$$\left[-\frac{\mu_3}{({}^{\mu_1}s_0)}(a_{22}({}^{\mu_1}s_0) + 2a_{12})\mu_1\sinh(\mu_1 L) + (a_{22}\mu_3^2 + 2a_{12}({}^{\mu_3}p_0))\sinh(\mu_3 L)\right]$$
$$-\left[-\frac{({}^{\mu_3}p_0)}{\mu_1}(a_{22}({}^{\mu_1}s_0) + 2a_{12})\mu_1\cosh(\mu_1 L) + (a_{22}\mu_3^2 + 2a_{12}({}^{\mu_3}p_0))\cosh(\mu_3 L)\right] \times$$
$$\left[-\frac{\mu_3}{({}^{\mu_1}s_0)}(b_{11}\mu_1^2 + 2b_{12}({}^{\mu_1}s_0))\cosh(\mu_1 L) + (b_{11}({}^{\mu_3}p_0) + 2b_{12})\mu_3\cosh(\mu_3 L)\right] = 0.$$

We can see that $\mu_3 = 0$ is a solution on the previous equation. Then dividing the previous equation by μ_3^2 because in this case $\mu_3 \neq 0$, we have

$$\left[-\frac{(p_0^*)\mu_3}{\mu_1}(b_{11}\mu_1^2 + 2b_{12}({}^{\mu_1}s_0))\sinh(\mu_1 L) + (b_{11}(p_0^*\mu_3^2) + 2b_{12})\sinh(\mu_3 L)\right] \times$$
$$\left[-\frac{1}{({}^{\mu_1}s_0)}(a_{22}({}^{\mu_1}s_0) + 2a_{12})\mu_1\sinh(\mu_1 L) + (a_{22} + 2a_{12}(p_0^*))\mu_3\sinh(\mu_3 L)\right]$$
$$-\left[-\frac{(p_0^*)\mu_3}{\mu_1}(a_{22}({}^{\mu_1}s_0) + 2a_{12})\mu_1\cosh(\mu_1 L) + (a_{22} + 2a_{12}(p_0^*))\mu_3\cosh(\mu_3 L)\right] \times$$
$$\left[-\frac{1}{({}^{\mu_1}s_0)}(b_{11}\mu_1^2 + 2b_{12}({}^{\mu_1}s_0))\cosh(\mu_1 L) + (b_{11}(p_0^*\mu_3^2) + 2b_{12})\cosh(\mu_3 L)\right] = 0.$$

We can use identities for hyperbolic functions and obtain the following master equation

$$\begin{aligned}
&-\mu_3\left[\frac{p_0^*}{({}^{\mu_1}s_0)}(b_{11}\mu_1^2 + 2b_{12}({}^{\mu_1}s_0))(a_{22}({}^{\mu_1}s_0) + 2a_{12}) + (a_{22} + 2a_{12}(p_0^*))(b_{11}(p_0^*\mu_3^2) + 2b_{12})\right] \\
&-\left[\frac{\mu_3^2 p_0^*}{\mu_1}(a_{22} + 2a_{12}(p_0^*))(b_{11}\mu_1^2 + 2b_{12}({}^{\mu_1}s_0)) + \frac{\mu_1}{({}^{\mu_1}s_0)}(b_{11}(p_0^*\mu_3^2) + 2b_{12})(a_{22}({}^{\mu_1}s_0) + 2a_{12})\right] \\
&\times\sinh(\mu_1 L)\sinh(\mu_3 L) \\
&+\left[\frac{\mu_3}{({}^{\mu_1}s_0)}(a_{22} + 2a_{12}(p_0^*))(b_{11}\mu_1^2 + 2b_{12}({}^{\mu_1}s_0)) + \mu_3 p_0^*(b_{11}(p_0^*\mu_3^2) + 2b_{12})(a_{22}({}^{\mu_1}s_0) + 2a_{12})\right] \\
&\times\cosh(\mu_1 L)\cosh(\mu_3 L) = 0.
\end{aligned} \tag{4.18}$$

Dividing (4.18) by μ_3 and multiplying by $({}^{\mu_1}s_0)$ we obtain

$$\begin{aligned}
&-\left[p_0^*(b_{11}\mu_1^2 + 2b_{12}({}^{\mu_1}s_0))(a_{22}({}^{\mu_1}s_0) + 2a_{12}) + ({}^{\mu_1}s_0)(a_{22} + 2a_{12}(p_0^*))(b_{11}(p_0^*\mu_3^2) + 2b_{12})\right] \\
&-\left[\frac{\mu_3 p_0^*({}^{\mu_1}s_0)}{\mu_1}(a_{22} + 2a_{12}(p_0^*))(b_{11}\mu_1^2 + 2b_{12}({}^{\mu_1}s_0)) + \frac{\mu_1}{\mu_3}(b_{11}(p_0^*\mu_3^2) + 2b_{12})(a_{22}({}^{\mu_1}s_0) + 2a_{12})\right] \\
&\times\sinh(\mu_1 L)\sinh(\mu_3 L) \\
&+\left[(a_{22} + 2a_{12}(p_0^*))(b_{11}\mu_1^2 + 2b_{12}({}^{\mu_1}s_0)) + ({}^{\mu_1}s_0)p_0^*(b_{11}(p_0^*\mu_3^2) + 2b_{12})(a_{22}({}^{\mu_1}s_0) + 2a_{12})\right] \\
&\times\cosh(\mu_1 L)\cosh(\mu_3 L) = 0.
\end{aligned} \tag{4.19}$$

4.2 Case $b_{22} < \Upsilon < a_{11}$

We divide the case in three cases:

(a) $\Upsilon_2 < \Upsilon < a_{11}$.

(b) $\Upsilon = \Upsilon_2$.

(c) $b_{22} < \Upsilon < \Upsilon_2$.

4.2.1 Case $\Upsilon_2 < \Upsilon < a_{11}$.

The characteristic equation is:

$$a\mu^4 + b\mu^2 + c = 0,$$

where $a = a_{22}b_{11} = (\gamma + H''(\lambda))\varepsilon^{-2}\gamma > 0$ and

$$c = (a_{11} - \Upsilon)(b_{22} - \Upsilon) \to \begin{cases} c > 0 & \text{if} \quad \Upsilon \in (-\infty, b_{22}) \cup (a_{11}, +\infty), \\ c < 0 & \text{if} \quad \Upsilon \in (b_{22}, a_{11}). \end{cases}$$

Since $(\Upsilon_2, a_{11}) \subset (b_{22}, a_{11})$ then $c < 0$. In the other hand since $\Upsilon > \Upsilon_2$ then $b > 0$. The discriminant

$$\Delta(\Upsilon) = b^2 - 4ac > 0$$

and

$$\sqrt{\Delta(\Upsilon)} = b\sqrt{1 + \frac{4a|c|}{b^2}},$$

since $\frac{4a|c|}{b^2} \sim \mathcal{O}(\varepsilon^{-2})$, then $\sqrt{\Delta(\Upsilon)} \gg b$. Now, the roots

$$\begin{cases} \mu_{3,4}^2 = \dfrac{-b(\Upsilon) + \sqrt{\Delta(\Upsilon)}}{2a} > 0 \\ \mu_{1,2}^2 = \dfrac{-b(\Upsilon) - \sqrt{\Delta(\Upsilon)}}{2a} < 0. \end{cases}$$

therefore

$$\begin{cases} \mu_{3,4} = \pm\sqrt{\dfrac{-b(\Upsilon) + \sqrt{\Delta(\Upsilon)}}{2a}}, \\ \mu_{1,2} = \pm\theta i, \end{cases} \tag{4.20}$$

where $\theta = \sqrt{\dfrac{b(\Upsilon) + \sqrt{\Delta(\Upsilon)}}{2a}}$. Since

$$\frac{-b(\Upsilon) + \sqrt{\Delta(\Upsilon)}}{2a} \sim \frac{\sqrt{4|a||c|}}{2a} \sim \sqrt{\frac{|c|}{a}} = \varepsilon\sqrt{\frac{|c|}{\gamma(\gamma + H''(\lambda))}} \sim \mathcal{O}(\varepsilon),$$

then both θ and $\mu_{3,4}$ will be $\mathcal{O}(\sqrt{\varepsilon})$.

In the other hand, we have the following identities

$$\begin{cases} \sinh(\mu_{1,2}L) = \pm i\sin(\theta L), \\ \sinh(\mu_1 L) = i\sin(\theta L), \\ \cosh(\mu_1 L) = \cos(\theta). \end{cases} \tag{4.21}$$

4.2.2 Case $\Upsilon = \Upsilon_2$.

The characteristic equation is:

$$a\mu^4 + b\mu^2 + c = 0,$$

where $a = a_{22}b_{11} = (\gamma + H''(\lambda))\varepsilon^{-2}\gamma > 0$ and

$$c = (a_{11} - \Upsilon)(b_{22} - \Upsilon) \to \begin{cases} c > 0 & \text{if} \quad \Upsilon \in (-\infty, b_{22}) \cup (a_{11}, +\infty), \\ c < 0 & \text{if} \quad \Upsilon \in (b_{22}, a_{11}). \end{cases}$$

Since $\Upsilon_2 \subset (b_{22}, a_{11})$ then $c < 0$. In the other hand since $\Upsilon = \Upsilon_2$ then $b = 0$. The discriminant

$$\Delta(\Upsilon) = -4ac > 0.$$

Now, the roots

$$\begin{cases} \mu_{3,4}^2 = \dfrac{+\sqrt{-ac}}{a} > 0 \\ \mu_{1,2}^2 = \dfrac{-\sqrt{-ac}}{a} < 0. \end{cases}$$

therefore

$$\begin{cases} \mu_{3,4} = \pm\theta, \\ \mu_{1,2} = \pm\theta i, \end{cases} \tag{4.22}$$

where $\theta = \sqrt{\dfrac{\sqrt{-ac}}{a}}$. The identities in (4.21) holds again.

4.2.3 Case $b_{22} < \Upsilon < \Upsilon_2$.

The characteristic equation is:

$$a\mu^4 + b\mu^2 + c = 0,$$

where $a = a_{22}b_{11} = (\gamma + H''(\lambda))\varepsilon^{-2}\gamma > 0$ and

$$c = (a_{11} - \Upsilon)(b_{22} - \Upsilon) \to \begin{cases} c > 0 & \text{if} \quad \Upsilon \in (-\infty, b_{22}) \cup (a_{11}, +\infty), \\ c < 0 & \text{if} \quad \Upsilon \in (b_{22}, a_{11}). \end{cases}$$

Since $(b_{22}, \Upsilon_2) \subset (b_{22}, a_{11})$ then $c < 0$. In the other hand since $\Upsilon < \Upsilon_2$ then $b < 0$. The discriminant

$$\Delta(\Upsilon) = b^2 - 4ac > 0$$

and

$$\sqrt{\Delta(\Upsilon)} = b\sqrt{1 + \frac{4a|c|}{b^2}},$$

since $\frac{4a|c|}{b^2} \sim \mathcal{O}(\varepsilon^{-2})$, then $\sqrt{\Delta(\Upsilon)} \gg b$. Now, the roots

$$\begin{cases} \mu_{3,4}^2 = \dfrac{-b(\Upsilon) + \sqrt{\Delta(\Upsilon)}}{2a} > 0 \\ \mu_{1,2}^2 = \dfrac{-b(\Upsilon) - \sqrt{\Delta(\Upsilon)}}{2a} < 0. \end{cases}$$

therefore

$$\begin{cases} \mu_{3,4} = \pm\sqrt{\dfrac{-b(\Upsilon)+\sqrt{\Delta(\Upsilon)}}{2a}}, \\ \mu_{1,2} = \pm\theta i, \end{cases} \tag{4.23}$$

where $\theta = \sqrt{\dfrac{b(\Upsilon)+\sqrt{\Delta(\Upsilon)}}{2a}}$. Since

$$\frac{-b(\Upsilon)+\sqrt{\Delta(\Upsilon)}}{2a} \sim \frac{\sqrt{4|a||c|}}{2a} \sim \sqrt{\frac{|c|}{a}} = \varepsilon\sqrt{\frac{|c|}{\gamma(\gamma+H''(\lambda))}} \sim \mathcal{O}(\varepsilon),$$

then both θ and $\mu_{3,4}$ will be $\mathcal{O}(\sqrt{\varepsilon})$. The identities in (4.21) holds again.

4.2.4 Conclusion for $b_{22} < \Upsilon < a_{11}$

We have that for $\Upsilon \in (b_{22}, \Upsilon_2) \cup (\Upsilon_2, a_{11})$ the solutions are of the form

$$\begin{cases} \mu_{3,4} = \pm\sqrt{\dfrac{-b(\Upsilon)+\sqrt{\Delta(\Upsilon)}}{2a}}, \\ \mu_{1,2} = \pm\theta i, \end{cases}$$

where $\theta = \sqrt{\dfrac{b(\Upsilon)+\sqrt{\Delta(\Upsilon)}}{2a}}$. Then by virtue of the identities in (4.21) applied in the master equation (4.18) we have

$$\begin{aligned}
&-\mu_3\left[\frac{p_0^*}{({}^{\mu_1}s_0)}(-b_{11}\theta^2+2b_{12}({}^{\mu_1}s_0))(a_{22}({}^{\mu_1}s_0)+2a_{12})+(a_{22}+2a_{12}(p_0^*))(b_{11}(p_0^*\mu_3^2)+2b_{12})\right] \\
&-\left[\frac{\mu_3^2p_0^*}{\theta i}(a_{22}+2a_{12}(p_0^*))(-b_{11}\theta^2+2b_{12}({}^{\mu_1}s_0))+\frac{\theta i}{({}^{\mu_1}s_0)}(b_{11}(p_0^*\mu_3^2)+2b_{12})(a_{22}({}^{\mu_1}s_0)+2a_{12})\right] \\
&\times i\sin(\theta L)\sinh(\mu_3 L) \\
&+\left[\frac{\mu_3}{({}^{\mu_1}s_0)}(a_{22}+2a_{12}(p_0^*))(-b_{11}\theta^2+2b_{12}({}^{\mu_1}s_0))+\mu_3p_0^*(b_{11}(p_0^*\mu_3^2)+2b_{12})(a_{22}({}^{\mu_1}s_0)+2a_{12})\right] \\
&\times\cos(\theta L)\cosh(\mu_3 L) = 0.
\end{aligned}$$

then

$$\begin{aligned}
&-\mu_3\left[\frac{p_0^*}{({}^{\mu_1}s_0)}(-b_{11}\theta^2+2b_{12}({}^{\mu_1}s_0))(a_{22}({}^{\mu_1}s_0)+2a_{12})+(a_{22}+2a_{12}(p_0^*))(b_{11}(p_0^*\mu_3^2)+2b_{12})\right] \\
&-\left[\frac{\mu_3^2p_0^*}{\theta}(a_{22}+2a_{12}(p_0^*))(-b_{11}\theta^2+2b_{12}({}^{\mu_1}s_0))-\frac{\theta}{({}^{\mu_1}s_0)}(b_{11}(p_0^*\mu_3^2)+2b_{12})(a_{22}({}^{\mu_1}s_0)+2a_{12})\right] \\
&\times\sin(\theta L)\sinh(\mu_3 L) \\
&+\left[\frac{\mu_3}{({}^{\mu_1}s_0)}(a_{22}+2a_{12}(p_0^*))(-b_{11}\theta^2+2b_{12}({}^{\mu_1}s_0))+\mu_3p_0^*(b_{11}(p_0^*\mu_3^2)+2b_{12})(a_{22}({}^{\mu_1}s_0)+2a_{12})\right] \\
&\times\cos(\theta L)\cosh(\mu_3 L) = 0,
\end{aligned} \tag{4.24}$$

where

$$({}^{\mu_1}s_0) = \frac{-b_{11}\theta^2 - a_{11} + \Upsilon}{-2(b_{12} - a_{12})},$$

$$p_0^* = \frac{-2(b_{12} - a_{12})}{b_{11}\mu_3^2 - a_{11} + \Upsilon},$$

$$\theta = \sqrt{\frac{b(\Upsilon) + \sqrt{\Delta(\Upsilon)}}{2a}},$$

$$\mu_3 = \sqrt{\frac{-b(\Upsilon) + \sqrt{\Delta(\Upsilon)}}{2a}}.$$

Finally,

$$\begin{aligned}
&- \left[p_0^*(-b_{11}\theta^2 + 2b_{12}({}^{\mu_1}s_0))(a_{22}({}^{\mu_1}s_0) + 2a_{12}) + ({}^{\mu_1}s_0)(a_{22} + 2a_{12}(p_0^*))(b_{11}(p_0^*\mu_3^2) + 2b_{12})\right] \\
&- \left[\frac{\mu_3 p_0^*({}^{\mu_1}s_0)}{\theta}(a_{22} + 2a_{12}(p_0^*))(-b_{11}\theta^2 + 2b_{12}({}^{\mu_1}s_0)) - \frac{\theta}{\mu_3}(b_{11}(p_0^*\mu_3^2) + 2b_{12})(a_{22}({}^{\mu_1}s_0) + 2a_{12})\right] \\
&\times \sin(\theta L)\sinh(\mu_3 L) \\
&+ \left[(a_{22} + 2a_{12}(p_0^*))(-b_{11}\theta^2 + 2b_{12}({}^{\mu_1}s_0)) + ({}^{\mu_1}s_0)p_0^*(b_{11}(p_0^*\mu_3^2) + 2b_{12})(a_{22}({}^{\mu_1}s_0) + 2a_{12})\right] \\
&\times \cos(\theta L)\cosh(\mu_3 L) = 0,
\end{aligned} \tag{4.25}$$

where

$$({}^{\mu_1}s_0) = \frac{-b_{11}\theta^2 - a_{11} + \Upsilon}{-2(b_{12} - a_{12})},$$

$$p_0^* = \frac{-2(b_{12} - a_{12})}{b_{11}\mu_3^2 - a_{11} + \Upsilon},$$

$$\theta = \sqrt{\frac{b(\Upsilon) + \sqrt{\Delta(\Upsilon)}}{2a}},$$

$$\mu_3 = \sqrt{\frac{-b(\Upsilon) + \sqrt{\Delta(\Upsilon)}}{2a}}.$$

For the case of $\Upsilon = \Upsilon_2$ we have

$$\begin{aligned}
&- \theta\left[\frac{p_0^*}{({}^{\mu_1}s_0)}(-b_{11}\theta^2 + 2b_{12}({}^{\mu_1}s_0))(a_{22}({}^{\mu_1}s_0) + 2a_{12}) + (a_{22} + 2a_{12}(p_0^*))(b_{11}(p_0^*\mu_3^2) + 2b_{12})\right] \\
&- \left[\frac{\mu_3^2 p_0^*}{\theta}(a_{22} + 2a_{12}(p_0^*))(-b_{11}\theta^2 + 2b_{12}({}^{\mu_1}s_0)) - \frac{\theta}{({}^{\mu_1}s_0)}(b_{11}(p_0^*\mu_3^2) + 2b_{12})(a_{22}({}^{\mu_1}s_0) + 2a_{12})\right] \\
&\times \sin(\theta L)\sinh(\mu_3 L) \\
&+ \left[\frac{\mu_3}{({}^{\mu_1}s_0)}(a_{22} + 2a_{12}(p_0^*))(-b_{11}\theta^2 + 2b_{12}({}^{\mu_1}s_0)) + \mu_3 p_0^*(b_{11}(p_0^*\mu_3^2) + 2b_{12})(a_{22}({}^{\mu_1}s_0) + 2a_{12})\right] \\
&\times \cos(\theta L)\cosh(\mu_3 L) = 0,
\end{aligned}$$

where

$$
\begin{aligned}
({}^{\mu_1}s_0) &= \frac{-b_{11}\theta^2 - a_{11} + \Upsilon}{-2(b_{12}-a_{12})},\\
p_0^* &= \frac{-2(b_{12}-a_{12})}{b_{11}\theta^2 - a_{11} + \Upsilon},\\
\mu_1 &= \theta i,\\
\mu_3 &= \theta := \sqrt{\frac{\sqrt{-ac}}{a}}.
\end{aligned}
$$

therefore

$$
\begin{aligned}
&-\theta\left[(p_0^*)^2(-b_{11}\theta^2 + 2b_{12}({}^{\mu_1}s_0))(a_{22}({}^{\mu_1}s_0) + 2a_{12}) + (a_{22} + 2a_{12}(p_0^*))(b_{11}(p_0^*\mu_3^2) + 2b_{12})\right]\\
&-\left[\theta p_0^*(a_{22} + 2a_{12}(p_0^*))(-b_{11}\theta^2 + 2b_{12}({}^{\mu_1}s_0)) - \frac{\theta}{({}^{\mu_1}s_0)}(b_{11}(p_0^*\theta^2) + 2b_{12})(a_{22}({}^{\mu_1}s_0) + 2a_{12})\right]\\
&\times\sin(\theta L)\sinh(\theta L)\\
&+\left[\frac{\theta}{({}^{\mu_1}s_0)}(a_{22} + 2a_{12}(p_0^*))(-b_{11}\theta^2 + 2b_{12}({}^{\mu_1}s_0)) + \mu_3 p_0^*(b_{11}(p_0^*\theta^2) + 2b_{12})(a_{22}({}^{\mu_1}s_0) + 2a_{12})\right]\\
&\times\cos(\theta L)\cosh(\theta L) = 0, \qquad (4.26)
\end{aligned}
$$

4.3 Case $\Upsilon = b_{22}$

Since

$$c = (a_{11} - \Upsilon)(b_{22} - \Upsilon) = 0$$

then $\Delta(\Upsilon) = b^2$. In the other hand since $b_{22} < \Upsilon_2$ then $b(\Upsilon) < 0$, therefore $|b| = -b$. Now, the roots are as follow

$$
\begin{cases}
\mu_{3,4}^2 = \dfrac{-b(\Upsilon) + \sqrt{\Delta(\Upsilon)}}{2a} = \dfrac{-b + |b|}{2a} = \dfrac{-b - b}{2a} = -\dfrac{b}{a} > 0,\\
\mu_{1,2}^2 = \dfrac{-b(\Upsilon) - \sqrt{\Delta(\Upsilon)}}{2a} = \dfrac{-b - |b|}{2a} = \dfrac{-b + b}{2a} = 0.
\end{cases}
$$

Finally,

$$
\begin{cases}
\mu_{3,4} = \pm\sqrt{-\dfrac{b}{a}},\\
\mu_{1,2} = 0.
\end{cases}
\qquad (4.27)
$$

4.4 Case $\Upsilon = a_{11}$

Since

$$c = (a_{11} - \Upsilon)(b_{22} - \Upsilon) = 0$$

then $\Delta(\Upsilon) = b^2$. In the other hand since $b_{22} > \Upsilon_2$ then $b(\Upsilon) > 0$, therefore $|b| = b$. Now, the roots are as follow

$$\begin{cases} \mu_{3,4}^2 = \dfrac{-b(\Upsilon) + \sqrt{\Delta(\Upsilon)}}{2a} = \dfrac{-b + |b|}{2a} = \dfrac{-b + b}{2a} = 0, \\ \mu_{1,2}^2 = \dfrac{-b(\Upsilon) - \sqrt{\Delta(\Upsilon)}}{2a} = \dfrac{-b - |b|}{2a} = \dfrac{-b - b}{2a} = -\dfrac{b}{a} < 0. \end{cases}$$

Finally,

$$\begin{cases} \mu_{3,4} = 0, \\ \mu_{1,2} = \pm\theta i, \quad \text{where} \quad \theta := \sqrt{\dfrac{b}{a}}. \end{cases} \tag{4.28}$$

4.5 Case $\Upsilon = \Upsilon_0$ or $\Upsilon = \Upsilon_1$

Since $C_2 > 0$ and

$$\sqrt{\Delta(\Upsilon)} = \sqrt{C_2(\Upsilon - \Upsilon_0)(\Upsilon - \Upsilon_1)},$$

then $\sqrt{\Delta(\Upsilon_{0,1})} = 0$. In the other hand since $\Upsilon < \Upsilon_2$ then $b < 0$. Now, the roots are as follow

$$\begin{cases} \mu_{3,4}^2 = \dfrac{-b(\Upsilon) + \sqrt{\Delta(\Upsilon)}}{2a} = \dfrac{-b}{2a} > 0, \\ \mu_{1,2}^2 = \dfrac{-b(\Upsilon) - \sqrt{\Delta(\Upsilon)}}{2a} = \dfrac{-b}{2a} > 0. \end{cases}$$

Finally, for $\Upsilon = \Upsilon_0$ we have

$$\mu_{3,4} = \mu_{1,2} = \pm\sqrt{\frac{-b(\Upsilon_0)}{2a}} = \pm\sqrt{\frac{1}{2}\left(\frac{1}{a_{11}} + \frac{1}{b_{11}}\right)}\sqrt{(\Upsilon_2 - \Upsilon_0)}, \tag{4.29}$$

we have two pair of real repeated roots. Similarly, for $\Upsilon = \Upsilon_1$ we have

$$\mu_{3,4} = \mu_{1,2} = \pm\sqrt{\frac{-b(\Upsilon_1)}{2a}} = \pm\sqrt{\frac{1}{2}\left(\frac{1}{a_{11}} + \frac{1}{b_{11}}\right)}\sqrt{(\Upsilon_2 - \Upsilon_1)} \tag{4.30}$$

again we have two pair of real repeated roots.

4.6 Case $\Upsilon_0 < \Upsilon < \Upsilon_1$

We have that

$$\Delta(\Upsilon) = C_2(\Upsilon - \Upsilon_0)(\Upsilon - \Upsilon_1) < 0,$$

since $\Upsilon < \Upsilon_2$ then $b < 0$.

Now, the roots are as follow

$$\begin{cases} \mu_{3,4}^2 = \dfrac{-b(\Upsilon) + \sqrt{\Delta(\Upsilon)}}{2a} = \dfrac{-b(\Upsilon) + \sqrt{-\Delta(\Upsilon)}i}{2a}, \\ \mu_{1,2}^2 = \dfrac{-b(\Upsilon) - \sqrt{\Delta(\Upsilon)}}{2a} = \dfrac{-b(\Upsilon) - \sqrt{-\Delta(\Upsilon)}i}{2a}. \end{cases}$$

Therefore,

$$\begin{cases} \mu_{3,4} = \pm\sqrt{z_1 - z_2 i}, \\ \mu_{1,2} = \pm\sqrt{z_1 + z_2 i}, \quad \text{where} \quad z_2 := -\dfrac{\sqrt{-\Delta(\Upsilon)}}{2a},\ z_1 := -\dfrac{b(\Upsilon)}{2a} \neq 0. \end{cases}$$

Finally,

$$\begin{cases} \mu_{3,4} = \pm\left(\sqrt{\dfrac{|z| + z_1}{2}} - \dfrac{z_2}{|z_2|}\sqrt{\dfrac{|z| - z_1}{2}}i\right), \\ \mu_{1,2} = \pm\left(\sqrt{\dfrac{|z| + z_1}{2}} + \dfrac{z_2}{|z_2|}\sqrt{\dfrac{|z| - z_1}{2}}i\right), \end{cases} \tag{4.31}$$

where

$$z_2 := -\frac{\sqrt{-\Delta(\Upsilon)}}{2a},\ z_1 := -\frac{b(\Upsilon)}{2a},\ |z| = \sqrt{z_1^2 + z_2^2}.$$

We can compute explicitly the form of the roots as follows:

$$\begin{aligned} |z| &= \sqrt{\frac{b^2}{4a^2} + \frac{-\Delta(\Upsilon)}{4a^2}} = \sqrt{\frac{b^2 - (b^2 - 4ac)}{4a^2}} \\ &= \sqrt{\frac{4ac}{4a^2}} = \sqrt{\frac{c}{a}}, \end{aligned}$$

and $c = (a_{11} - \Upsilon)(b_{22} - \Upsilon) > 0$ since $(\Upsilon_0, \Upsilon_1) \subset (-\infty, b_{22}) \cup (a_{11}, +\infty)$. In the other hand

$$\frac{z_2}{|z_2|} = \frac{-\dfrac{\sqrt{-\Delta(\Upsilon)}}{2a}}{\left|-\dfrac{\sqrt{-\Delta(\Upsilon)}}{2a}\right|} = -1.$$

We have now that

$$\begin{cases} \mu_{3,4} = \pm\left(\sqrt{\dfrac{\sqrt{\frac{c}{a}} - \frac{b}{2a}}{2}} + \sqrt{\dfrac{\sqrt{\frac{c}{a}} + \frac{b}{2a}}{2}}i\right), \\ \mu_{1,2} = \pm\left(\sqrt{\dfrac{\sqrt{\frac{c}{a}} - \frac{b}{2a}}{2}} - \sqrt{\dfrac{\sqrt{\frac{c}{a}} + \frac{b}{2a}}{2}}i\right). \end{cases}$$

Finally, the eigenvalues are of the form

$$\begin{cases} \mu_{3,4} = \pm(\alpha + \beta i), \\ \mu_{1,2} = \pm(\alpha - \beta i), \end{cases} \tag{4.32}$$

where,

$$\alpha = \sqrt{\frac{\sqrt{\frac{c}{a}} - \frac{b}{2a}}{2}}, \tag{4.33}$$

$$\beta = \sqrt{\frac{\sqrt{\frac{c}{a}} + \frac{b}{2a}}{2}}, \tag{4.34}$$

Now, by the classical standard theory for ODE we have that the solutions are of the form

$$\begin{aligned}\begin{pmatrix} p(y) \\ s(y) \end{pmatrix} &= A_1[B_1\cos(\beta y) - B_2\sin(\beta y)]e^{\alpha y} + A_2[B_2\cos(\beta y) + B_1\sin(\beta y)]e^{\alpha y} \\ &+ A_3[B_3\cos(\beta y) - B_4\sin(\beta y)]e^{-\alpha y} + A_4[B_4\cos(\beta y) + B_3\sin(\beta y)]e^{-\alpha y}, \end{aligned} \tag{4.35}$$

where

$$B_1 = \mathfrak{Re}(K_1),\ B_2 = \mathfrak{Im}(K_1),$$
$$B_3 = \mathfrak{Re}(K_2),\ B_4 = \mathfrak{Im}(K_2),$$

and where

$$K_1 = \begin{pmatrix} {}^{\mu_3}p_0 \\ {}^{\mu_3}s_0 \end{pmatrix}$$
$$K_2 = \begin{pmatrix} {}^{\mu_2}p_0 \\ {}^{\mu_2}s_0 \end{pmatrix}$$

are the eigenvectors associated to the complex eigenvalues $\mu_3 = \alpha + \beta i$ and $\mu_2 = -\alpha + \beta i$ respectively. By (4.10) we have similarly that

$$({}^{\mu_3}p_0, {}^{\mu_3}s_0) = \left(\mu_3,\ \frac{b_{11}\mu_3^2 - a_{11} + \Upsilon}{-2(b_{12} - a_{12})}\right),$$
$$({}^{\mu_2}p_0, {}^{\mu_2}s_0) = \left(\mu_2,\ \frac{b_{11}\mu_2^2 - a_{11} + \Upsilon}{-2(b_{12} - a_{12})}\right).$$

Then, we have explicits forms for K_1 and K_2

$$K_1 = \begin{pmatrix} \alpha \\ \alpha^* \end{pmatrix} + \begin{pmatrix} \beta \\ \beta^* \end{pmatrix} i,$$
$$K_2 = \begin{pmatrix} -\alpha \\ \alpha^* \end{pmatrix} + \begin{pmatrix} \beta \\ -\beta^* \end{pmatrix} i,$$

where

$$\alpha^* = \frac{b_{11}(\alpha^2-\beta^2) - a_{11} + \Upsilon}{-2(b_{12}-a_{12})},$$
$$\beta^* = -\frac{\alpha\beta b_{11}}{b_{12}-a_{12}}.$$

Now, (4.35) becomes explicitly into

$$\begin{aligned} p(y) = {} & A_1[\alpha\cos(\beta y) - \beta\sin(\beta y)]e^{\alpha y} + A_2[\beta\cos(\beta y) + \alpha\sin(\beta y)]e^{\alpha y} \\ & + A_3[-\alpha\cos(\beta y) - \beta\sin(\beta y)]e^{-\alpha y} + A_4[\beta\cos(\beta y) - \alpha\sin(\beta y)]e^{-\alpha y}, \end{aligned} \tag{4.36}$$

and

$$\begin{aligned} s(y) = {} & A_1[\alpha^*\cos(\beta y) - \beta^*\sin(\beta y)]e^{\alpha y} + A_2[\beta^*\cos(\beta y) + \alpha^*\sin(\beta y)]e^{\alpha y} \\ & + A_3[\alpha^*\cos(\beta y) + \beta^*\sin(\beta y)]e^{-\alpha y} + A_4[-\beta^*\cos(\beta y) + \alpha^*\sin(\beta y)]e^{-\alpha y}, \end{aligned} \tag{4.37}$$

We apply the boundary conditions (4.3) to obtain the 4×4 system with unknows A_i to find conditions for stability. Therefore, we have that $p(0)=0$ becomes into

$$\alpha A_1 + \beta A_2 - \alpha A_3 + \beta A_4 = 0. \tag{4.38}$$

The boundary condition $s(0)=0$ becomes into

$$\alpha^* A_1 + \beta^* A_2 + \alpha^* A_3 - \beta^* A_4 = 0. \tag{4.39}$$

The boundary condition $b_{11}p'(L) + 2b_{12}s(L) = 0$ becomes into

$$\begin{aligned} & e^{\alpha L}\left[(b_{11}(\alpha^2-\beta^2) + 2b_{12}\alpha^*)\cos(\beta L) + (-2b_{11}\alpha\beta - 2b_{12}\beta^*)\sin(\beta L)\right] A_1 \\ & + e^{\alpha L}\left[(2b_{11}\alpha\beta + 2b_{12}\beta^*)\cos(\beta L) + (b_{11}(\alpha^2-\beta^2) + 2b_{12}\alpha^*)\sin(\beta L)\right] A_2 \\ & + e^{-\alpha L}\left[(b_{11}(\alpha^2-\beta^2) + 2b_{12}\alpha^*)\cos(\beta L) + (2b_{11}\alpha\beta + 2b_{12}\beta^*)\sin(\beta L)\right] A_3 \\ & + e^{-\alpha L}\left[(-2b_{11}\alpha\beta - 2b_{12}\beta^*)\cos(\beta L) + (b_{11}(\alpha^2-\beta^2) + 2b_{12}\alpha^*)\sin(\beta L)\right] A_4 = 0. \end{aligned} \tag{4.40}$$

Finally, the boundary condition $a_{22}s'(L) + 2a_{12}p(L) = 0$ becomes into

$$\begin{aligned} & e^{\alpha L}\left[(a_{22}(\alpha\alpha^* - \beta\beta^*) + 2a_{12}\alpha)\cos(\beta L) + (-a_{22}(\alpha\beta^* + \beta\alpha^*) - 2a_{12}\beta)\sin(\beta L)\right] A_1 \\ & + e^{\alpha L}\left[(a_{22}(\alpha\beta^* + \beta\alpha^*) + 2a_{12}\beta)\cos(\beta L) + (a_{22}(\alpha\alpha^* - \beta\beta^*) + 2a_{12}\alpha)\sin(\beta L)\right] A_2 \\ & + e^{-\alpha L}\left[-(a_{22}(\alpha\alpha^* - \beta\beta^*) + 2a_{12}\alpha)\cos(\beta L) + (-a_{22}(\alpha\beta^* + \beta\alpha^*) - 2a_{12}\beta)\sin(\beta L)\right] A_3 \\ & + e^{-\alpha L}\left[(a_{22}(\alpha\beta^* + \beta\alpha^*) + 2a_{12}\beta)\cos(\beta L) + (-a_{22}(\alpha\alpha^* - \beta\beta^*) - 2a_{12}\alpha)\sin(\beta L)\right] A_4 = 0. \end{aligned} \tag{4.41}$$

Let be,

$$\begin{aligned} \rho_1 &= b_{11}(\alpha^2-\beta^2) + 2b_{12}\alpha^*, \\ \rho_2 &= -2b_{11}\alpha\beta - 2b_{12}\beta^*, \\ \eta_1 &= a_{22}(\alpha\alpha^* - \beta\beta^*) + 2a_{12}\alpha, \\ \eta_2 &= -a_{22}(\alpha\beta^* + \beta\alpha^*) - 2a_{12}\beta, \end{aligned}$$

then we have that (4.40) and (4.41) become into

$$\begin{aligned}
&e^{\alpha L}\left[\rho_1\cos(\beta L)+\rho_2\sin(\beta L)\right]A_1+e^{\alpha L}\left[-\rho_2\cos(\beta L)+\rho_1\sin(\beta L)\right]A_2\\
&+e^{-\alpha L}\left[\rho_1\cos(\beta L)-\rho_2\sin(\beta L)\right]A_3+e^{-\alpha L}\left[\rho_2\cos(\beta L)+\rho_1\sin(\beta L)\right]A_4=0,
\end{aligned} \tag{4.42}$$

and

$$\begin{aligned}
&e^{\alpha L}\left[\eta_1\cos(\beta L)+\eta_2\sin(\beta L)\right]A_1+e^{\alpha L}\left[-\eta_2\cos(\beta L)+\eta_1\sin(\beta L)\right]A_2\\
&+e^{-\alpha L}\left[-\eta_1\cos(\beta L)+\eta_2\sin(\beta L)\right]A_3+e^{-\alpha L}\left[-\eta_2\cos(\beta L)-\eta_1\sin(\beta L)\right]A_4=0,
\end{aligned} \tag{4.43}$$

respectively.

Finally the 4×4 system can be written as follows

$$MA=0,$$

, where $A=\begin{bmatrix}A_1\\A_2\\A_3\\A_4\end{bmatrix}$ and

$$M(L)=\begin{bmatrix}
\alpha & \beta & -\alpha & \beta\\
\alpha^* & \beta^* & \alpha^* & -\beta^*\\
e^{\alpha L}\left[\rho_1\cos(\beta L)+\rho_2\sin(\beta L)\right] & e^{\alpha L}\left[-\rho_2\cos(\beta L)+\rho_1\sin(\beta L)\right] & e^{-\alpha L}\left[\rho_1\cos(\beta L)-\rho_2\sin(\beta L)\right] & e^{-\alpha L}\left[\rho_2\cos(\beta L)+\rho_1\sin(\beta L)\right]\\
e^{\alpha L}\left[\eta_1\cos(\beta L)+\eta_2\sin(\beta L)\right] & e^{\alpha L}\left[-\eta_2\cos(\beta L)+\eta_1\sin(\beta L)\right] & e^{-\alpha L}\left[-\eta_1\cos(\beta L)+\eta_2\sin(\beta L)\right] & e^{-\alpha L}\left[-\eta_2\cos(\beta L)-\eta_1\sin(\beta L)\right]
\end{bmatrix}. \tag{4.44}$$

Then, the unstable condition

$$\det[M(L)]=0. \tag{4.45}$$

For this we have

$$M=\begin{bmatrix}
\alpha & \beta & -\alpha & \beta\\
\alpha^* & \beta^* & \alpha^* & -\beta^*\\
m & n & o & p^*\\
r & q & t & s^*
\end{bmatrix}, \tag{4.46}$$

where

$$m=e^{\alpha L}\big(\rho_1\cos(\beta L)+\rho_2\sin(\beta L)\big), \tag{4.47}$$
$$n=e^{\alpha L}\big(-\rho_2\cos(\beta L)+\rho_1\sin(\beta L)\big), \tag{4.48}$$
$$o=e^{-\alpha L}\big(\rho_1\cos(\beta L)-\rho_2\sin(\beta L)\big), \tag{4.49}$$
$$p^*=e^{-\alpha L}\big(\rho_2\cos(\beta L)+\rho_1\sin(\beta L)\big), \tag{4.50}$$
$$r=e^{\alpha L}\big(\eta_1\cos(\beta L)+\eta_2\sin(\beta L)\big), \tag{4.51}$$
$$q=e^{\alpha L}\big(-\eta_2\cos(\beta L)+\eta_1\sin(\beta L)\big), \tag{4.52}$$
$$t=e^{-\alpha L}\big(-\eta_1\cos(\beta L)+\eta_2\sin(\beta L)\big), \tag{4.53}$$
$$s^*=e^{-\alpha L}\big(-\eta_2\cos(\beta L)-\eta_1\sin(\beta L)\big). \tag{4.54}$$

Then the unstable condition is

$$
\begin{aligned}
&m\big(q(\alpha\beta^* - \beta\alpha^*) + s^*(\alpha\beta^* + \beta\alpha^*) + 2t\beta\beta^*\big) - n\big(r(\alpha\beta^* - \beta\alpha^*) + 2s^*\alpha\alpha^* + t(\alpha\beta^* + \beta\alpha^*)\big) \\
&+ o\big(q(\alpha\beta^* + \beta\alpha^*) - 2r\beta\beta^* + s^*(\alpha\beta^* - \beta\alpha^*)\big) + p^*\big(2q\alpha\alpha^* - r(\alpha\beta^* + \beta\alpha^*) + t(\beta\alpha^* - \alpha\beta^*)\big) = 0
\end{aligned} \tag{4.55}
$$

Short-wavelength

In the other hand, since $\alpha > 0$ we have that as $L \to \infty$ then $e^{-\alpha L} \to 0$ and therefore

$$
M(L) = \begin{bmatrix} \alpha & \beta & -\alpha & \beta \\ \alpha^* & \beta^* & \alpha^* & -\beta^* \\ e^{\alpha L}\left[\rho_1 \cos(\beta L) + \rho_2 \sin(\beta L)\right] & e^{\alpha L}\left[-\rho_2 \cos(\beta L) + \rho_1 \sin(\beta L)\right] & 0 & 0 \\ e^{\alpha L}\left[\eta_1 \cos(\beta L) + \eta_2 \sin(\beta L)\right] & e^{\alpha L}\left[-\eta_2 \cos(\beta L) + \eta_1 \sin(\beta L)\right] & 0 & 0 \end{bmatrix}. \tag{4.56}
$$

Then, the unstable condition (4.45) implies that

$$
\begin{aligned}
e^{2\alpha L}&\left\{\left[\rho_1 \cos(\beta L) + \rho_2 \sin(\beta L)\right]\left[-\eta_2 \cos(\beta L) + \eta_1 \sin(\beta L)\right]\right. \\
&\left.- \left[-\rho_2 \cos(\beta L) + \rho_1 \sin(\beta L)\right]\left[\eta_1 \cos(\beta L) + \eta_2 \sin(\beta L)\right]\right\}(\alpha\beta^* - \beta\alpha^*) = 0.
\end{aligned}
$$

Now, we have that

$$
\begin{aligned}
&\left\{\left[\rho_1 \cos(\beta L) + \rho_2 \sin(\beta L)\right]\left[-\eta_2 \cos(\beta L) + \eta_1 \sin(\beta L)\right]\right. \\
&\quad\left.- \left[-\rho_2 \cos(\beta L) + \rho_1 \sin(\beta L)\right]\left[\eta_1 \cos(\beta L) + \eta_2 \sin(\beta L)\right]\right\} = 0 \qquad (4.57)
\end{aligned}
$$
$$ \tag{4.58} $$

or

$$
(\alpha\beta^* - \beta\alpha^*) = 0, \tag{4.59}
$$

where

$$
\begin{aligned}
\alpha^* &= \frac{b_{11}(\alpha^2 - \beta^2) - a_{11} + \Upsilon}{-2(b_{12} - a_{12})}, \\
\beta^* &= -\frac{\alpha\beta b_{11}}{b_{12} - a_{12}}.
\end{aligned}
$$

4.7 Case $\Upsilon > a_{11}$

The characteristic equation is:

$$
a\mu^4 + b\mu^2 + c = 0,
$$

where $a = a_{22}b_{11} = (\gamma + H''(\lambda))\varepsilon^{-2}\gamma > 0$ and

$$
c = (a_{11} - \Upsilon)(b_{22} - \Upsilon) \to \begin{cases} c > 0 & \text{if} \quad \Upsilon \in (-\infty, b_{22}) \cup (a_{11}, +\infty), \\ c < 0 & \text{if} \quad \Upsilon \in (b_{22}, a_{11}). \end{cases}
$$

Since $(a_{11}, +\infty) \subset ((-\infty, b_{22}) \cup (a_{11}, +\infty)$ then $c > 0$. In the other hand since $\Upsilon > \Upsilon_2$ then $b > 0$. The discriminant

$$\Delta(\Upsilon) = b^2 - 4ac = C_2(\Upsilon - \Upsilon_0)(\Upsilon - \Upsilon_1) > 0,$$

since $\Upsilon > \Upsilon_1$. Now, the roots

$$\begin{cases} \mu_{3,4}^2 = \dfrac{-b(\Upsilon) + \sqrt{\Delta(\Upsilon)}}{2a} < 0 \\ \mu_{1,2}^2 = \dfrac{-b(\Upsilon) - \sqrt{\Delta(\Upsilon)}}{2a} < 0. \end{cases}$$

therefore

$$\begin{cases} \mu_{3,4} = \pm\theta_3 i, , \\ \mu_{1,2} = \pm\theta_1 i, \end{cases} \tag{4.60}$$

where $\theta_1 = \sqrt{\dfrac{b(\Upsilon) + \sqrt{\Delta(\Upsilon)}}{2a}}$ and $\theta_3 = \sqrt{\dfrac{b(\Upsilon) - \sqrt{\Delta(\Upsilon)}}{2a}}$.

In the other hand, we have the following identities

$$\begin{cases} \sinh(\mu_{1,2}L) = \pm i \sin(\theta_1 L), \\ \sinh(\mu_1 L) = i \sin(\theta_1 L), \\ \cosh(\mu_1 L) = \cos(\theta_1). \end{cases} \tag{4.61}$$

and

$$\begin{cases} \sinh(\mu_{3,4}L) = \pm i \sin(\theta_3 L), \\ \sinh(\mu_3 L) = i \sin(\theta_3 L), \\ \cosh(\mu_3 L) = \cos(\theta_3). \end{cases} \tag{4.62}$$

Then by virtue of the identities in (4.61)-(4.62) applied in the master equation (4.18) we have

$$\begin{aligned} &- \theta_3 i \left[\frac{p_0^*}{({}^{\mu_1}s_0)}(-b_{11}\theta_1^2 + 2b_{12}({}^{\mu_1}s_0))(a_{22}({}^{\mu_1}s_0) + 2a_{12}) + (a_{22} + 2a_{12}(p_0^*))(b_{11}(-p_0^*\theta_3^2) + 2b_{12}) \right] \\ &- \left[\frac{-\theta_3^2 p_0^*}{\theta_1 i}(a_{22} + 2a_{12}(p_0^*))(-b_{11}\theta_1^2 + 2b_{12}({}^{\mu_1}s_0)) + \frac{\theta_1 i}{({}^{\mu_1}s_0)}(b_{11}(-p_0^*\theta_3^2) + 2b_{12})(a_{22}({}^{\mu_1}s_0) + 2a_{12}) \right] \\ &\times i \sin(\theta_1 L) i \sin(\theta_3 L) \\ &+ \left[\frac{\theta_3 i}{({}^{\mu_1}s_0)}(a_{22} + 2a_{12}(p_0^*))(-b_{11}\theta_1^2 + 2b_{12}({}^{\mu_1}s_0)) + \theta_3 i p_0^*(-b_{11}(p_0^*\theta_3^2) + 2b_{12})(a_{22}({}^{\mu_1}s_0) + 2a_{12}) \right] \\ &\times \cos(\theta_1 L)\cos(\theta_3 L) = 0. \end{aligned}$$

then dividing by i we have

$$
\begin{aligned}
&- \theta_3 \left[\frac{p_0^*}{({}^{\mu_1}s_0)} (-b_{11}\theta_1^2 + 2b_{12}({}^{\mu_1}s_0))(a_{22}({}^{\mu_1}s_0) + 2a_{12}) + (a_{22} + 2a_{12}(p_0^*))(b_{11}(-p_0^*\theta_3^2) + 2b_{12}) \right] \\
&+ \left[\frac{\theta_3^2 p_0^*}{\theta_1} (a_{22} + 2a_{12}(p_0^*))(-b_{11}\theta_1^2 + 2b_{12}({}^{\mu_1}s_0)) + \frac{\theta_1}{({}^{\mu_1}s_0)} (b_{11}(-p_0^*\theta_3^2) + 2b_{12})(a_{22}({}^{\mu_1}s_0) + 2a_{12}) \right] \\
&\times \sin(\theta_1 L)\sin(\theta_3 L) \\
&+ \left[\frac{\theta_3}{({}^{\mu_1}s_0)} (a_{22} + 2a_{12}(p_0^*))(-b_{11}\theta_1^2 + 2b_{12}({}^{\mu_1}s_0)) + \theta_3 p_0^* (-b_{11}(p_0^*\theta_3^2) + 2b_{12})(a_{22}({}^{\mu_1}s_0) + 2a_{12}) \right] \\
&\times \cos(\theta_1 L)\cos(\theta_3 L) = 0, \qquad (4.63)
\end{aligned}
$$

where

$$
\begin{aligned}
({}^{\mu_1}s_0) &= \frac{-b_{11}\theta_1^2 - a_{11} + \Upsilon}{-2(b_{12} - a_{12})}, \\
p_0^* &= \frac{-2(b_{12} - a_{12})}{-b_{11}\theta_3^2 - a_{11} + \Upsilon}, \\
\theta_1 &= \sqrt{\frac{b(\Upsilon) + \sqrt{\Delta(\Upsilon)}}{2a}}, \\
\theta_3 &= \sqrt{\frac{b(\Upsilon) - \sqrt{\Delta(\Upsilon)}}{2a}}.
\end{aligned}
$$

Asymptotic analysis of the master equation as $\varepsilon \to 0$ (October 13)

We consider the master equation

$$
\begin{aligned}
&- \Big[p_0^*(-b_{11}\theta^2 + 2b_{12}({}^{\mu_1}s_0))(a_{22}({}^{\mu_1}s_0) + 2a_{12}) + ({}^{\mu_1}s_0)(a_{22} + 2a_{12}p_0^*)(b_{11}(p_0^*\mu_3^2) + 2b_{12}) \Big] \\
&- \Big[\frac{\mu_3 p_0^*({}^{\mu_1}s_0)}{\theta} (a_{22} + 2a_{12}p_0^*)(-b_{11}\theta^2 + 2b_{12}({}^{\mu_1}s_0)) - \frac{\theta}{\mu_3} (b_{11}(p_0^*\mu_3^2) + 2b_{12})(a_{22}({}^{\mu_1}s_0) + 2a_{12}) \Big] \\
&\qquad \times \sin(\theta L)\sinh(\mu_3 L) \\
&+ \Big[(a_{22} + 2a_{12}p_0^*)(-b_{11}\theta^2 + 2b_{12}({}^{\mu_1}s_0)) + ({}^{\mu_1}s_0)p_0^*(b_{11}(p_0^*\mu_3^2) + 2b_{12})(a_{22}({}^{\mu_1}s_0) + 2a_{12}) \Big] \\
&\qquad \times \cos(\theta L)\cosh(\mu_3 L) = 0, \qquad (4.64)
\end{aligned}
$$

where

$$({}^{\mu_1}s_0) = \frac{-b_{11}\theta^2 - a_{11} + \Upsilon}{-2(b_{12} - a_{12})},$$

$$p_0^* = \frac{-2(b_{12} - a_{12})}{b_{11}\mu_3^2 - a_{11} + \Upsilon},$$

$$\theta = \sqrt{\frac{b(\Upsilon) + \sqrt{\Delta(\Upsilon)}}{2a}},$$

$$\mu_3 = \sqrt{\frac{-b(\Upsilon) + \sqrt{\Delta(\Upsilon)}}{2a}},$$

and

$$a = a_{22}b_{11}, \tag{4.65}$$

$$b(\Upsilon) = (-b_{11}b_{22} + b_{11}\Upsilon - a_{11}a_{22} + a_{22}\Upsilon + 4(b_{12} - a_{12})^2), \tag{4.66}$$

$$c(\Upsilon) = (a_{11} - \Upsilon)(b_{22} - \Upsilon). \tag{4.67}$$

The coefficients are given by

$$a_{11} = 4\pi^2(\gamma + \lambda^2 H''(\lambda)), \qquad a_{22} = \gamma + H''(\lambda), \qquad a_{12} = -\pi(\lambda H''(\lambda) + H'(\lambda)),$$

$$b_{11} = \varepsilon^{-2}\gamma, \qquad b_{22} = 4\pi^2\gamma\varepsilon^2, \qquad b_{12} = -\pi H'(\lambda).$$

1. Scaling as $\varepsilon \to 0$

From the definitions we have

$$b_{11} \sim O(\varepsilon^{-2}), \qquad b_{22} \sim O(\varepsilon^2), \qquad a = a_{22}b_{11} \sim O(\varepsilon^{-2}).$$

A standard asymptotic expansion of the roots of the quadratic characteristic equation yields

$$\theta^2 = \frac{b + \sqrt{\Delta}}{2a} = \frac{\Upsilon}{a_{22}} + O(\varepsilon^2), \tag{4.68}$$

$$\mu_3^2 = \frac{-b + \sqrt{\Delta}}{2a} = \varepsilon^2 \frac{a_{11} - \Upsilon}{\gamma} + O(\varepsilon^4). \tag{4.69}$$

Thus,

$$\theta \to \sqrt{\frac{\Upsilon}{a_{22}}} \quad \text{(finite)}, \qquad \mu_3 \sim \varepsilon\sqrt{\frac{a_{11} - \Upsilon}{\gamma}} \quad (\to 0).$$

The remaining quantities scale as

$${}^{\mu_1}s_0 = O(\varepsilon^{-2}), \qquad p_0^* = O(\varepsilon^{-2}), \qquad p_0^*\mu_3^2 = O(1).$$

2. Dominant balance in the master equation

Let us schematically denote

$$-A(\varepsilon) - B(\varepsilon)\sin(\theta L)\sinh(\mu_3 L) + C(\varepsilon)\cos(\theta L)\cosh(\mu_3 L) = 0.$$

Estimating the orders of magnitude:

$$A(\varepsilon) \sim O(\varepsilon^{-6}), \qquad B(\varepsilon)\sinh(\mu_3 L) \sim O(\varepsilon^{-6}), \qquad C(\varepsilon) \sim O(\varepsilon^{-8}).$$

Since $C(\varepsilon)$ dominates as $\varepsilon \to 0$, the leading order of the equation is

$$C(\varepsilon)\cos(\theta L)\cosh(\mu_3 L) \simeq 0.$$

Because $\cosh(\mu_3 L) \to 1$ and $C(\varepsilon) \neq 0$ generically, the only way to satisfy the equation at leading order is

$$\boxed{\cos(\theta L) = 0.}$$

3. Limiting eigenvalue condition

From the asymptotic relation $\theta^2 \to \Upsilon / a_{22}$, the limiting condition becomes

$$\sqrt{\frac{\Upsilon}{a_{22}}}\, L = \frac{\pi}{2} + n\pi, \qquad n \in \mathbb{Z}.$$

Hence,

$$\boxed{\Upsilon_n = a_{22}\left(\frac{(2n+1)\pi}{2L}\right)^2 = \big(\gamma + H''(\lambda)\big)\left(\frac{(2n+1)\pi}{2L}\right)^2, \qquad n = 0, 1, 2, \ldots}$$

4. Remarks

- The above derivation is generic; it assumes that no exact cancellations occur in the prefactor $C(\varepsilon)$. If such a cancellation takes place, one would need to retain the next order in ε to obtain a modified condition.
- The quantities ${}^{\mu_1}s_0$ and p_0^* behave like ε^{-2}, which confirms that the term proportional to $\cos(\theta L)\cosh(\mu_3 L)$ indeed dominates.
- Therefore, in the singular limit $\varepsilon \to 0$, the master equation reduces to the simple eigenvalue condition $\cos(\theta L) = 0$.

5. Summary

$$\boxed{\text{As } \varepsilon \to 0, \quad \text{the master equation reduces to } \cos(\theta L) = 0, \quad \theta^2 = \frac{\Upsilon}{a_{22}}.}$$

$$\boxed{\text{Hence the limiting eigenvalues are } \Upsilon_n = a_{22}\left(\frac{(2n+1)\pi}{2L}\right)^2.}$$

Asymptotic expansions for the master equation as $\varepsilon \to 0$

We start from the master equation (written schematically below in the same notation as in the statement):

$$\begin{aligned}
&-\Big[p_0^*(-b_{11}\theta^2 + 2b_{12}s_0)(a_{22}s_0 + 2a_{12}) + s_0(a_{22} + 2a_{12}p_0^*)(b_{11}(p_0^*\mu_3^2) + 2b_{12})\Big] \\
&\quad -\Big[\frac{\mu_3 p_0^* s_0}{\theta}(a_{22} + 2a_{12}p_0^*)(-b_{11}\theta^2 + 2b_{12}s_0) - \frac{\theta}{\mu_3}(b_{11}(p_0^*\mu_3^2) + 2b_{12})(a_{22}s_0 + 2a_{12})\Big] \\
&\qquad\quad \times \sin(\theta L)\sinh(\mu_3 L) \\
&\quad + \Big[(a_{22} + 2a_{12}p_0^*)(-b_{11}\theta^2 + 2b_{12}s_0) + s_0 p_0^*(b_{11}(p_0^*\mu_3^2) + 2b_{12})(a_{22}s_0 + 2a_{12})\Big] \\
&\qquad\quad \times \cos(\theta L)\cosh(\mu_3 L) = 0,
\end{aligned} \tag{4.70}$$

where, for brevity, we write $s_0 \equiv ({}^{\mu_1}s_0)$. The auxiliary definitions are

$$s_0 = \frac{-b_{11}\theta^2 - a_{11} + \Upsilon}{-2(b_{12} - a_{12})},$$

$$p_0^* = \frac{-2(b_{12} - a_{12})}{b_{11}\mu_3^2 - a_{11} + \Upsilon},$$

$$\theta = \sqrt{\frac{b(\Upsilon) + \sqrt{\Delta(\Upsilon)}}{2a}}, \qquad \mu_3 = \sqrt{\frac{-b(\Upsilon) + \sqrt{\Delta(\Upsilon)}}{2a}},$$

with

$$\begin{aligned}
a &= a_{22}b_{11}, \\
b(\Upsilon) &= -b_{11}b_{22} + b_{11}\Upsilon - a_{11}a_{22} + a_{22}\Upsilon + 4(b_{12} - a_{12})^2, \\
c(\Upsilon) &= (a_{11} - \Upsilon)(b_{22} - \Upsilon), \\
\Delta(\Upsilon) &= b(\Upsilon)^2 - 4a\,c(\Upsilon).
\end{aligned}$$

The parameter assignments are

$$a_{11} = 4\pi^2(\gamma + \lambda^2 H''(\lambda)), \qquad a_{22} = \gamma + H''(\lambda), \qquad a_{12} = -\pi(\lambda H''(\lambda) + H'(\lambda)),$$

$$b_{11} = \frac{\gamma}{\varepsilon^2}, \qquad b_{22} = 4\pi^2\gamma\,\varepsilon^2, \qquad b_{12} = -\pi H'(\lambda).$$

Scaling observations (leading orders)

As $\varepsilon \to 0$,

$$b_{11} = \gamma\varepsilon^{-2} = O(\varepsilon^{-2}), \qquad b_{22} = 4\pi^2\gamma\varepsilon^2 = O(\varepsilon^2), \qquad a = a_{22}b_{11} = O(\varepsilon^{-2}).$$

Using these scalings and expanding the quadratic-root formulas for θ^2 and μ_3^2 (standard singular perturbation expansion for the two roots of a quadratic with coefficients depending on b_{11}), one obtains the leading asymptotic forms

$$\theta^2 = \frac{\Upsilon}{a_{22}} + O(\varepsilon^2), \tag{4.71}$$

$$\mu_3^2 = \varepsilon^2 \frac{a_{11} - \Upsilon}{\gamma} + O(\varepsilon^4). \tag{4.72}$$

Hence θ remains $O(1)$ (finite) and $\mu_3 = O(\varepsilon) \to 0$.

Explicit expansion for s_0

From its definition,

$$s_0 = \frac{-b_{11}\theta^2 - a_{11} + \Upsilon}{-2(b_{12} - a_{12})}.$$

Insert the expansion $\theta^2 = \dfrac{\Upsilon}{a_{22}} + O(\varepsilon^2)$ and $b_{11} = \gamma\varepsilon^{-2}$. Then

$$-b_{11}\theta^2 = -\frac{\gamma}{\varepsilon^2}\left(\frac{\Upsilon}{a_{22}} + O(\varepsilon^2)\right) = -\frac{\gamma\Upsilon}{a_{22}}\varepsilon^{-2} + O(1). \tag{4.73}$$

Therefore the expansion of s_0 is

$$s_0 = \frac{\gamma\Upsilon}{2a_{22}(b_{12} - a_{12})}\,\varepsilon^{-2} + \frac{a_{11} - \Upsilon}{2(b_{12} - a_{12})} + O(\varepsilon^2). \tag{4.74}$$

So s_0 has a leading order $O(\varepsilon^{-2})$ term and an $O(1)$ term as shown.

Explicit expansion for μ_3^2 (repeat and comment)

From (4.72) we reproduce the explicit leading term:

$$\boxed{\mu_3^2 = \varepsilon^2 \frac{a_{11} - \Upsilon}{\gamma} + O(\varepsilon^4)\ .} \tag{4.75}$$

Consequently $\mu_3 = \varepsilon\sqrt{\dfrac{a_{11} - \Upsilon}{\gamma}} + O(\varepsilon^3)$ (for $a_{11} > \Upsilon$ so the root is real; if $a_{11} < \Upsilon$ the root is imaginary and similar statements hold with i factors).

Expansion and leading scaling of p_0^*

The definition is

$$p_0^* = \frac{-2(b_{12} - a_{12})}{b_{11}\mu_3^2 - a_{11} + \Upsilon}.$$

Now substitute the expansion $\mu_3^2 = \varepsilon^2 \frac{a_{11}-\Upsilon}{\gamma} + O(\varepsilon^4)$. Then

$$b_{11}\mu_3^2 = \frac{\gamma}{\varepsilon^2} \cdot \left(\varepsilon^2 \frac{a_{11} - \Upsilon}{\gamma} + O(\varepsilon^4)\right) = a_{11} - \Upsilon + O(\varepsilon^2).$$

Hence the denominator $b_{11}\mu_3^2 - a_{11} + \Upsilon$ *vanishes at leading order* and is $O(\varepsilon^2)$. This implies that p_0^* is large of order ε^{-2}. We write

$$p_0^* = P_{-2}\,\varepsilon^{-2} + O(1), \tag{4.76}$$

for some constant P_{-2} that depends on the *next* coefficient in the expansion of μ_3^2 (the $O(\varepsilon^4)$ term in μ_3^2). Concretely, if we write

$$\mu_3^2 = \varepsilon^2 \frac{a_{11} - \Upsilon}{\gamma} + \varepsilon^4 Q + O(\varepsilon^6)$$

for some (computable) $Q = Q(\Upsilon, \gamma, a_{22}, a_{11}, a_{12}, b_{12}, \ldots)$, then

$$b_{11}\mu_3^2 - a_{11} + \Upsilon = \gamma\varepsilon^{-2}\Big(\varepsilon^2 \frac{a_{11} - \Upsilon}{\gamma} + \varepsilon^4 Q\Big) - a_{11} + \Upsilon = \gamma\varepsilon^2 Q + O(\varepsilon^4).$$

Thus

$$P_{-2} \;=\; -\frac{2(b_{12} - a_{12})}{\gamma Q}, \tag{4.77}$$

and the result (4.76) follows. In words: the leading coefficient P_{-2} is *explicitly determined* by the coefficient Q of the ε^4 term in the expansion of μ_3^2. The calculation of Q requires carrying the asymptotic expansion for the discriminant $\Delta(\Upsilon)$ one order further than the leading balance shown above. That can be done algebraically (straightforward but a bit lengthy), and I can produce the explicit Q on request.

Dominant balance in the master equation and limiting condition

Using the leading scalings:

$$s_0 = O(\varepsilon^{-2}), \qquad p_0^* = O(\varepsilon^{-2}), \qquad \mu_3 = O(\varepsilon),$$

and inspecting the three grouped terms of the master equation (4.70), one finds that the coefficient multiplying $\cos(\theta L)\cosh(\mu_3 L)$ is of the highest algebraic order in ε (it scales like ε^{-8} generically), while the other grouped terms are of lower algebraic order. Consequently, for a *generic* parameter set (no accidental cancellation of the large prefactor), the leading-order balance enforces

$$\boxed{\cos(\theta L) = 0.}$$

Using $\theta^2 \to \dfrac{\Upsilon}{a_{22}}$ from (4.71) this yields the limiting eigenvalue condition

$$\sqrt{\frac{\Upsilon}{a_{22}}}\,L = \frac{\pi}{2} + n\pi, \qquad n \in \mathbb{Z},$$

or equivalently

$$\boxed{\Upsilon_n = a_{22}\Big(\frac{(2n+1)\pi}{2L}\Big)^2 = \big(\gamma + H''(\lambda)\big)\Big(\frac{(2n+1)\pi}{2L}\Big)^2, \qquad n = 0, 1, 2, \ldots.}$$

Summary of the expansions (compact)

$$\theta^2 = \frac{\Upsilon}{a_{22}} + O(\varepsilon^2),$$

$$\mu_3^2 = \varepsilon^2\,\frac{a_{11}-\Upsilon}{\gamma} + O(\varepsilon^4),$$

$$s_0 = \frac{\gamma\Upsilon}{2a_{22}(b_{12}-a_{12})}\,\varepsilon^{-2} + \frac{a_{11}-\Upsilon}{2(b_{12}-a_{12})} + O(\varepsilon^2),$$

$$p_0^* = P_{-2}\,\varepsilon^{-2} + O(1), \qquad P_{-2} = -\frac{2(b_{12}-a_{12})}{\gamma Q},$$

where Q is the coefficient of ε^4 in the expansion $\mu_3^2 = \varepsilon^2\frac{a_{11}-\Upsilon}{\gamma} + \varepsilon^4 Q + \cdots$. The constant Q can be written explicitly by expanding $\sqrt{\Delta(\Upsilon)}$ to the next order; this is an algebraic (but somewhat lengthy) expression in $(\Upsilon, \gamma, a_{11}, a_{22}, a_{12}, b_{12})$.

Final remark on degeneracies

The derivation above is *generic*: if the large prefactor that multiplies $\cos(\theta L)\cosh(\mu_3 L)$ happens to be exactly zero for some special parameter relations, the dominant balance changes and one must retain the next order in ε to obtain the correct limiting condition. In that degenerate case the procedure is the same but one must carry the expansion one order further (for instance computing Q explicitly and checking cancellations).

5 Computations October 14 of 2025

Detailed derivation of the asymptotic form of μ_3^2

Recall that

$$\theta^2 = \frac{b(\Upsilon)+\sqrt{\Delta(\Upsilon)}}{2a}, \qquad \mu_3^2 = \frac{-b(\Upsilon)+\sqrt{\Delta(\Upsilon)}}{2a},$$

where

$$a = a_{22}b_{11}, \qquad \Delta(\Upsilon) = b(\Upsilon)^2 - 4a\,c(\Upsilon),$$

and

$$b(\Upsilon) = -b_{11}b_{22} + b_{11}\Upsilon - a_{11}a_{22} + a_{22}\Upsilon + 4(b_{12}-a_{12})^2, \qquad c(\Upsilon) = (a_{11}-\Upsilon)(b_{22}-\Upsilon).$$

Step 1: Determine the ε-scaling of each term. Using the given coefficients:

$$b_{11} = \frac{\gamma}{\varepsilon^2}, \qquad b_{22} = 4\pi^2\gamma\,\varepsilon^2, \qquad b_{12} = -\pi H'(\lambda),$$

we find

$$a = a_{22}b_{11} = \frac{a_{22}\gamma}{\varepsilon^2}.$$

Let us expand each part in powers of ε:

$$
\begin{aligned}
b(\Upsilon) &= -b_{11}b_{22} + b_{11}\Upsilon - a_{11}a_{22} + a_{22}\Upsilon + 4(b_{12} - a_{12})^2 \\
&= -\frac{\gamma}{\varepsilon^2}(4\pi^2\gamma\varepsilon^2) + \frac{\gamma}{\varepsilon^2}\Upsilon - a_{11}a_{22} + a_{22}\Upsilon + 4(b_{12} - a_{12})^2 \\
&= \frac{\gamma\Upsilon}{\varepsilon^2} - 4\pi^2\gamma^2 - a_{11}a_{22} + a_{22}\Upsilon + 4(b_{12} - a_{12})^2.
\end{aligned}
$$

Hence, as $\varepsilon \to 0$,

$$
b(\Upsilon) = \frac{\gamma\Upsilon}{\varepsilon^2} + O(1).
$$

Similarly, for $c(\Upsilon)$:

$$
c(\Upsilon) = (a_{11} - \Upsilon)(b_{22} - \Upsilon) = (a_{11} - \Upsilon)(4\pi^2\gamma\varepsilon^2 - \Upsilon) = -(a_{11} - \Upsilon)\Upsilon + O(\varepsilon^2).
$$

Step 2: Compute the discriminant $\Delta(\Upsilon)$.

$$
\begin{aligned}
\Delta(\Upsilon) &= b(\Upsilon)^2 - 4a\,c(\Upsilon) \\
&= \left(\frac{\gamma\Upsilon}{\varepsilon^2} + O(1)\right)^2 - 4\frac{a_{22}\gamma}{\varepsilon^2}\left(-(a_{11} - \Upsilon)\Upsilon + O(\varepsilon^2)\right) \\
&= \frac{\gamma^2\Upsilon^2}{\varepsilon^4} + 4\frac{a_{22}\gamma}{\varepsilon^2}(a_{11} - \Upsilon)\Upsilon + O(\varepsilon^{-2}).
\end{aligned}
$$

Step 3: Expand the square root. We need $\sqrt{\Delta(\Upsilon)}$. Factor out the dominant term $\frac{\gamma^2\Upsilon^2}{\varepsilon^4}$:

$$
\sqrt{\Delta(\Upsilon)} = \frac{\gamma\Upsilon}{\varepsilon^2}\sqrt{1 + 4\frac{a_{22}}{\gamma}\varepsilon^2\frac{a_{11} - \Upsilon}{\Upsilon} + O(\varepsilon^4)}.
$$

Using the Taylor expansion $\sqrt{1+x} = 1 + \frac{1}{2}x + O(x^2)$, we obtain

$$
\sqrt{\Delta(\Upsilon)} = \frac{\gamma\Upsilon}{\varepsilon^2}\left[1 + 2\frac{a_{22}}{\gamma}\varepsilon^2\frac{a_{11} - \Upsilon}{\Upsilon} + O(\varepsilon^4)\right] = \frac{\gamma\Upsilon}{\varepsilon^2} + 2a_{22}(a_{11} - \Upsilon) + O(\varepsilon^2).
$$

Step 4: Substitute into the formula for μ_3^2. By definition,

$$
\mu_3^2 = \frac{-b(\Upsilon) + \sqrt{\Delta(\Upsilon)}}{2a}.
$$

Substitute the expansions:

$$
\begin{aligned}
-b(\Upsilon) + \sqrt{\Delta(\Upsilon)} &= -\left(\frac{\gamma\Upsilon}{\varepsilon^2} + O(1)\right) + \left(\frac{\gamma\Upsilon}{\varepsilon^2} + 2a_{22}(a_{11} - \Upsilon) + O(\varepsilon^2)\right) \\
&= 2a_{22}(a_{11} - \Upsilon) + O(\varepsilon^2).
\end{aligned}
$$

Then divide by $2a = 2a_{22}\gamma/\varepsilon^2$:

$$
\mu_3^2 = \frac{2a_{22}(a_{11} - \Upsilon)}{2(a_{22}\gamma/\varepsilon^2)} + O(\varepsilon^4) = \varepsilon^2\,\frac{a_{11} - \Upsilon}{\gamma} + O(\varepsilon^4).
$$

Step 5: Interpretation. Thus,

$$\boxed{\mu_3^2 = \varepsilon^2 \, \frac{a_{11} - \Upsilon}{\gamma} + O(\varepsilon^4).}$$

This means that $\mu_3 \sim \varepsilon$ and tends to zero linearly with ε, while θ remains finite. The physical interpretation is that the "hyperbolic" mode becomes very slowly varying in space as the small parameter ε tends to zero, which is typical of singular perturbation (boundary-layer) structures.

$$\begin{bmatrix} \dfrac{1}{2m(z-w)} & 0 & 0 & -\dfrac{2(z-w)}{r-x} \\ 0 & \dfrac{1}{ym^2} & 1 & 0 \\ \left(\dfrac{n}{2(z-w)} + \dfrac{2z}{ym^2}\right)\sinh(ml) & \left(\dfrac{n}{2(z-w)} + \dfrac{2z}{ym^2}\right)\cosh(ml) & 2z & 2zl \\ \left(\dfrac{1}{m} + \dfrac{w}{m(z-w)}\right)\cosh(ml) & \left(\dfrac{1}{m} + \dfrac{w}{m(z-w)}\right)\sinh(ml) & 0 & \left(r - \dfrac{4w(z-w)}{r-x}\right) \end{bmatrix}$$

6 October 16, Master equation for case: $\Upsilon = b_{22}$

We consider the formula

$$\mu_3 \;=\; \sqrt{\frac{a_{11}a_{22} - b_{22}a_{22} - 4(a_{12} - b_{12})^2}{a_{22}b_{11}}}.$$

Using the scalings

$$b_{11} = \frac{\gamma}{\varepsilon^2}, \qquad b_{22} = 4\pi^2\gamma\,\varepsilon^2, \qquad a_{22} = O(1),$$

we examine numerator and denominator separately.

Numerator.

$$\begin{aligned} N &:= a_{11}a_{22} - b_{22}a_{22} - 4(a_{12} - b_{12})^2 \\ &= a_{11}a_{22} - 4(a_{12} - b_{12})^2 \;+\; O(\varepsilon^2), \end{aligned}$$

because $b_{22}a_{22} = a_{22}\, 4\pi^2\gamma\,\varepsilon^2 = O(\varepsilon^2)$.

Denominator.

$$D := a_{22}b_{11} = a_{22}\,\frac{\gamma}{\varepsilon^2} = \frac{a_{22}\gamma}{\varepsilon^2}.$$

Conclusion (expansión principal). Dividing,

$$\mu_3^2 = \frac{N}{D} = \frac{a_{11}a_{22} - 4(a_{12}-b_{12})^2 + O(\varepsilon^2)}{a_{22}\gamma/\varepsilon^2}$$
$$= \varepsilon^2\, \frac{a_{11}a_{22} - 4(a_{12}-b_{12})^2}{a_{22}\gamma} \;+\; O(\varepsilon^4).$$

Therefore

$$\boxed{\mu_3 = \varepsilon\, \sqrt{\frac{a_{11}a_{22} - 4(a_{12}-b_{12})^2}{a_{22}\gamma}} \;+\; O(\varepsilon^3).}$$

The master equation is

$$\left[a_{22}\mu_3^2\left(\frac{b_{11}}{2(b_{12}-a_{12})}+\frac{2b_{12}}{a_{22}\mu_3^2}\right)\left(a_{22}-\frac{4a_{12}(b_{12}-a_{12})}{(b_{22}-a_{11})}\right)-8b_{12}\left(1+\frac{a_{12}}{(b_{12}-a_{12})}\right)\left(\frac{(b_{12}-a_{12})^2}{(b_{12}-a_{11})}\right)\right]$$
$$\times \cosh(\mu_3 L)$$
$$-\, b_{12}a_{22}\mu_3 L\left(1+\frac{a_{12}}{(b_{12}-a_{12})}\right)\sinh(\mu_3 L) - 2b_{12}\left(a_{22}-\frac{4a_{12}(b_{12}-a_{12})}{(b_{22}-a_{11})}\right)$$
$$+\, 4\mu_3^2 a_{22}\left(\frac{b_{11}}{2(b_{12}-a_{12})}+\frac{2b_{12}}{a_{22}\mu_3^2}\right)\frac{(b_{12}-a_{12})^2}{(b_{22}-a_{11})}\left(1+\frac{a_{12}}{(b_{12}-a_{12})}\right) = 0$$

Asymptotic evaluation of the given expression as $\varepsilon \to 0$

We start from

$$E(\varepsilon) = \left[a_{22}\mu_3^2\left(\frac{b_{11}}{2(b_{12}-a_{12})}+\frac{2b_{12}}{a_{22}\mu_3^2}\right)\left(a_{22}-\frac{4a_{12}(b_{12}-a_{12})}{(b_{22}-a_{11})}\right)-8b_{12}\left(1+\frac{a_{12}}{(b_{12}-a_{12})}\right)\left(\frac{(b_{12}-a_{12})^2}{(b_{12}-a_{11})}\right)\right]$$
$$\times \cosh(\mu_3 L) - b_{12}a_{22}\mu_3 L\left(1+\frac{a_{12}}{(b_{12}-a_{12})}\right)\sinh(\mu_3 L)$$
$$-\, 2b_{12}\left(a_{22}-\frac{4a_{12}(b_{12}-a_{12})}{(b_{22}-a_{11})}\right)$$
$$+\, 4\mu_3^2 a_{22}\left(\frac{b_{11}}{2(b_{12}-a_{12})}+\frac{2b_{12}}{a_{22}\mu_3^2}\right)\frac{(b_{12}-a_{12})^2}{(b_{22}-a_{11})}\left(1+\frac{a_{12}}{(b_{12}-a_{12})}\right) = 0.$$

Detailed order analysis of each term as $\varepsilon \to 0$

We recall the asymptotic scalings:

$$b_{11} = \frac{\gamma}{\varepsilon^2}, \qquad b_{22} = O(\varepsilon^2), \qquad \mu_3 = \varepsilon M + O(\varepsilon^3), \qquad \mu_3^2 = \varepsilon^2 M^2 + O(\varepsilon^4),$$

and all coefficients a_{ij}, b_{12} are $O(1)$ constants. We also set $\Delta = b_{12} - a_{12} = O(1)$.

1. Leading-order behaviour of the inner factor. The first composite term appearing in $E(\varepsilon)$ is

$$T_1 = \frac{b_{11}}{2\Delta} + \frac{2b_{12}}{a_{22}\mu_3^2}.$$

- The first part:

$$\frac{b_{11}}{2\Delta} = \frac{\gamma}{2\Delta}\varepsilon^{-2} \quad \Rightarrow \quad O(\varepsilon^{-2}).$$

- The second part:

$$\frac{2b_{12}}{a_{22}\mu_3^2} = \frac{2b_{12}}{a_{22}(\varepsilon^2 M^2)} = O(\varepsilon^{-2}).$$

Hence,

$$T_1 = O(\varepsilon^{-2}).$$

—

2. Multiply by $a_{22}\mu_3^2$.

$$a_{22}\mu_3^2 T_1 = a_{22}(\varepsilon^2 M^2) \times O(\varepsilon^{-2}) = O(1).$$

Therefore, the combination

$$S := a_{22}\mu_3^2 \left(\frac{b_{11}}{2\Delta} + \frac{2b_{12}}{a_{22}\mu_3^2} \right)$$

has a finite nonzero limit as $\varepsilon \to 0$. This is the main *balanced* term of order $O(1)$.

—

3. Behaviour of denominators $(b_{22} - a_{11})$ and $(b_{12} - a_{11})$.

$$b_{22} - a_{11} = O(\varepsilon^2) - a_{11} = -a_{11} + O(\varepsilon^2) \quad \Rightarrow \quad O(1),$$

so these denominators are $O(1)$ and contribute no additional powers of ε.

Thus, each of the following factors is $O(1)$:

$$\left(a_{22} - \frac{4a_{12}\Delta}{b_{22} - a_{11}} \right) = O(1), \qquad \frac{(b_{12} - a_{12})^2}{b_{12} - a_{11}} = O(1).$$

—

4. Hyperbolic functions. Because $\mu_3 = O(\varepsilon)$, we expand:

$$\cosh(\mu_3 L) = 1 + \frac{1}{2}\mu_3^2 L^2 + O(\varepsilon^4) = 1 + O(\varepsilon^2),$$

$$\sinh(\mu_3 L) = \mu_3 L + \frac{1}{6}\mu_3^3 L^3 + O(\varepsilon^5) = O(\varepsilon).$$

Hence: - $\cosh(\mu_3 L)$ contributes $O(1)$, - $\sinh(\mu_3 L)$ contributes $O(\varepsilon)$.

—

5. Term-by-term scaling in $E(\varepsilon)$. We now identify the asymptotic order of each large bracket in $E(\varepsilon)$.

$$
\begin{aligned}
E(\varepsilon) = \Big[&\underbrace{a_{22}\mu_3^2(\cdots)(\cdots)}_{O(1)} - \underbrace{8b_{12}\Big(1+\frac{a_{12}}{\Delta}\Big)\frac{\Delta^2}{(b_{12}-a_{11})}}_{O(1)}\Big]\underbrace{\cosh(\mu_3 L)}_{O(1)} \\
&- \underbrace{b_{12}a_{22}\mu_3 L\Big(1+\frac{a_{12}}{\Delta}\Big)\sinh(\mu_3 L)}_{O(\varepsilon^2)} \\
&- \underbrace{2b_{12}\Big(a_{22} - \frac{4a_{12}\Delta}{(b_{22}-a_{11})}\Big)}_{O(1)} \\
&+ \underbrace{4\mu_3^2 a_{22}\Big(\frac{b_{11}}{2\Delta} + \frac{2b_{12}}{a_{22}\mu_3^2}\Big)\frac{\Delta^2}{(b_{22}-a_{11})}\Big(1+\frac{a_{12}}{\Delta}\Big)}_{O(1)}.
\end{aligned}
$$

—

6. Collecting orders. - The **first**, **third**, and **fourth** lines contain $O(1)$ terms. These survive in the limit $\varepsilon \to 0$.

- The **second** line (with sinh) behaves as

$$\mu_3 L \sinh(\mu_3 L) \sim \mu_3^2 L^2 = O(\varepsilon^2),$$

so it vanishes in the limit.

Hence,

$$E(\varepsilon) = E_0 + O(\varepsilon^2),$$

where E_0 is the finite algebraic expression that remains.

—

7. Computation of the finite limit. Let us write explicitly the leading part:

$$
\begin{aligned}
E_0 = \Big[&S\Big(a_{22} - \frac{4a_{12}\Delta}{b_{22}-a_{11}}\Big) - 8b_{12}\Big(1+\frac{a_{12}}{\Delta}\Big)\frac{\Delta^2}{(b_{12}-a_{11})}\Big] - 2b_{12}\Big(a_{22} - \frac{4a_{12}\Delta}{(b_{22}-a_{11})}\Big) \\
&+ 4S\frac{\Delta^2}{(b_{22}-a_{11})}\Big(1+\frac{a_{12}}{\Delta}\Big).
\end{aligned}
$$

Replacing $b_{22} - a_{11} \to -a_{11}$ and using the explicit form

$$S = \frac{a_{11}a_{22} - 4\Delta^2}{2\Delta} + 2b_{12},$$

we finally obtain

$$\boxed{\begin{aligned}E_0 = \frac{a_{22}(a_{11}a_{22} - 8\Delta^2)}{2\Delta} - 8b_{12}\Big(1 + \frac{a_{12}}{\Delta}\Big)\frac{\Delta^2}{(b_{12} - a_{11})} \\ + \frac{8}{a_{11}}\Big(\Delta^3 - b_{12}\Delta^2 - b_{12}a_{12}\Delta\Big).\end{aligned}}$$

Therefore,

$$E(\varepsilon) = E_0 + O(\varepsilon^2),$$

and the limiting equation as $\varepsilon \to 0$ becomes

$$E_0 = 0.$$

1. Asymptotic scalings. We use the known relations:

$$b_{11} = \frac{\gamma}{\varepsilon^2}, \qquad b_{22} = O(\varepsilon^2), \qquad \mu_3 = \varepsilon M + O(\varepsilon^3), \qquad \mu_3^2 = \varepsilon^2 M^2 + O(\varepsilon^4),$$

with

$$M^2 = \frac{a_{11}a_{22} - 4(b_{12} - a_{12})^2}{a_{22}\gamma}.$$

Let $\Delta = b_{12} - a_{12}$ for brevity. Then

$$\frac{b_{11}}{2\Delta} + \frac{2b_{12}}{a_{22}\mu_3^2} = \varepsilon^{-2}\left(\frac{\gamma}{2\Delta} + \frac{2b_{12}}{a_{22}M^2}\right) + O(1).$$

Multiplying by $a_{22}\mu_3^2$ cancels the ε^{-2} factor:

$$a_{22}\mu_3^2\left(\frac{b_{11}}{2\Delta} + \frac{2b_{12}}{a_{22}\mu_3^2}\right) \to a_{22}M^2\left(\frac{\gamma}{2\Delta} + \frac{2b_{12}}{a_{22}M^2}\right) =: S.$$

All remaining terms are $O(1)$ as $\varepsilon \to 0$, since

$$b_{22} - a_{11} = -a_{11} + O(\varepsilon^2), \quad \cosh(\mu_3 L) = 1 + O(\varepsilon^2), \quad \sinh(\mu_3 L) = \mu_3 L + O(\mu_3^3) = O(\varepsilon).$$

Hence, the term with $\sinh(\mu_3 L)$ vanishes as $O(\varepsilon^2)$.

2. Limit of S. Using $M^2 = \dfrac{a_{11}a_{22} - 4\Delta^2}{a_{22}\gamma}$, we obtain

$$\boxed{S \xrightarrow{\varepsilon\to 0} \frac{a_{11}a_{22} - 4\Delta^2}{2\Delta} + 2b_{12}.}$$

3. Limit of the full expression. Neglecting $O(\varepsilon)$ terms and replacing $b_{22} - a_{11} \to -a_{11}$, we find

$$\lim_{\varepsilon\to 0} E(\varepsilon) = \Big[S\Big(a_{22} + \frac{4a_{12}\Delta}{a_{11}}\Big) - 8b_{12}\Big(1 + \frac{a_{12}}{\Delta}\Big)\frac{\Delta^2}{(b_{12} - a_{11})}\Big] - 2b_{12}\Big(a_{22} + \frac{4a_{12}\Delta}{a_{11}}\Big) - \frac{4S\Delta^2}{a_{11}}\Big(1 + \frac{a_{12}}{\Delta}\Big).$$

After straightforward algebraic simplifications, this limit can be written as

$$\boxed{\begin{aligned}\lim_{\varepsilon\to 0} E(\varepsilon) = {} & \frac{a_{22}(a_{11}a_{22} - 8\Delta^2)}{2\Delta} - 8b_{12}\Big(1 + \frac{a_{12}}{\Delta}\Big)\frac{\Delta^2}{(b_{12} - a_{11})} \\ & + \frac{8}{a_{11}}\Big(\Delta^3 - b_{12}\Delta^2 - b_{12}a_{12}\Delta\Big).\end{aligned}}$$

—

4. Limiting condition. Since the full expression equals zero, in the limit $\varepsilon \to 0$ we obtain the algebraic condition

$$\boxed{\frac{a_{22}(a_{11}a_{22} - 8\Delta^2)}{2\Delta} - 8b_{12}\Big(1 + \frac{a_{12}}{\Delta}\Big)\frac{\Delta^2}{(b_{12} - a_{11})} + \frac{8}{a_{11}}\Big(\Delta^3 - b_{12}\Delta^2 - b_{12}a_{12}\Delta\Big) = 0,}$$

where $\Delta = b_{12} - a_{12}$.

—

5. Remarks.

- The term with $\sinh(\mu_3 L)$ vanishes as $O(\varepsilon^2)$.
- If $\Delta = 0$ or $b_{12} - a_{11} = 0$, the above expansion breaks down and higher-order terms in ε are required.
- If $M^2 < 0$, μ_3 is purely imaginary but the scaling $\mu_3 \sim \varepsilon$ remains valid; the algebraic limit still holds (possibly in complex form).

Thus, the limit of the given equation as $\varepsilon \to 0$ is a finite algebraic relation among the parameters a_{ij} and b_{ij} given above.

7 Scalings for the case $\Upsilon_0 < \Upsilon < \Upsilon_1$

We consider the equation

$$\begin{aligned} & m\big(q(\alpha\beta^* - \beta\alpha^*) + s^*(\alpha\beta^* + \beta\alpha^*) + 2t\beta\beta^*\big) \\ & \quad - n\big(r(\alpha\beta^* - \beta\alpha^*) + 2s^*\alpha\alpha^* + t(\alpha\beta^* + \beta\alpha^*)\big) \\ & \quad + o\big(q(\alpha\beta^* + \beta\alpha^*) - 2r\beta\beta^* + s^*(\alpha\beta^* - \beta\alpha^*)\big) \\ & \quad + p^*\big(2q\alpha\alpha^* - r(\alpha\beta^* + \beta\alpha^*) + t(\beta\alpha^* - \alpha\beta^*)\big) = 0. \end{aligned} \tag{7.1}$$

Known scalings (recordatorio). Tomamos las escalas ya usadas antes:

$$b_{11} = \frac{\gamma}{\varepsilon^2}, \qquad b_{22} = O(\varepsilon^2), \qquad b_{12} = O(1),$$

y los coeficientes a_{ij} son $O(1)$. Definimos $\Delta := b_{12} - a_{12} = O(1)$.

1. Orden de α, β. Los valores α, β vienen de

$$\alpha = \sqrt{\frac{\sqrt{c/a} - b/(2a)}{2}}, \qquad \beta = \sqrt{\frac{\sqrt{c/a} + b/(2a)}{2}}.$$

Con $a = a_{22}b_{11} = O(\varepsilon^{-2})$ y $c = (a_{11} - \Upsilon)(b_{22} - \Upsilon) = O(1) \cdot O(\varepsilon^2) + O(1)$ se obtiene

$$\sqrt{\frac{c}{a}} = O(\varepsilon), \qquad \frac{b}{2a} = O(1).$$

Por tanto α y β quedan $O(1)$ (finito, no dependen en potencia negativa de ε):

$$\boxed{\alpha = O(1), \qquad \beta = O(1).}$$

2. Orden de α^*, β^*. Por las fórmulas

$$\alpha^* = \frac{b_{11}(\alpha^2 - \beta^2) - a_{11} + \Upsilon}{-2\Delta}, \qquad \beta^* = -\frac{\alpha\beta\, b_{11}}{\Delta},$$

y usando $b_{11} = O(\varepsilon^{-2})$ y $\alpha, \beta = O(1)$,

$$\boxed{\alpha^* = O(\varepsilon^{-2}), \qquad \beta^* = O(\varepsilon^{-2}).}$$

3. Orden de $\rho_1, \rho_2, \eta_1, \eta_2$. Recordando

$$\begin{aligned} \rho_1 &= b_{11}(\alpha^2 - \beta^2) + 2b_{12}\alpha^*, \\ \rho_2 &= -2b_{11}\alpha\beta - 2b_{12}\beta^*, \\ \eta_1 &= a_{22}(\alpha\alpha^* - \beta\beta^*) + 2a_{12}\alpha, \\ \eta_2 &= -a_{22}(\alpha\beta^* + \beta\alpha^*) - 2a_{12}\beta, \end{aligned}$$

cada término que involucra b_{11} o α^*, β^* aporta un factor ε^{-2}. Por tanto

$$\boxed{\rho_1 = O(\varepsilon^{-2}), \quad \rho_2 = O(\varepsilon^{-2}), \quad \eta_1 = O(\varepsilon^{-2}), \quad \eta_2 = O(\varepsilon^{-2}).}$$

4. Orden de $m, n, o, p^*, r, q, t, s^*$. Las definiciones son combinaciones lineales de ρ_i, η_i multiplicadas por $e^{\pm\alpha L}$ y funciones trigonométricas de βL. Dado que los exponenciales y las funciones trig. son $O(1)$,

$$\boxed{m, n, o, p^*, r, q, t, s^* = O(\varepsilon^{-2}).}$$

5. Órdenes de las combinaciones interiores (p. ej. $\alpha\beta^* - \beta\alpha^*$, $\alpha\alpha^*$, $\beta\beta^*$, $\alpha\beta^* + \beta\alpha^*$, etc.). Como $\alpha, \beta = O(1)$ y $\alpha^*, \beta^* = O(\varepsilon^{-2})$,

$$\boxed{\alpha\beta^*,\ \beta\alpha^*,\ \alpha\alpha^*,\ \beta\beta^*,\ \alpha\beta^* \pm \beta\alpha^* = O(\varepsilon^{-2}).}$$

6. Orden dentro de cada paréntesis de la ecuación principal. Cada paréntesis (por ejemplo $q(\alpha\beta^* - \beta\alpha^*) + s^*(\alpha\beta^* + \beta\alpha^*) + 2t\beta\beta^*$) combina factores q, s^*, t que son $O(\varepsilon^{-2})$ con combinaciones como $\alpha\beta^*$ que son $O(\varepsilon^{-2})$. Por tanto cada término del paréntesis es

$$O(\varepsilon^{-2}) \times O(\varepsilon^{-2}) = O(\varepsilon^{-4}),$$

y la suma dentro del paréntesis es del mismo orden:

$$\boxed{\text{cada paréntesis} = O(\varepsilon^{-4}).}$$

7. Orden de cada término completo (producto por m, n, o, p^*). Como $m, n, o, p^* = O(\varepsilon^{-2})$ y cada paréntesis es $O(\varepsilon^{-4})$,

$$\text{cada término tipo } m(\cdots),\ n(\cdots),\ o(\cdots),\ p^*(\cdots) \text{ es } O(\varepsilon^{-2}) \cdot O(\varepsilon^{-4}) = O(\varepsilon^{-6}).$$

Por tanto

$$\boxed{\text{término dominante global } \sim O(\varepsilon^{-6}).}$$

8. Conclusión principal. La expresión entera de la ecuación (4.55) escala como $O(\varepsilon^{-6})$ cuando $\varepsilon \to 0$. Más precisamente,

$$\text{Left Hand Side of (4.55)} = \varepsilon^{-6}\, C_0 + O(\varepsilon^{-4}),$$

donde C_0 es una combinación algebraica finita (independiente de ε) de los *leading coefficients*:

$$\begin{aligned}
&\text{leading values of } \rho_i, \eta_i \text{ (coefficients of } \varepsilon^{-2}),\\
&\text{leading values of } \alpha^*, \beta^* \text{ (coefficients of } \varepsilon^{-2}),\\
&\text{valores finitos de } \alpha, \beta,\ e^{\pm\alpha L},\ \cos(\beta L),\ \sin(\beta L).
\end{aligned}$$

La condición de la ecuación (igualar a cero) en el límite $\varepsilon \to 0$ impone entonces la condición de cancelación del coeficiente principal:

$$\boxed{C_0 = 0.}$$

Es decir, la ecuación produce una *condición algebraica no trivial* entre los coeficientes principales (los términos de orden ε^{-2} en $\rho_i, \eta_i, \alpha^*, \beta^*$).

7.1 Explicit expression for C_0 (leading coefficient of ε^{-6})

We extract the leading ε^{-6}-coefficient C_0 of the left-hand side of (4.55). First define

$$\Delta := b_{12} - a_{12}.$$

Leading coefficients of α^* and β^*. Because $b_{11} = \gamma\varepsilon^{-2}$ is the large coefficient, the leading parts of α^*, β^* (coefficients of ε^{-2}) are

$$\alpha^*_{(-2)} = -\frac{\gamma(\alpha^2 - \beta^2)}{2\Delta}, \qquad \beta^*_{(-2)} = -\frac{\gamma\alpha\beta}{\Delta}.$$

Leading coefficients of ρ_i and η_i. Using the definitions of ρ_i, η_i and keeping only the ε^{-2} pieces we obtain

$$\rho_{1,(-2)} = \gamma(\alpha^2 - \beta^2) + 2b_{12}\, \alpha^*_{(-2)} = -\frac{\gamma a_{12}}{\Delta}\,(\alpha^2 - \beta^2),$$

$$\rho_{2,(-2)} = -2\gamma\alpha\beta - 2b_{12}\, \beta^*_{(-2)} = \frac{2\gamma a_{12}}{\Delta}\,\alpha\beta,$$

and

$$\eta_{1,(-2)} = a_{22}\big(\alpha\alpha^*_{(-2)} - \beta\beta^*_{(-2)}\big) = \frac{\gamma a_{22}\,\alpha}{2\Delta}\big(-\alpha^2 + 3\beta^2\big),$$

$$\eta_{2,(-2)} = -a_{22}\big(\alpha\beta^*_{(-2)} + \beta\alpha^*_{(-2)}\big) = \frac{\gamma a_{22}\,\beta}{2\Delta}\big(3\alpha^2 - \beta^2\big).$$

Leading coefficients of $m, n, o, p^*, r, q, t, s^*$. Define the trigonometric shorthands

$$C := \cos(\beta L), \qquad S := \sin(\beta L).$$

Then the ε^{-2} leading parts of the exponential–linear combinations are

$$m_0 := e^{\alpha L}\big(\rho_{1,(-2)}C + \rho_{2,(-2)}S\big),$$

$$n_0 := e^{\alpha L}\big(-\rho_{2,(-2)}C + \rho_{1,(-2)}S\big),$$

$$o_0 := e^{-\alpha L}\big(\rho_{1,(-2)}C - \rho_{2,(-2)}S\big),$$

$$p_0 := e^{-\alpha L}\big(\rho_{2,(-2)}C + \rho_{1,(-2)}S\big),$$

and

$$r_0 := e^{\alpha L}\big(\eta_{1,(-2)}C + \eta_{2,(-2)}S\big),$$

$$q_0 := e^{\alpha L}\big(-\eta_{2,(-2)}C + \eta_{1,(-2)}S\big),$$

$$t_0 := e^{-\alpha L}\big(-\eta_{1,(-2)}C + \eta_{2,(-2)}S\big),$$

$$s_0 := e^{-\alpha L}\big(-\eta_{2,(-2)}C - \eta_{1,(-2)}S\big).$$

Leading small combinations built with α^*, β^*. Using the leading α^*, β^* we obtain the following leading-order combinations (all are of order ε^{-2}):

$$A := \alpha\beta^* - \beta\alpha^* = -\frac{\gamma\beta}{2\Delta}\,(\alpha^2+\beta^2),$$

$$B := \alpha\beta^* + \beta\alpha^* = -\frac{\gamma\beta}{2\Delta}\,(3\alpha^2-\beta^2),$$

$$\alpha\alpha^* = -\frac{\gamma\alpha(\alpha^2-\beta^2)}{2\Delta},$$

$$\beta\beta^* = -\frac{\gamma\alpha\beta^2}{\Delta},$$

$$D := \beta\alpha^* - \alpha\beta^* = -A = \frac{\gamma\beta}{2\Delta}(\alpha^2+\beta^2).$$

Inner parentheses leading constants. Define the four leading inner parentheses that multiply m, n, o, p^* respectively:

$$P := q_0\,A + s_0\,B + 2\,t_0\,(\beta\beta^*),$$

$$Q := r_0\,A + 2\,s_0\,(\alpha\alpha^*) + t_0\,B,$$

$$R := q_0\,B - 2\,r_0\,(\beta\beta^*) + s_0\,A,$$

$$S := 2\,q_0\,(\alpha\alpha^*) - r_0\,B + t_0\,D.$$

Each of P, Q, R, S is *explicitly* computable from the previously defined quantities $m_0, n_0, \ldots$.

Final assembly of C_0. The coefficient C_0 multiplying ε^{-6} is the combination

$$\boxed{C_0 \;=\; m_0\,P \;-\; n_0\,Q \;+\; o_0\,R \;+\; p_0\,S.}$$

Expanding P, Q, R, S in terms of the primitive leading blocks gives a completely explicit (but long) formula. For completeness we write C_0 fully substituted (grouped by the outer coefficients m_0, n_0, o_0, p_0):

$$\begin{aligned}
C_0 = m_0\Big(&q_0\Big[-\frac{\gamma\beta}{2\Delta}(\alpha^2+\beta^2)\Big] + s_0\Big[-\frac{\gamma\beta}{2\Delta}(3\alpha^2-\beta^2)\Big] + 2t_0\Big[-\frac{\gamma\alpha\beta^2}{\Delta}\Big]\Big)\\
&- n_0\Big(r_0\Big[-\frac{\gamma\beta}{2\Delta}(\alpha^2+\beta^2)\Big] + 2s_0\Big[-\frac{\gamma\alpha(\alpha^2-\beta^2)}{2\Delta}\Big] + t_0\Big[-\frac{\gamma\beta}{2\Delta}(3\alpha^2-\beta^2)\Big]\Big)\\
&+ o_0\Big(q_0\Big[-\frac{\gamma\beta}{2\Delta}(3\alpha^2-\beta^2)\Big] - 2r_0\Big[-\frac{\gamma\alpha\beta^2}{\Delta}\Big] + s_0\Big[-\frac{\gamma\beta}{2\Delta}(\alpha^2+\beta^2)\Big]\Big)\\
&+ p_0\Big(2q_0\Big[-\frac{\gamma\alpha(\alpha^2-\beta^2)}{2\Delta}\Big] - r_0\Big[-\frac{\gamma\beta}{2\Delta}(3\alpha^2-\beta^2)\Big] + t_0\Big[\frac{\gamma\beta}{2\Delta}(\alpha^2+\beta^2)\Big]\Big).
\end{aligned}$$

Each occurrence of $m_0, n_0, o_0, p_0, r_0, q_0, t_0, s_0$ in the last display must be replaced by the exponential–trig combinations given above (for instance $m_0 = e^{\alpha L}(\rho_{1,(-2)}C + \rho_{2,(-2)}S)$, etc.), and each $\rho_{i,(-2)}, \eta_{i,(-2)}$ must be replaced by the closed formulas supplied earlier. After these substitutions one obtains a single algebraic expression depending only on

$$\alpha,\ \beta,\ \gamma,\ \Delta,\ a_{22},\ a_{12},\ b_{12},\ L$$

and on the trigonometric factors $C = \cos(\beta L),\ S = \sin(\beta L)$.

$$\begin{bmatrix} -\mu & 1 & 0 & 0 \\ \dfrac{x-\Upsilon}{n} & -\mu & 0 & \dfrac{2(z-w)}{n} \\ 0 & 0 & -\mu & 1 \\ 0 & \dfrac{2(w-z)}{m} & \dfrac{r-\Upsilon}{m} & -\mu \end{bmatrix}$$

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